# Tax Policy and Capital Maintenance

Jackson Mejia*

November 10, 2026

I study the theoretical, empirical, and quantitative implications of differential taxation of capital maintenance and capital investment. In standard models, depreciation is exogenous and firms cannot maintain their capital. Relaxing these assumptions yields two new theoretical results across a broad class of models. First, when firms choose maintenance optimally to reduce depreciation, tax cuts induce them to maintain existing capital less intensively, causing depreciation to rise endogenously. As a result, the tax elasticity of investment exceeds the tax elasticity of capital. Second, the first-order effect of a tax cut on capital accumulation, and hence on output and wages, is attenuated by the maintenance share of user cost. To assess these mechanisms empirically, I leverage two datasets. First, I digitize the financial filings of large railroads. Second, I leverage industry-level corporate tax returns from the Statistics of Income. Across both datasets, I show that maintenance expenditures are the same order of magnitude as investment, and the price elasticity of maintenance is large and significant. Finally, I show that these estimates materially change how to interpret the quantitative effects of tax reforms. Under a conservative calibration, I show that the tax elasticity of investment is roughly 30% larger than the tax elasticity of the capital stock. This implies that the standard fixed depreciation approach to scoring the 2017 Tax Cuts and Jobs Act would overstate the effects of the reform by a similar magnitude.

JEL-Classification: D21, D24, E22, H25, L92

Keywords: Corporate taxation, capital maintenance, investment, capital accumulation, user cost of capital.

*University of Texas at Austin, jackson.mejia@austin.utexas.edu. I am especially grateful to Jim Poterba and Chris Wolf for invaluable guidance throughout this project. Additionally, Daron Acemoglu, Martin Beraja, Raj Chetty, Tomás Caravello, Janice Eberly, Andrew Johnston, Sam Jordan-Wood, Pedro Martinez-Bruera, Ellen McGrattan, Chelsea Mitchell, Giuditta Perinelli, Adriano Rampini, Jordan Richmond, Antoinette Schoar, Matt Smith, Juan Carlos Suarez-Serrato, Iván Werning, Brian Wheaton, and participants in numerous seminars provided helpful comments and discussions. I am also grateful to ChatGPT for research assistance. This material is based upon work supported by the NSF under Grant No. 1122374, the George and Obie Shultz Fund, and the Lynde and Harry Bradley Foundation.

# 1 Introduction

Investment flows are often treated as a sufficient statistic for how capital—and hence, output and wages—respond to different kinds of shocks. That is particularly salient in the context of tax policy, where policymakers have repeatedly cut corporate taxes and subsidized investment to raise growth and wages (Romer and Romer 2010), with evidence for their effectiveness resting on the response of investment (Hassett and Hubbard 2002; Hartley, Hassett, and Rauh 2025). That inference rests on the fact that, in both popular parlance and standard models, the observed investment can be mapped one-for-one into capital and, under further general equilibrium assumptions, into output and wages (Hall and Jorgenson 1967; Chodorow-Reich et al. 2025). In this paper, I analyze how the transmission of tax cuts to capital accumulation changes when there is an additional, observable input to the capital stock—capital maintenance—and it is differentially taxed from investment. Echoing an earlier literature from Feldstein and Rothschild (1974) and McGrattan and Schmitz Jr. (1999), I show theoretically, empirically, and quantitatively that capital maintenance is too big to ignore.

Firms spend considerable resources on ensuring that their existing capital lives longer, and that causes the same firms to exhibit relatively small responses to changes in tax policy. In corporate tax data, the maintenance rate is the same order of magnitude as investment, while maintenance exceeds investment for certain types of long-lived equipment like freight cars and locomotives. This fact remains underappreciated in the context of tax policy. In standard models, the user cost of capital, which determines the optimal capital stock, depends on the after-tax return of a unit of new investment. In the Hall and Jorgenson (1967) model, for example, a firm's capital stock is determined by the opportunity cost of capital—determined by the discount rate—plus depreciation, adjusted for the marginal effective tax rate. However, in contrast to new investment, which adds new capital to the stock and is typically not immediately tax-deductible, maintenance has always been immediately tax-deductible. That neglects the cost of *maintaining* any existing capital, which, as any homeowner or car owner would know and as corporate tax returns demonstrate, is nontrivial. This carries an important implication. In any model in which the optimal capital stock is determined by the user cost of capital—which is practically all models of investment—changes in tax policy directly affect only the investment component of user cost. As I show in the context of a simple neoclassical model, the larger the maintenance share of user cost, the less tax-elastic user cost and the capital stock are. For instance, if the maintenance share of user cost is 50%, then the tax elasticity of the capital stock would be half as large as commonly understood. I call this a *capital preservation effect*.

Differential taxation of maintenance and investment, which is typically a fundamental feature of the tax system, also means that the tax elasticity of investment need not be a sufficient statistic

for the tax elasticity of capital. If some mix of maintenance and investment produces an identically valuable capital stock, then variation in the relative price of maintenance to investment should, under certain conditions, change the optimal mix of maintenance and investment.[1] Common business tax reforms, like corporate tax cuts or investment subsidies, make investment cheaper relative to maintenance, causing firms to shift their optimal input mix toward acquiring more new capital and allowing existing capital to depreciate faster. Since steady-state investment is the depreciation rate times the capital stock, this means that the tax elasticity of investment is equal to the sum of the tax elasticities of depreciation and capital. This implies that, as long as maintenance is tax-elastic, then depreciation is too, which means that the tax elasticities of capital and investment are not equal. In other words, the standard inference that the investment response to tax shocks can be read as the capital response does not hold.

The gap between investment and capital elasticities also alters how we should understand the standard approach to assessing business tax reforms: regressions of investment on the user cost. Since investment flows naturally correspond to additions to the capital stock in workhorse models, researchers in public finance and macroeconomics often assess how investment responds to tax reforms to evaluate their efficacy (Cummins, Hassett, and Hubbard 1994; Hassett and Hubbard 2002; Zwick and Mahon 2017; Chodorow-Reich, Zidar, and Zwick 2024; Hartley, Hassett, and Rauh 2025). Maintenance breaks this in two ways. First, I show that regressions of investment on user cost tend to be biased downward by omitting maintenance from the right-hand side of such regressions. Since the user cost of capital is relatively tax-inelastic when accounting for maintenance, regressions of investment on user cost tend to understate how much investment responds to tax reforms. On the other hand, that finding loses some significance when maintenance is tax-elastic; even if the investment response is larger, it cannot tell us what happens to the capital stock when maintenance, and hence depreciation, responds to the same reforms.

These alterations to the transmission of tax cuts to capital accumulation and its assessment are mediated by two variables: the maintenance share of the user cost of capital, and the elasticity of maintenance demand. Both are, to my knowledge, underexplored in the United States.[2] I estimate these two variables using two different datasets. First, I rely on a representative sample

1. This was particularly evident during the pandemic, when it became practically impossible for firms to acquire new airplanes—driving the effective price of investment goods very high—resulting in airlines ramping up spending on maintaining an older fleet of aircraft (Pfeifer 2023).

2. McGrattan and Schmitz Jr. (1999) shows that maintenance is large relative to investment in Canada, which Bachmann, Caballero, and Engel (2013) uses to calibrate the maintenance-investment ratio as roughly 40 percent. Moreover, Goolsbee (1998b) shows that the decision to *replace* capital is price-sensitive, but we lack a direct estimate of the demand elasticity. In related work, Goolsbee (1998b) provides important precedent for the mechanism I study. He shows that airlines effectively increase their depreciation rates by retiring planes more quickly in response to investment subsidies. Although the scrappage and resale decision is a distinct margin from maintenance, the economic insight is similar. It is simply that investment incentives can endogenously raise depreciation, breaking the standard one-to-one mapping from investment to capital.

of corporate tax returns from the Internal Revenue Service's Statistics of Income (SOI). Second, I construct a new dataset of firm-specific freight car and locomotive maintenance using the regulatory filings of large Class I freight railroads.

When firms file tax returns, they report maintenance expenditures separately as a line item. Matched to industry-specific tax wedges and depreciation rates from the Bureau of Economic Analysis and discount rates from Gormsen and Huber (2022), I find that the maintenance share of user cost is roughly 20% when averaged across industries. This implies that, compared to standard models, the steady-state tax elasticity of capital is roughly 80 percent as large as commonly understood. However, this is an underestimate because maintenance is labor-intensive, and any internal wages to maintenance workers would be reported on a separate line item. For instance, a large airline like Delta typically spends billions of dollars annually on maintenance, but nearly half of those expenditures would be reported as labor expenditures rather than maintenance.

To estimate the maintenance demand elasticity, which captures the investment-capital elasticity gap, I follow the empirical *investment* literature (Zwick and Mahon 2017; Curtis et al. 2021). In particular, I use pre-existing cross-sectional variation in exposure to exogenous investment subsidies throughout the 2000s and 2010s—bonus depreciation and the 2017 Tax Cuts and Jobs Act—to identify the *maintenance* demand elasticity. Because industries differ in their capital composition, these national policies create differential, plausibly exogenous shocks to the after-tax wedge between new investment and maintenance. This yields an estimated elasticity of approximately 2.6, and the elasticity is positive only for taxable firms, exactly as theory predicts. Thus, maintenance is both large and elastic.

To confirm the aggregate findings, I construct a data source unused in economics since Bitros (1976). Class I freight railroads—those with revenue over roughly $1 billion—must submit detailed accounts of their financial and real activities to the Surface Transportation Board (STB). These can be thought of as extraordinarily detailed versions of corporate 10-K filings. I hand-collect and digitize these going back several decades. These data provide an unparalleled window into firms' maintenance and investment behavior at the asset level, allowing me to observe the quantity and value of each firm's locomotives and freight cars, what they spend maintaining them, and how much of that maintenance is done in-house rather than contracted out. By contrast, Compustat, a common source of firm-level investment data, contains no data on maintenance and the National Income and Product Accounts do not either. To study maintenance, I focus on the period 1999-2023.[3]

In the freight rail data, I find that the maintenance share of user cost is far larger than the

3. Before the late 1990s, there was considerable regulatory upheaval and consolidation in the freight rail industry. This makes it difficult to study. Maintenance behavior may differ if firms expect to be acquired, and part of my data only begin in the late 1990s. See (Saunders 2003) for a history of the period.

aggregate SOI estimates suggest. In particular, I find that the maintenance share of user cost is about 55% across freight cars and locomotives, implying standard user cost formulas overstate tax responsiveness by a factor of 2.2. Moreover, maintenance spending typically exceeds investment in new assets. That may be a function of how long-lived this equipment is. Additionally, I find that about two-thirds of maintenance is performed in-house, with labor costs accounting for 40% of internal maintenance expenditures and 25% of total maintenance expenditures.

Next, to identify the maintenance demand elasticity, I construct a Bartik-style instrument that leverages variation in the geographic distribution and labor composition of maintenance costs by firm. By interacting pre-determined, firm-specific maintenance labor cost shares with plausibly exogenous state-level shocks to maintenance wages, this strategy isolates cost-driven shifts in the relative price of maintenance from unobserved firm-level demand shocks. The analysis yields a maintenance demand elasticity of approximately 4.8. Crucially, the estimated response is concentrated in adjustments to in-house maintenance, while the point estimate for outsourced maintenance is near zero. Thus, in both the SOI and R-1 data, maintenance demand is large and elastic.

The empirical findings—positive and elastic maintenance demand—alter how we interpret existing evidence on tax reform and how we conduct counterfactual analysis of it.

To quantify these forces, I embed the SOI-based estimated maintenance demand function into several different general equilibrium models calibrated to the U.S. economy, following the framework of Chodorow-Reich et al. (2025). For each model closure, I compare the response of relevant macroeconomic variables to the 2017 Tax Cuts and Jobs Act (TCJA) between a model with maintenance and a model without it. The model closures include a partial equilibrium variant, and general equilibrium variants including one with an inelastic supply of labor, crowding out, and an intermediate producer of maintenance. The results confirm the first-order importance of the maintenance margin. I find that the response of capital is roughly 50-75% as large as the response of investment, which suggests that the response of investment to tax reforms should not be read as a stand-in for the response of capital to tax reforms. Moreover, across a variety of model closures, the steady-state responses of capital are roughly 70-80% as large as in the standard framework. The response of output is less robust, with most models agreeing with the capital response. However, when maintenance is endogenously provided by intermediates, output rises relative to the standard model since intermediate inputs are reallocated to the final goods sector following a tax reform. The quantitative lesson, however, is more general than a haircut to the neoclassical model. Since maintenance acts through user cost, and the user cost is central to practically all model closures in this literature, the maintenance mechanism carries over.

Finally, I apply the first-order correction implied by the SOI maintenance share to investment

coefficients from the existing literature. The correction raises their magnitudes by roughly 22%, moving the Hassett and Hubbard (2002) consensus range from approximately 0.5–1.0 to 0.61–1.22. Despite that adjustment of investment upward, the resulting coefficients do not imply that the response of capital is any larger.

In sum, the maintenance margin is quantitatively important in the context of tax reform. Since tax policy has historically favored maintenance over investment, the economics of this paper suggest that firms have, as a result, been induced to maintain capital for longer than an undistorted equilibrium would prescribe. As a result, capital is older than it otherwise would be.

**Related Literature.** This paper relates to four main strands of literature.

First, my paper builds on the literature on multiple margins of capital adjustment. The closest paper is McGrattan and Schmitz Jr. (1999). They develop a simple neoclassical model with endogenous depreciation and asymmetric tax treatment of maintenance and investment, showing that a tax cut leads to an endogenous response of maintenance, and hence, depreciation. I build on that model in several ways. First, I extend it to the capital preservation effect. Second, I point out that the model implies a divergence between investment and capital elasticities. Finally, I discuss how this alters standard investment regression coefficients and their conceptual interpretation. While they rely on aggregate Canadian survey data to establish the magnitude of maintenance, I provide the first estimates of the maintenance demand function for the United States, then trace their implications for the aggregate effects of the TCJA. Other papers, including Kabir, Tan, and Vardishvili (2024), Boucekkine, Fabbri, and Gozzi (2010), and Albonico, Kalyvitis, and Pappa (2014), study maintenance and capital utilization in non-tax settings. Feldstein and Rothschild (1974) and Cooley, Greenwood, and Yorukoglu (1997) study how tax policy shapes capital replacement in vintage capital settings, with broadly similar logic: tax cuts can lower the value of existing capital relative to newer vintages and so accelerate replacement.

Second, maintenance's empirical relevance is well documented for housing but not for firms. A large literature documents the determinants and effects of residential housing maintenance, showing it is economically large (Knight and Sirmans 1996; Harding, Rosenthal, and Sirmans 2007; Hernandez and Trupkin 2021).[4] On the firm side, tax cuts appear to induce firms to replace old, higher-maintenance capital with younger, lower-maintenance capital. For example, Goolsbee (1998b) shows that airlines retire their airplanes more quickly when tax policy makes it favorable to buy new ones. Similarly, Goolsbee (2004) shows that firms buy capital with lower mainte-

4. The investment-maintenance distortion goes the other way in the housing tax code. Whereas improvements are deductible from the capital gains tax basis, maintenance is not, which creates a distortion in favor of the former. There is no direct evidence of the importance of that margin, but Cunningham and Engelhardt (2008) and Shan (2011) show that the 1997 Taxpayer Relief Act, which lowered the capital gains tax, increased housing mobility, which is akin to increasing the renewal rate of housing.

nance requirements following tax cuts. Some studies—which abstract from tax distortions—rely on maintenance data from India (Kabir, Tan, and Vardishvili 2024; Kabir and Tan 2024) or Canada (Albonico, Kalyvitis, and Pappa 2014; Angelopoulou and Kalyvitis 2012), but none, to my knowledge, estimate a maintenance demand function or use U.S. data. This lack of direct evidence has forced prior models to rely on assumptions. My work provides the first empirical estimate of the U.S. maintenance demand function.

Third, the user cost of capital is the primary transmission channel for business tax policy in virtually all modern frameworks, whatever complications they add—explicit demographics (PWBM 2019), heterogeneous capital (Barro and Furman 2018), heterogeneous firms (Sedlacek and Sterk 2019; Zeida 2022), lumpy adjustment (Winberry 2021), financial frictions (Occhino 2023), or global tax considerations (Chodorow-Reich et al. 2025). Since maintenance alters user cost directly, the maintenance mechanism I identify using the neoclassical benchmark similarly carries over to these richer settings.

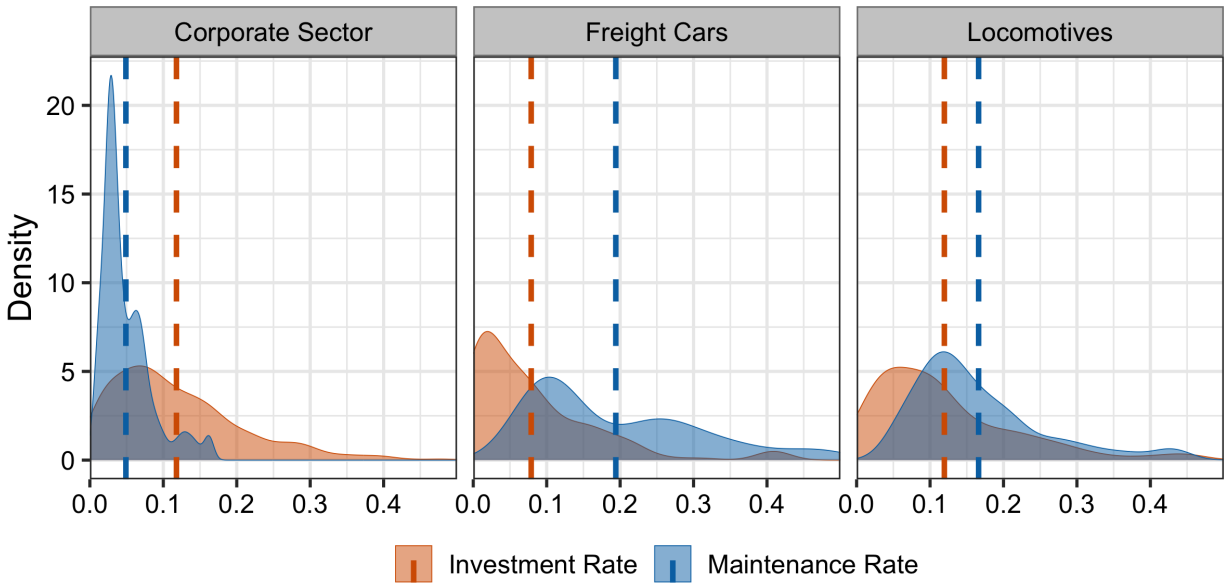
Finally, I contribute to the investment literature through an interpretive lens. Since Hall and Jorgenson (1967) and Summers (1981), that literature has produced a consensus short-run semi-elasticity of the investment rate to the user cost of roughly $-0.5$ to $-1$ (Hassett and Hubbard 2002; Chodorow-Reich, Zidar, and Zwick 2024), with modern quasi-experimental studies finding robust, sometimes larger, responses (Zwick and Mahon 2017; Kennedy et al. 2023; Chodorow-Reich et al. 2025; Hartley, Hassett, and Rauh 2025). Recently, Chodorow-Reich (2025) stresses that, for theoretical reasons, many of these regressions are conceptually distinct from one another. I make a broadly similar point about both the estimation and interpretation of these elasticities. First, since maintenance reduces the tax elasticity of the user cost of capital and these studies omit maintenance, the resulting coefficients are too small. However, since maintenance is tax-elastic, one also cannot interpret the corrected investment response as producing an equivalent amount of new capital. These mechanisms do not hinge on adopting a particular estimate of the investment elasticity.

## 2 Maintenance vs. Investment

Firms engage in a wide range of activities to change the capital stock. Those activities are distinguished by maintenance, which preserves or restores capital services without improving quality, and investment, which creates or upgrades capital. For example, a ground shipping company may make its vehicle fleet live longer through diligent attention to routine maintenance like oil changes or proactively changing the tires early to avoid worse damage through a highway tire blowout. By contrast, investment would be purchasing an entirely new fleet of vehicles or re-

placing the engine in an existing vehicle.[5] Thus, the key distinction between investment and maintenance is that the former adds new capital to the stock, while the latter simply keeps old capital around for longer in its existing quality. This distinction is not merely semantic. It maps directly to firm expenditures and how the tax code treats them.

Figure 1: Density plots for maintenance and investment rates




**Note:** Each density plot is constructed with beginning of period book capital in the denominator. The dashed lines are mean maintenance and investment rates. The rolling stock data span from 1999-2023, with 172 observations per asset type. There are 1116 observations in the SOI data, covering 49 major industries from 1999-2021.

Before turning to institutional detail about the tax treatment of maintenance, it is useful to compare maintenance and investment quantitatively. Using data that I discuss in more detail in Section 4, Figure 1 plots the distribution of maintenance and investment rates at the industry level across the corporate sector (left panel), and across two major equipment types at the firm-by-asset level. In each case, maintenance rates are substantial, suggesting that firms allocate considerable resources toward preserving existing capital, on the same order of magnitude as investment.

The tax code reinforces the distinction between maintenance and investment through differential treatment under Sections 162 and 263 of the Internal Revenue Code. In general, maintenance is immediately deductible as an operating expense, while investment must be capitalized and depreciated over time. That matters for a firm deciding whether to invest an extra dollar

5. This distinction is between *observed* investment and maintenance. However, firms may maintain their capital in ways that do not require expenditures. For example, to shield a truck from weather, it may proactively move it under shelter. That requires no expenditure, but it is nevertheless a form of maintenance. The remainder of this paper focuses solely on expenditure-based maintenance.

in new capital or spend the dollar on maintaining existing capital. Letting $\tau^c$ denote the firm's tax rate, the after-tax cost of maintenance is $1 - \tau^c$ because it is deductible. On the other hand, a new capital expenditure must be depreciated over time.[6] Let $z$ denote the net present value of the depreciation deductions, and $c$ denote any additional investment incentives like investment tax credits. The after-tax cost of investment is $1 - \tau^c z - c$. This implies a wedge in the maintenance-investment decision given by

$$\text{Wedge} = \frac{\text{After-tax Cost of \$1 of Maintenance}}{\text{After-tax Cost of \$1 of Investment}} = \frac{1 - \tau^c}{1 - \tau^c z - c}. \tag{1}$$

By shifting the wedge, tax reforms can shift the incentive to maintain existing capital, or invest in new capital. I make that statement precise in the following section.[7]

# 3 A Model of Capital Maintenance

This section develops a model where firms choose both investment and maintenance to manage their capital stock, building on Hall and Jorgenson (1967) and McGrattan and Schmitz Jr. (1999). I start by discussing the baseline neoclassical model, then show the implications of accounting for maintenance. For simplicity, I focus on time-invariant policies throughout this section.

## 3.1 The Baseline Neoclassical Model

Consider the Hall and Jorgenson (1967) model, which extends the neoclassical model of investment to incorporate a realistic tax system. This model is the appropriate benchmark because it is widely used across public finance in academia, the private sector, and government to assess the effects of tax reform. Even in more complex models featuring adjustment costs, irreversibility, or even lumpiness, the core transmission mechanism remains the same.

In this setting, a representative firm in discrete time produces output with a concave production function $F(K_t)$. Capital accumulates according to the standard law of motion

$$K_{t+1} = I_t + (1 - \delta)K_t, \tag{2}$$

in which capital $K_{t+1}$ in period $t + 1$ is the sum of undepreciated capital $(1 - \delta)K_t$ and new in-

6. Firms cannot easily game this distinction because routine maintenance expenditures typically fall below a *de minimis* safe-harbor threshold, while capital improvements do not, and established case law enforces consistent application of the rules. For example, in the 2003 court case *FedEx Corp. v. United States*, FedEx disputed whether \$66M worth of expenditures on maintaining existing aircraft engines in 1993 should be considered maintenance or investment. The question decided ten percent of their tax bill for that year.

7. Appendix Figure A.1 shows how the wedge varies by asset and over time.

vestment $I_t$. Here, tax policy enters cash flows in two distinct ways. First, profits are taxed at the corporate tax rate $\tau^c$. Second, investment is subsidized via an investment tax credit $c$, as well as a stream of depreciation deductions. That stream has present value $\tau^c z$. Discounting the future at rate $r$, the firm chooses a path for capital to maximize the discounted stream of profits subject to (2), taking tax policy and the price of investment $p^I$ as given:

$$\max_{\{I_t, K_{t+1}\}} \sum_{t=0}^{\infty} \left(\frac{1}{1+r}\right)^t \left\{(1-\tau^c)F(K_t) - p_t^I(1-\tau^c z - c)I_t\right\} \quad \text{s.t.} \quad (2). \tag{3}$$

I focus on the steady-state solution to (3), which is defined by two equations:

$$F'(K) = p^I(r+\delta)\frac{1-\tau^c z - c}{1-\tau^c} \equiv \text{UCC}^{\text{HJ}}, \tag{4}$$

$$I = \delta K. \tag{5}$$

The first equation relates the marginal product of capital to its user cost, defined as $\text{UCC}^{\text{HJ}}$, while the second relates investment to capital.

Together, these equations have two key implications. First, the effect of a tax cut on investment and capital are identical. Let $\tau$ denote the marginal tax rate, defined by

$$1-\tau \equiv \frac{1-\tau^c}{1-\tau^c z - c},$$

so that $1-\tau$ is the maintenance-investment tax wedge defined in (1). Throughout this section, I define tax elasticities with respect to $1-\tau$:

$$\varepsilon_{X,\tau} \equiv \frac{\partial \log X}{\partial \log(1-\tau)}.$$

Then for any tax reform, whether it stems from a change in the corporate tax rate $\tau^c$ or a change to investment subsidy policy through $c$ or $\tau^c z$, the tax elasticities of investment and capital are identical:

$$\varepsilon_{I,\tau} = \varepsilon_{K,\tau}. \tag{6}$$

This identity allows us to infer the effects of a tax reform on the capital stock through its effects on investment. With additional assumptions about production and equilibrium, the capital response likewise determines how output and wages respond to tax reform. That leads to the second key implication. In particular, we can, as many have done in the investment literature, learn directly about the effects of tax reform on these aggregate variables by running a regression like

$$\log I = \alpha + \beta \log \text{UCC}^{\text{HJ}} + e, \tag{7}$$

which provides a log-linear approximation to the steady-state investment demand relationship. This specification illustrates the logic underlying a large empirical literature that estimates investment responses to changes in tax incentives (Hassett and Hubbard 2002; Chodorow-Reich 2025).

Two features of the benchmark support these two implications: the Hall and Jorgenson user cost summarizes the incentive to acquire capital, and fixed depreciation makes steady-state investment proportional to capital. As I show in the next section, introducing maintenance substantially changes the implications of the model.

## 3.2 Extending to Endogenous Depreciation

Now, consider a simple extension to the Hall and Jorgenson model incorporating maintenance services, $M_t$. Following McGrattan and Schmitz Jr. (1999), let the law of motion for capital become

$$K_{t+1} = I_t + (1 - \delta(m_t))K_t, \tag{8}$$

where $m_t = M_t/K_t$ denotes maintenance intensity and $\delta(m_t)$ is depreciation as a function of how intensively firms maintain their capital. I make two assumptions on the depreciation technology $\delta(m)$.

**Assumption 1** (Depreciation Technology)**.** *The depreciation function $\delta : [0, \bar{m}) \to (0, 1)$ satisfies:*

*(i) $\delta'(m) < 0$ for all $m > 0$: maintenance reduces depreciation;*

*(ii) $\delta''(m) > 0$ for all $m > 0$: diminishing returns to maintenance.*

For the sake of parsimony, I assume that both maintenance and investment are supplied perfectly elastically here. In the appendix, I allow for maintenance to be supplied by a competitive intermediate sector. The differential tax treatment of investment and maintenance is summarized by the wedge in (1). This asymmetry is the key feature of the model. Let $p_t^M$ denote the price of maintenance services. The firm chooses sequences $\{I_t, M_t, K_{t+1}\}_{t=0}^{\infty}$ to maximize:

$$\max \sum_{t=0}^{\infty} \left(\frac{1}{1+r}\right)^t \left\{(1-\tau^c)\left[F(K_t) - p_t^M M_t\right] - (1 - \tau^c z - c)p_t^I I_t\right\} \quad \text{s.t.} \quad (8). \tag{9}$$

In steady state, the solution satisfies

$$F'(K) = p^I \frac{1 - \tau^c z - c}{1 - \tau^c}\left[r + \delta(m) - m\delta'(m)\right], \tag{10}$$

$$(1 - \tau^c)p^M = (1 - \tau^c z - c)p^I[-\delta'(m)], \tag{11}$$

$$I = \delta(m)K. \tag{12}$$

Compared to the benchmark, the maintenance-augmented system features a new optimality condition relating the marginal benefit of additional maintenance—lower depreciation—to its marginal cost, the after-tax price of maintenance. Maintenance also fundamentally alters the other, more familiar equations, resulting in a different transmission mechanism of tax policy. I discuss the implications of that in the following subsections.

## 3.3 The Capital Preservation Effect

In the baseline model, tax cuts transit to capital directly. When accounting for maintenance, there is an extra term which makes the tax elasticity of user cost—and hence of capital—less sensitive to tax reform. To see that, consider (10), which relates the marginal product of capital to its user cost. It differs from the baseline model through the additional term $-m\delta'(m)$. Separating this term and substituting the maintenance condition (11) gives

$$\begin{aligned} F'(K) &= p^I \frac{1-\tau^c z - c}{1-\tau^c}\left[r + \delta(m) - m\delta'(m)\right] \\ &= p^I (r+\delta(m)) \frac{1-\tau^c z - c}{1-\tau^c} + \frac{p^I(1-\tau^c z - c)}{1-\tau^c}[-m\delta'(m)] \\ &= \mathrm{UCC}^{\mathrm{HJ}} + p^M m \equiv \mathrm{UCC}. \end{aligned} \tag{13}$$

Here, $\mathrm{UCC}^{\mathrm{HJ}}$ is the investment component of user cost, evaluated at the depreciation rate $\delta(m)$, and $p^M m$ is the maintenance component.

The maintenance component of the user cost of capital has a straightforward interpretation. Holding total maintenance fixed, an additional unit of capital lowers maintenance intensity $m = M/K$. This raises depreciation on the existing capital stock, increasing the user cost of capital through the positive term $-m\delta'(m)$. Alternatively, maintaining the same intensity requires $m$ additional units of maintenance for each additional unit of capital. The maintenance condition makes these two ways of evaluating the additional cost equivalent at the optimum, yielding the term $p^M m$. This has two effects. First, it alters the *level* of user cost. Comparing the two models at the same depreciation rate, $\delta = \delta(m)$, user cost is higher with maintenance, leading to a lower optimal capital stock.

Second, the maintenance-augmented model exhibits a lower tax elasticity of capital. To characterize this, define the maintenance share of user cost as

$$s_m \equiv \frac{p^M m}{\mathrm{UCC}},$$

which results in an investment share given by

$$1 - s_m = \frac{\text{UCC}^{\text{HJ}}}{\text{UCC}}.$$

With that, it is straightforward to define the capital preservation effect as measuring how attenuated the tax elasticity of user cost is relative to the benchmark.

**Proposition 1** (Capital Preservation Effect)**.** *Holding interest rates and input prices fixed, and evaluating the Hall and Jorgenson benchmark at the same depreciation rate, $\delta = \delta(m)$,*

$$\varepsilon_{\text{UCC},\tau} = (1 - s_m)\varepsilon^{\text{HJ}}_{\text{UCC},\tau} = -(1 - s_m). \tag{14}$$

*The elasticities with respect to $r$ and $p^I$ are similarly attenuated by $1 - s_m$, while the elasticity with respect to $p^M$ is $\varepsilon_{\text{UCC},p^M} = s_m$. Since $\varepsilon_{K,\tau} \propto \varepsilon_{\text{UCC},\tau}$, the tax elasticity of capital is similarly attenuated. Proof: Appendix B.1.*

The proposition follows from the envelope theorem. At the optimum, the maintenance condition implies

$$\frac{\partial \text{UCC}}{\partial m} = \frac{p^I \delta'(m)}{1 - \tau} + p^M = 0.$$

Consequently, the induced change in maintenance has no first-order effect on user cost: the saving in maintenance expenditure offsets the additional replacement cost from higher depreciation. Only the investment component responds directly to tax policy. Since that component accounts for share $1 - s_m$ of total user cost, the tax elasticity is attenuated by precisely that share. The implication for capital follows from $F'(K) = \text{UCC}$. For a common elasticity of capital demand with respect to user cost,

$$\varepsilon_{K,\tau} = (1 - s_m)\varepsilon^{\text{HJ}}_{K,\tau}.$$

For example, if maintenance accounts for half of user cost, the capital response is halved. This is particularly important for long-lived assets requiring heavy maintenance. House (2014) and Koby and Wolf (2020) note that long-lived assets should be counterfactually more elastic to tax policy than we typically observe. However, if those assets require substantial maintenance, that could reverse the conclusion given by the standard model.

## 3.4 The Investment-Capital Gap

In the maintenance model, tax policy transmits differently to investment. When there are two flexible inputs to capital production, investment and capital no longer have the same tax elasticity. In the standard model, depreciation is fixed at $\delta$, so steady-state investment is $I = \delta K$ and $\varepsilon_{I,\tau} =$

$\varepsilon_{K,\tau}$. With endogenous maintenance, this identity no longer holds. To see why, consider the firm's optimal maintenance choice. Because maintenance is chosen each period without adjustment costs—i.e., there is no "stock of repair" that accumulates over time—the decision is static. The firm equates the marginal benefit of maintenance (reduced depreciation) with its marginal cost:

$$-\delta'(m) = \frac{p^M(1-\tau^c)}{p^I(1-\tau^c z - c)} = \frac{p^M}{p^I}(1-\tau).$$

There are three comparative statics which follow immediately from the first-order condition.

1. $\partial m/\partial p^M < 0$: higher maintenance prices reduce maintenance intensity.
2. $\partial m/\partial p^I > 0$: higher investment prices increase maintenance intensity.
3. $\partial m/\partial(1-\tau) < 0$: tax cuts reduce maintenance intensity.

Part (iii) is central. When investment becomes relatively cheaper through higher $z$ or $c$, or its after-tax return rises through a lower $\tau^c$, that raises $1-\tau$, causing the firm to substitute away from maintenance toward investment. This raises the depreciation rate $\delta(m)$ endogenously. The consequence for the investment-capital relationship is immediate. In steady state, investment replaces depreciated capital: $I = \delta(m)K$. Now, the depreciation rate depends on tax policy through maintenance, which implies a wedge in the tax elasticities of the capital stock and investment:

$$\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau}.$$

As long as demand for maintenance declines when taxes fall, depreciation rises and induces a gap between the tax elasticities of capital and investment. Because a tax cut raises both capital and depreciation, their tax elasticities have the same sign.

**Proposition 2** (Investment-Capital Gap)**.** *Suppose* $\varepsilon_{\delta,\tau} \neq 0$. *Then*

$$|\varepsilon_{I,\tau}| = |\varepsilon_{K,\tau}| + |\varepsilon_{\delta,\tau}| > |\varepsilon_{K,\tau}|.$$

*Proof: Appendix B.1.*

In the standard model, $\varepsilon_{\delta,\tau} = 0$ and the proposition collapses to $\varepsilon_{I,\tau} = \varepsilon_{K,\tau}$. That allows researchers to map investment elasticities into capital elasticities. However, with maintenance, some of the investment induced by tax cuts and investment subsidies reflects capital churn rather than growth in the capital stock.[8] The magnitude of the gap depends on the elasticity of maintenance demand. If maintenance demand is highly inelastic, then depreciation does not move very

8. Firms also have a scrappage margin, which Goolsbee (1998b) shows is empirically important for tax reform. Scrappage provides another channel through which tax policy can endogenously raise depreciation.

much when taxes change and the measured investment elasticity is a good proxy for the response of capital. If, on the other hand, maintenance demand is relatively elastic, then a substantial fraction of observed investment replaces endogenously higher depreciation rather than building new capital. Appendix B.2 discusses the interpretation of these effects in general equilibrium.

## 3.5 Bias in Investment Regressions

The third implication of the maintenance-augmented model relates to the conceptual interpretation of empirical investment regressions. Such regressions are typically theoretically motivated. Historically, it is common to run regressions of investment—either as an investment rate or in log-levels—on some measure of the cost of capital. That follows from the first-order condition for capital in the Hall and Jorgenson model since user cost is the primitive change in the incentive to acquire capital. However, omitting maintenance can bias the resulting regressions. Consider, for example, a regression

$$f(I_{i,t}, K_{i,t-1}) = \widehat{\beta} \log \mathrm{UCC}_{i,t}^{\mathrm{HJ}} + X_{i,t} + u_{i,t}, \tag{15}$$

where $f(\cdot)$ is either $\log I_{i,t}$ or the investment rate $I_{i,t}/K_{i,t-1}$, and $X_{i,t}$ includes controls and fixed effects.[9] If the user cost actually contains maintenance, the corresponding specification is

$$f(I_{i,t}, K_{i,t-1}) = \beta \log\left[\mathrm{UCC}_{i,t}^{\mathrm{HJ}} + p_{i,t}^{M} m_{i,t}\right] + X_{i,t} + u_{i,t}.$$

The conventional measure holds depreciation fixed, while the investment component in the full user-cost measure is evaluated at endogenous depreciation, as in (13).

**Proposition 3** (Bias in Investment Regressions)**.** *For tax-induced variation in user cost, omitting maintenance biases the investment coefficient toward zero:*

$$\beta \approx \frac{\widehat{\beta}}{1 - s_m}. \tag{16}$$

*Proof: Appendix B.1.*

The proposition follows from the capital preservation effect, with a correction factor independent of the choice of $f(\cdot)$ on the left-hand side because it operates through the measurement of user cost on the right-hand side. The intuition is straightforward. The econometrician, by

9. See Cummins, Hassett, and Hubbard (1996), Desai and Goolsbee (2004), Edgerton (2010), and Hartley, Hassett, and Rauh (2025) for examples. See Chodorow-Reich (2025) for a discussion of the theoretical interpretation of these sorts of regressions.

excluding the maintenance component, measures a user cost which moves more than it actually did. But since this has no effect on measured investment, the coefficient on the user-cost term must be larger in absolute value when the measure is corrected.[10] The correction can be substantial. If maintenance accounts for 25% of user cost, the corrected coefficient is one-third larger in absolute value than the estimated coefficient.

While the coefficient correction would typically inflate coefficients, it is independent of classic and ongoing debates about their equilibrium interpretation raised by Chodorow-Reich (2025). Regardless of whether the regression should be interpreted as holding output fixed or allowing it to adjust, the coefficient is mechanically larger by virtue of the right-hand side changing less than we would typically think. Nevertheless, the resulting observed change is not sufficient to then map into a capital response to tax reform.

## 3.6 Extensions and Robustness

The maintenance mechanism rests on three features: firms can make capital live longer through higher maintenance intensity, tax policy treating maintenance expenditures differently from investment, and the incentive to accumulate capital governed by something like its user cost. These features are not specific to the baseline neoclassical model. This section considers the robustness of the results to richer settings and extensions.

First, I consider what, if anything, changes when prices can adjust in general equilibrium. That could happen through changes in investment good prices $p^I$, maintenance prices $p^M$, interest rates $r$, or, when there is labor in the model, through wages. The last mechanism is particularly important because maintenance is labor-intensive and, in principle, business tax cuts should affect wages. In Appendix B.2, I show how these general equilibrium forces change how the maintenance mechanism propagates. I show that, for matched initial steady states, GE forces amplify the attenuation when maintenance prices respond more strongly to capital accumulation than investment good prices. This is plausible when maintenance is labor-intensive and locally supplied while investment goods are globally traded. When investment prices respond more strongly, GE forces weaken the attenuation. In Appendix F.3, I explicitly write down a competitive maintenance sector supplying maintenance services through a combination of labor and materials to a private sector. When taxes decline, lower demand for maintenance releases labor for produc-

10. This parallels Goolsbee (1998a), who shows that regressing investment on tax terms alone implicitly assumes perfectly elastic supply of investment goods. When the supply of investment goods is not perfectly elastic, then prices respond to tax cuts, and the resulting measured investment demand elasticity understates how elastic demand is. The logic of the maintenance bias can be explained in similar terms. By omitting maintenance from user cost, standard regressions overstate how elastic the user cost of capital is to a tax cut, biasing estimated demand elasticities downward.

tion and reduces materials expenditures, which can increase output.[11] Later, in an application to the 2017 Tax Cuts and Jobs Act, I quantify the magnitude of these forces and show that capital attenuation persists across model closures.

In another extension in Appendix F.1, I examine if the transmission mechanism substantially changes when there is a secondary market for capital goods. One can imagine that tax cuts may induce firms to sell their capital and upgrade to new capital, which may require lower maintenance. As a result, we could observe for certain firms that they are cutting their maintenance in response to a tax reform, while aggregate maintenance does not change substantially. In equilibrium, this attenuates the capital-investment elasticity gap—since maintenance may not be as tax-elastic in this closure—but it retains the capital preservation mechanism.

Finally, I move beyond steady-state comparisons to ask how maintenance may alter transition dynamics. In Appendix G, I examine how maintenance changes transition dynamics when capital adjusts gradually. Under the conditions established in the appendix, a larger long-run investment response combined with the same short-run response implies slower convergence in the maintenance model. This matters because most policy evaluation is done in a ten-year budget window, which is typically before the model economy has reached its steady state. In the same appendix, I quantify how much this matters in the context of TCJA.

## 3.7 Taking the Model to Data

Omitting maintenance has clear consequences for both theory and empirical work, and the consequences all flow from two parameters: the level of maintenance, and its demand elasticity. I work with a constant-elasticity demand function to illustrate each one in the empirical sections which follow this one. Suppose

$$m^* = \gamma \left( \frac{p^M}{p^I} (1 - \tau) \right)^{-\omega},$$

where $\gamma > 0$ is a demand shifter and $\omega > 1$ is the maintenance elasticity of demand. Importantly, holding other parameters fixed, $\gamma$ effectively determines $s_m$—and hence the capital preservation effect and omitted variable bias—while $\omega$ determines the tax elasticity of depreciation (and the wedge between the capital and investment tax elasticities). It can be shown that this demand function drops out of a depreciation technology satisfying Assumption 1:

$$\delta(m) = \delta_0 - \frac{\gamma^{1/\omega}}{1 - 1/\omega} m^{1-1/\omega} \tag{17}$$

11. In another extension, I examine what happens when firms can maintain capital using their own labor and materials or by purchasing services. With multiple maintenance inputs, the capital preservation effect operates through the total maintenance share of user cost, while changes in the inputs jointly determine depreciation and the investment-capital gap (Appendix F.2).

where $\delta_0 > 0$ is baseline depreciation without maintenance. While it is straightforward to see how $\gamma$ determines $s_m$, it is useful to show the relationship between parameters and the tax elasticity of depreciation, evaluated around $\tau = 0$ and $p^M/p^I = 1$, as:

$$\varepsilon_{\delta,\tau} \approx \frac{\gamma\omega}{\delta_0 + \frac{\gamma\omega}{1-\omega}}. \tag{18}$$

Under the required parameter region for the demand elasticity ($\omega > 0$), the tax elasticity of depreciation is increasing in the elasticity of demand $\omega$ and the level parameter $\gamma$. The following sections are devoted to understanding what these parameters plausibly are in the data.

# 4 Data

This section introduces two novel sources of maintenance data for the United States. First, I digitize regulatory filings from Class I freight railroads spanning 1999-2023, providing firm-asset level detail on maintenance expenditures, capital stocks, and input costs. To my knowledge, this is the first use of R-1 data in modern economics since Bitros (1976). Second, I construct industry-level measures from IRS Statistics of Income (SOI) corporate tax returns covering 1999-2021. While the SOI has been used to study investment (e.g., Zwick and Mahon (2017)), this is the first application to maintenance. Given the theory, each dataset must provide two objects: the maintenance share of user cost $s_m$, which requires data on maintenance expenditures, capital stocks, depreciation rates, interest rates, investment prices, and tax parameters; and variation in relative prices sufficient to estimate the elasticity of maintenance demand $\omega$. These datasets provide complementary evidence: the railroad data provide detailed variation within firms and asset types, while the SOI provides economy-wide coverage.

## 4.1 R-1 Data from the Surface Transportation Board

Class I freight railroads—defined as having revenue exceeding roughly $1 billion—account for about 40% of U.S. freight transportation. Following industry consolidation in the 1980s and early 1990s, the sector has consisted of approximately seven large firms in stable competitive equilibrium since the late 1990s.[12] By regulation, these firms must file annual R-1 reports with the Surface Transportation Board (STB). These reports detail the size and composition of firms' capital in value and quantities, trackage by state, taxes paid, capital expenditures, and maintenance

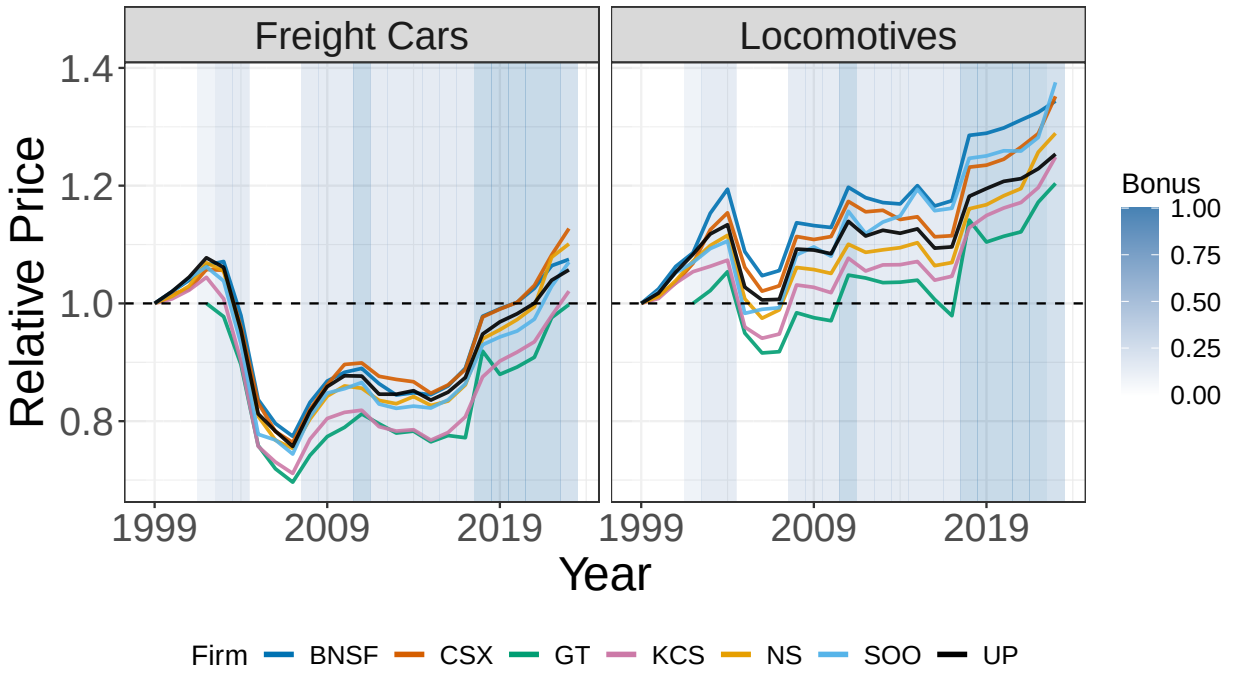
12. The seven firms are CSX, Burlington Northern & Santa Fe, Union Pacific, Norfolk Southern, Kansas City Southern, Soo Line, and Grand Trunk (operated by Canadian National Railway). I end the sample in 2023 due to subsequent industry consolidation.

expenditures broken down by capital type and input source (materials, labor, or purchased services). Reports are independently audited and provide an unparalleled window into firm capital structure and maintenance behavior. The data span 1999-2023.

For this paper, I maintain a narrow focus on freight cars and locomotives. These assets offer two advantages for studying maintenance demand. First, maintenance activities and associated prices can be straightforwardly identified in the data, whereas this is not true for other capital types. Second, track maintenance is strictly regulated by the Federal Railroad Administration, while locomotive and freight car maintenance faces considerably less regulatory constraint. This provides meaningful variation in firm maintenance choices, which is essential for identification.

The main measure of the maintenance rate is the ratio of maintenance expenditures to lagged book capital. This is the empirical analogue to the model object. I also construct two alternative metrics: a physical maintenance rate (maintenance expenditures per horsepower for locomotives or per freight ton capacity for freight cars), and separate internal and external maintenance rates. The distribution of maintenance rates for both asset types appears in Figure 1.

Figure 2: After-tax relative price of maintenance to investment




**Notes:** I construct the after-tax relative price of maintenance to investment as described in the main text and in Appendix C.1. Background shading is for bonus depreciation.

Next, I construct firm- and asset-specific relative prices as

$$P_{i,j,t} = \frac{p^M_{i,j,t}}{p^I_{j,t}} \frac{1 - \tau^c_{i,t}}{1 - \tau^c_{i,t} z_t}, \tag{19}$$

where $p^M_{i,j,t}$ is the pre-tax maintenance price of capital good $j$ for firm $i$ at time $t$, $p^I_{j,t}$ is the investment price of asset $j$, and $\tau^c_{i,t}$ is the firm-specific tax rate.

For internal maintenance, I combine firm-specific maintenance wages from Wage Form A&B with the BLS rolling-stock parts index in a Törnqvist index, using firm-by-asset labor and materials cost shares. I proxy external maintenance prices with the expenditure-weighted internal price across railroads and combine internal and external prices using each firm-asset's purchased-services share. Investment prices $p^I_{j,t}$ come from the BLS locomotive and freight car price series. Due to data availability, only the pre-tax price of maintenance varies by firm and capital type, whereas tax rates vary by firm and investment prices by capital type. Depreciation allowances do not vary between assets because the IRS places locomotives and freight cars in the same depreciation category. Further details on data construction and summary statistics are in Appendix C.1.

Figure 2 plots the after-tax relative price of maintenance to investment for all Class I railroads since 1999. The relative price of maintaining locomotives has persistently increased since 2000, while the pattern is more U-shaped for freight cars. Background shading indicates periods of bonus depreciation, which reduced the relative price of investment and thus increased the maintenance-investment wedge. However, bonus depreciation follows a common national schedule for both asset types, while tax treatment can still differ across firms. The variation I later exploit for the railroad data comes from relative input prices, $p^M/p^I$, which vary with firm-specific maintenance wages, input cost shares, and asset-specific investment prices. Section 6.2 uses geographic wage variation to instrument for relative prices and estimate the elasticity.

To construct the maintenance share of user cost, I also require depreciation rates and a cost of capital. I use economic depreciation rates for railroad equipment (4% for locomotives, 3% for freight cars) and the industry-average discount rate for NAICS 48 from Gormsen and Huber (2022). Appendix C.1 provides details.

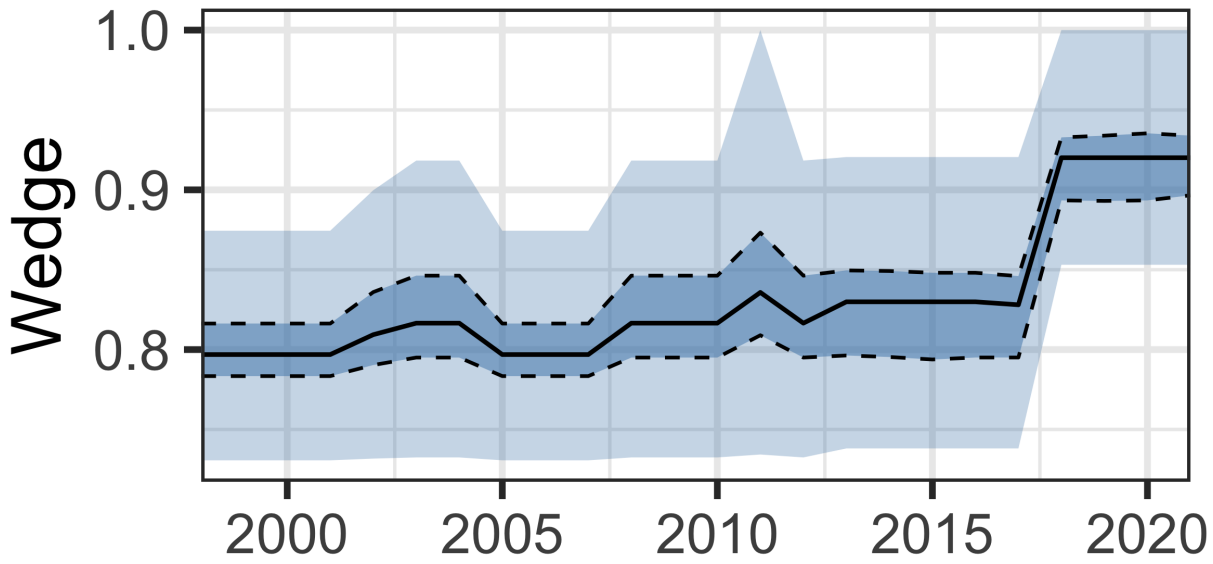
## 4.2 Industry Data from the Statistics of Income

The IRS Statistics of Income (SOI) aggregates corporate tax returns into industry-level samples at approximately the three-digit NAICS level. This is the only economy-wide collection of maintenance data at annual frequency in the United States. Corporations report maintenance expenditures on the repairs and maintenance line of Form 1120 (line 14) and book capital on Schedule L,

which the SOI aggregates across firms within industries. I use SOI data from 1999-2021, aggregating to 49 industries to correspond with Bureau of Economic Analysis (BEA) classifications.[13] For some analyses, I separate the sample into taxable firms (those with positive net income) and non-taxable firms.

The SOI maintenance rate—the ratio of maintenance expenditures to lagged book capital—is noisier than the R-1 measure for two reasons. First, because labor expenditures appear as a separate line item on tax forms (line 13 of Form 1120), reported maintenance may omit these wages and understate total maintenance costs.[14] Second, the capital stock denominator uses book values, which may not accurately reflect economic capital stocks. Despite these measurement issues, the resulting maintenance rates are similar to those observed in aggregate Canadian data.[15] Figure 1 plots the distribution of SOI maintenance rates across industries and years.

Figure 3: Distribution of industry-specific wedges over time




**Notes:** Every line is a quartile of the industry-specific wedge defined in the main text.

Because there is no easily identifiable measure of relative prices at the industry level, my identification strategy for the SOI data relies on the policy wedge defined in equation 1. I construct

13. I exclude filings made with Forms 1120S, 1120-REIT, and 1120-RIC, and remove the financial sector. See Appendix C.2 for details on the SOI-BEA industry mapping and why I use the BEA definition.

14. The labor cost share in the maintenance and repair sector (NAICS 811) is typically 30%. If omitted labor accounts for this share of total maintenance costs, the corrected user cost is $\text{UCC}^{\text{HJ}} + p^M m/0.7$. This adjustment raises the average SOI maintenance share of user cost from about 18% to 24%. I retain the unadjusted measure in the main analysis, making the maintenance-share estimates and the implied capital preservation effect and investment-coefficient corrections conservative. Appendix C.2 provides details.

15. See Appendix Figure C.5 for a comparison.

asset-specific present values of depreciation allowances for every BEA asset using the mapping from assets to the tax code in House and Shapiro (2008). I then aggregate these allowances using pre-reform capital weights and construct an industry-specific tax wedge, where the weights reflect each industry's capital composition. Industries with more equipment-intensive capital structures face larger policy-driven changes in the maintenance-investment wedge when tax incentives vary.[16]

Figure 3 plots the distribution of industry-specific wedges over time. Tax reforms throughout the 2000s—particularly expansions of bonus depreciation—tended to increase the wedge proportionally more for equipment-intensive industries, while structures-intensive industries saw little change until the TCJA in 2017. There is significant variation in the incentive to substitute between maintenance and investment over time, which is exactly what I require to estimate the elasticity of maintenance demand.

To construct the maintenance share of user cost, I use industry-specific depreciation rates from the BEA Fixed Asset Tables, the BEA investment deflator, and industry-specific discount rates from Gormsen and Huber (2022). Appendix C.2 provides details.

# 5 The Maintenance Share of User Cost

Standard models of capital taxation assume the entire user cost responds to tax incentives. In practice, only the investment component responds directly to investment tax incentives at given interest rates and input prices, while the maintenance component is already fully expensed. The maintenance share of user cost $s_m \equiv p^M m^*/\text{UCC}$ thus governs how much investment tax incentives actually reduce the cost of capital services. If maintenance accounts for half of user cost, the direct tax elasticity of user cost is half as large as standard models predict. Figure 4 presents estimates from both the SOI and freight rail datasets.

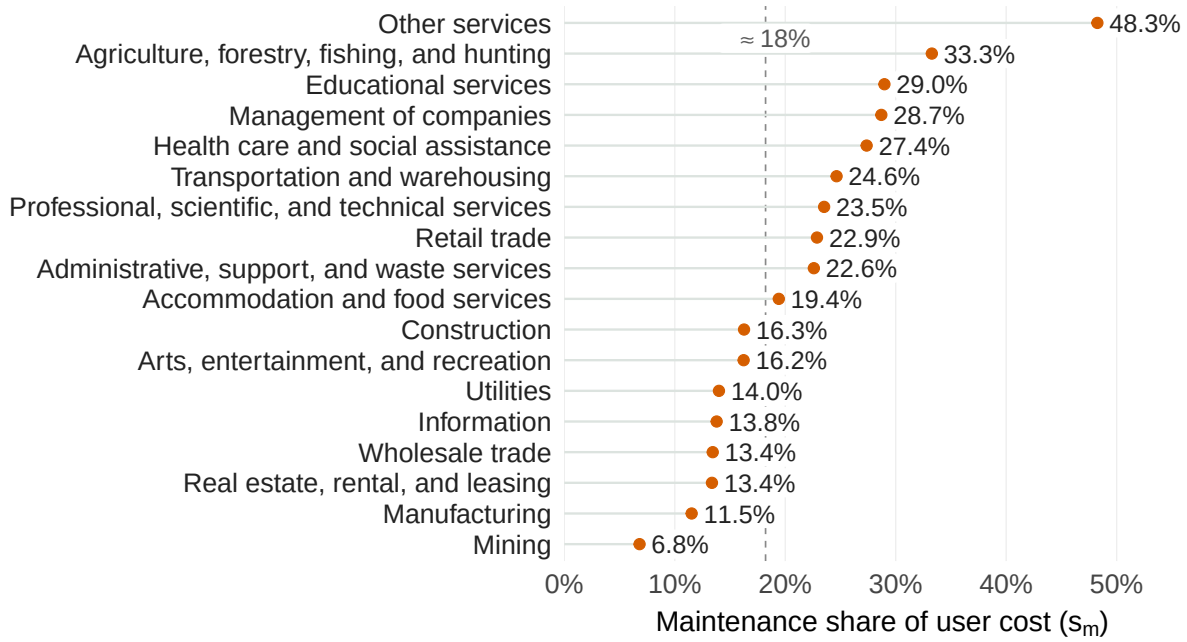
The more aggregated SOI data show a maintenance share of around 18%, though this estimate is likely a lower bound. That is because the SOI maintenance expenditures likely exclude internal labor costs, which may appear on reported forms as wages. We can assess how much bigger the maintenance share could be using external sources. Appendix Figure C.2 plots the labor share of costs in the maintenance repair sector (NAICS 811) from the Economic Census alongside the labor cost share of rolling stock maintenance. The former is generally around 30%, while the latter is 35–40%. If omitted labor accounts for 30% of total maintenance costs, correcting both maintenance expenditures and total user cost raises the average maintenance share to about 24%. Assuming omitted labor shares of 35–40% would instead raise it to 25–26%. For the remainder of

16. Appendix C.2 provides details on asset-level wedge construction and capital weighting.

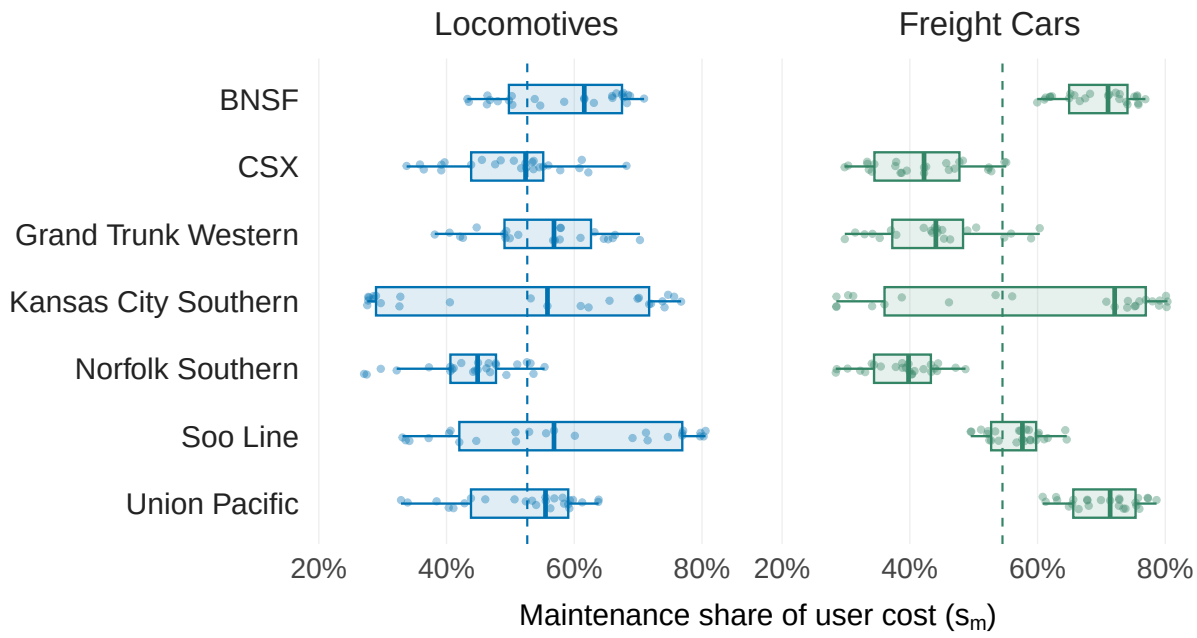
this paper, I use the more conservative estimate of 18%, which implies an 18% attenuation of the direct tax elasticity of user cost relative to the Hall and Jorgenson benchmark.

Figure 4: The Maintenance Share of User Cost

(a) Corporate sector: broad industries




(b) Railroad equipment: within firm-asset distributions




**Note:** Panel (a) shows equal-weighted averages of industry means within broad industry groups, using SOI data for 1999–2021. The dashed line is the mean across industry-year observations (18%). Panel (b) shows annual R-1 observations for 1999–2023. Boxes show the interquartile range and median within each firm–asset pair; dots are annual observations. Dashed lines are asset-specific means: 53% for locomotives and 55% for freight cars. See Appendix C for construction details.

The average also conceals substantial variation across industries. Maintenance accounts for roughly 7% of user cost in mining and 12% in manufacturing, compared with 33% in agriculture and 48% in other services. The implied attenuation of the tax elasticity of user cost thus varies considerably across industries.

The railroad data show maintenance shares averaging 53% for locomotives and 55% for freight cars. These large shares reflect both substantial maintenance expenditures and low depreciation rates. Locomotives and freight cars are long-lived assets, so they have low depreciation rates. With a maintenance share of 55%, standard models would overstate the tax elasticity of user cost by a factor of roughly 2.2.[17]

Maintenance shares also vary across railroads and over time within firm–asset pairs. Kansas City Southern and Soo Line exhibit particularly wide distributions for locomotives, while freight-car maintenance shares are persistently higher at BNSF and Union Pacific. Even within a narrowly defined asset class, the average maintenance share conceals substantial heterogeneity.

# 6 Estimating The Maintenance Demand Elasticity

The elasticity of maintenance demand governs the investment-capital gap. I estimate it from two complementary sources of variation. My primary estimate is economy-wide: in industry-level tax data, differential exposure to changes in investment subsidies and the corporate tax rate generates policy-driven variation in the tax wedge between maintenance and investment. I then corroborate the economy-wide estimate with more granular data by evaluating how exogenous shocks to the relative price of maintenance affect the maintenance intensity of locomotives and freight cars. The two designs deliver large, positive elasticities of comparable magnitude.

## 6.1 Evidence from the Statistics of Income

My primary estimate of the maintenance elasticity comes from the Statistics of Income (SOI), the only economy-wide record of maintenance expenditures in the United States, reported annually at the industry level. Two features make it the right place to recover the parameter the model needs. First, the data span the whole corporate sector. Second, aggregation to the industry level nets out the reallocation of capital within industries. I use the SOI estimate in the quantitative exercise because of its broad coverage. Appendix F.1 treats the netting-out formally.

17. The depreciation rates used here reflect average depreciation given average maintenance behavior over the sample period, making them appropriate for roughly measuring the level of $s_m$ at current policy.

**Identification Strategy.** The SOI has no direct measure of the relative price of maintenance, so I identify the elasticity from policy-driven variation in the tax wedge of equation 1. Industries use different types of capital, and that variation leads to differences in their pre-existing exposure to bonus depreciation. Following Zwick and Mahon (2017), I use this variation to measure how more-exposed industries respond to repeated phase-ins and -outs of bonus, as well as to the 2017 Tax Cuts and Jobs Act. Bonus was enacted at 30% in 2002, allowed to lapse over 2005–2007, and reinstated in 2008, while the 2017 Tax Cuts and Jobs Act raised bonus to 100% and cut the corporate rate from 35% to 21%. There is substantial evidence that both kinds of reforms boosted investment (House and Shapiro 2008; Zwick and Mahon 2017; Garrett, Ohrn, and Suárez Serrato 2020; Kennedy et al. 2023; Chodorow-Reich et al. 2025). Following Zwick and Mahon (2017), I estimate the maintenance demand elasticity using a continuous-intensity difference-in-differences design. Each industry's pre-reform equipment intensity supplies the cross-sectional exposure, while the bonus schedule supplies the time variation.

In particular, I estimate:

$$\log m_{i,t} = \alpha_i + \lambda_t - \omega \log\left(\frac{1-\tau_t^c}{1-\tau_t^c z_{i,t}}\right) + \text{Controls} + \eta_{i,t}, \tag{20}$$

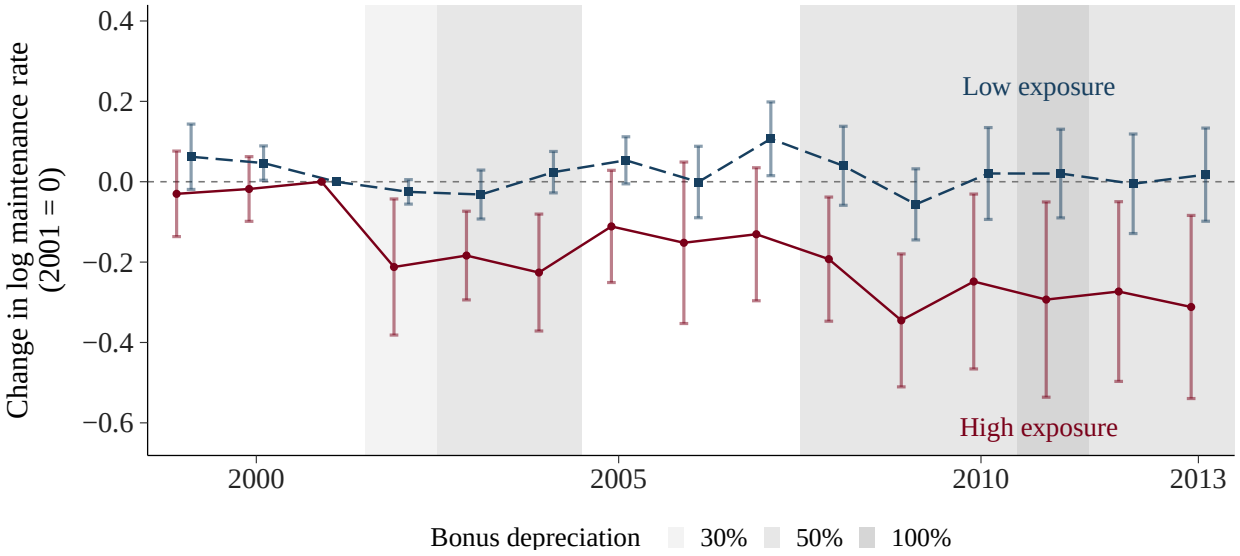
where $\alpha_i$ is an industry fixed effect and $\lambda_t$ is a time fixed effect, and the tax term is the wedge from equation 1. Bonus depreciation raises $z_{i,t}$ differentially depending on each industry's pre-existing capital stock. I construct $z_{i,t}$ using pre-reform capital weights, so any resulting variation over time reflects changes in the tax schedule rather than contemporaneous changes in the composition of the capital stock. This captures the economically relevant margin: firms are differentially exposed based on their existing stock of capital, which is what they maintain.[18]

The identifying assumption is parallel trends: absent bonus, industries with different exposure would have followed parallel maintenance trends, conditional on the controls. I provide some support for this assumption by grouping industries into the top and bottom terciles of exposure and running an event study comparing maintenance behavior as bonus was turned on and off. Figure 5 plots the adjusted maintenance paths for the two groups from 1999–2013, each normalized to 2001. The pre-reform differences are statistically indistinguishable from zero. Maintenance in the high-exposure group then falls relative to the low-exposure group when bonus is introduced in 2002. The gap narrows in 2005 and 2006 but does not disappear throughout the lapse, and generally widens after bonus is reinstated in 2008. This comparison follows the event-study

18. One could also consider an investment-weighted scheme, as is standard in investment elasticity studies (e.g., Zwick and Mahon (2017)). In those studies, $z$ is constructed by taking a weighted average of recent investment flows. This captures the elasticity for recently acquired assets. However, maintenance applies to the entire installed capital base, not just recent purchases, making investment weights conceptually misaligned. I run a variant of the baseline regression with investment weights in Appendix Table D.8 and get null results.

approach in Zwick and Mahon (2017).

Figure 5: Calendar Difference-in-Differences




**Note:** The figure plots regression-adjusted changes in log maintenance rates for the top and bottom terciles of baseline (2001) exposure $z_i$, each normalized to zero in 2001. The estimation uses data from 1999–2013, a capital-age control, and industry and year fixed effects. Error bars are 95% confidence intervals clustered by industry. Shading indicates the bonus depreciation rate. The unshaded 2005–2007 window is the lapse.

**Results.** Equation (20), estimated on the continuous wedge, quantifies the pattern in Figure 5. Table 1 reports estimates for three samples (all firms, tax-paying firms, and untaxed firms), each with and without a control for capital age (the ratio of gross to net book capital, since older capital may require more maintenance). All columns include linear and quadratic two-digit NAICS trends. The trends absorb smooth differences in maintenance paths across broad industry groups. In the full sample the elasticity is $\omega = 2.61$ (column 1) and is little changed by the addition of a control for age.

I provide further validation of the mechanism by splitting the sample into firms with positive net income ("taxable") and firms without positive income ("untaxable").[19] In principle, the direct tax incentive to substitute away from maintenance is weaker for untaxable firms, so a larger response among taxable firms is consistent with the model. Columns 3–6 of Table 1 show the estimates for each subsample. The taxable firm elasticity rises to 3.07 and becomes more statistically significant compared to the full sample. On the other hand, the untaxable elasticities are centered at zero. Thus, as a placebo test for the theory, the sample split provides sound evidence in favor of the hypotheses advanced in Section 3.

19. The mapping is rough because some firms in the taxable sample carry net operating losses forward, so they are not actually taxable.

Table 1: The SOI Maintenance Elasticity

| | All | | Tax-paying | | Untaxed | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $\hat{\omega}$ | 2.61** | 2.71** | 3.07** | 3.16*** | −0.03 | −0.09 |
| | (1.01) | (1.04) | (1.15) | (1.13) | (1.69) | (1.70) |
| $N$ (industry-years) | 1,116 | 1,113 | 1,051 | 1,051 | 1,049 | 1,049 |
| Industry and year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Capital age control | | ✓ | | ✓ | | ✓ |
| Two-digit NAICS trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

**Notes:** Estimates of (20); $\hat{\omega}$ is minus the wedge coefficient. All columns divide maintenance by lagged capital and include industry and year fixed effects and quadratic two-digit NAICS trends, with industry-clustered standard errors. Columns (2), (4), and (6) add capital age; columns (3)–(4) use taxable firms and columns (5)–(6) use untaxable firms.

**Robustness and Validation.** The continuous-intensity design invites three concerns about the estimate, which I take up in turn. The first is that the result is an artifact of how the regression is specified. However, Appendix Table D.6 shows that across alternative trend specifications, sample windows, and capital measures, the elasticity stays between roughly 1.3 and 3.2. Every alternative estimator is significant at least at the 10% level except for when replacing the SOI measure with BEA capital, which yields an elasticity of 1.5 with a standard error of 1.11. However, the BEA pools corporate and non-corporate capital in their industry capital measures, which makes it difficult to interpret. The second issue may be that the difference-in-differences is invalid, either because exposed and unexposed industries were already diverging or because the two-way fixed-effects estimator is biased by treatment-effect heterogeneity. While standard robustness checks using group-time estimators based on staggered adoption are infeasible because all industries face the national policy changes simultaneously, Appendix Table D.7 reports a variety of diagnostics. I show that pretrends are flat, that a Yatchew (1997) test does not reject linearity, and that the elasticity response is monotonic in the residualized within-fixed-effects dose.

The third concern is the most specific. House (2014) and Koby and Wolf (2020) argue that long-lived capital is, in principle, most price-elastic, so long-lived equipment is both the most exposed to bonus and the most price-sensitive. That means cross-sectional exposure could pick up a

pre-existing elasticity gradient rather than the average response. Appendix Figure D.4 examines this possibility. The interaction between equipment intensity and the wedge is small and statistically insignificant, providing no evidence that the maintenance elasticity varies with equipment intensity.

Finally, I examine the dynamic response of maintenance to tax policy. If the response accumulates over time, that suggests some adjustment costs. In Appendix Figure D.5, I show, using local projections, that maintenance responds essentially contemporaneously, with the static response at the peak.

## 6.2 Evidence from Freight Railroads

The SOI estimate is economy-wide, but it aggregates across firms within industries. To see how maintenance responds to relative prices at the level of individual assets at the firm level, I turn to a hand-collected panel of the locomotives and freight cars owned by seven Class I railroads from 1999 to 2023. In this case, I rely on variation in the after-tax relative price of maintenance to investment. Let $P_{i,j,t}$ denote the firm $i$ by asset $j$ by year $t$ after-tax relative price of maintenance to investment defined in Section 4. I estimate the maintenance elasticity of demand $\omega$ with

$$\log m_{i,j,t} = \alpha_{ij} + \lambda_t - \omega \log P_{i,j,t} + \text{Controls} + u_{i,j,t}, \tag{21}$$

where $\alpha_{ij}$ is a firm-by-capital type fixed effect and $\lambda_t$ is a time fixed effect. Since firms differ in their mix of internal and external maintenance, as well as in the labor intensity of their maintenance, they face variation in relative prices. I cluster standard errors by firm to account for correlation in maintenance decisions within firms. For example, a firm may have correlated maintenance strategies for locomotives and freight cars if they decide to upgrade their fleet.

**Identification Strategy.** The relative price $P_{i,j,t}$ is likely endogenous. If firms experiencing higher maintenance demand also have systematically higher input costs or choose different factor mixes, then OLS estimates will be biased toward zero, understating the maintenance demand elasticity. For example, unexpected repair needs may require more overtime or higher-paid workers, raising both maintenance expenditures and the measured price of maintenance. On the other hand, regional expansions may raise both wages and the demand for freight, leading firms to defer maintenance to maximize utilization. In this case, the elasticity would be biased upward.

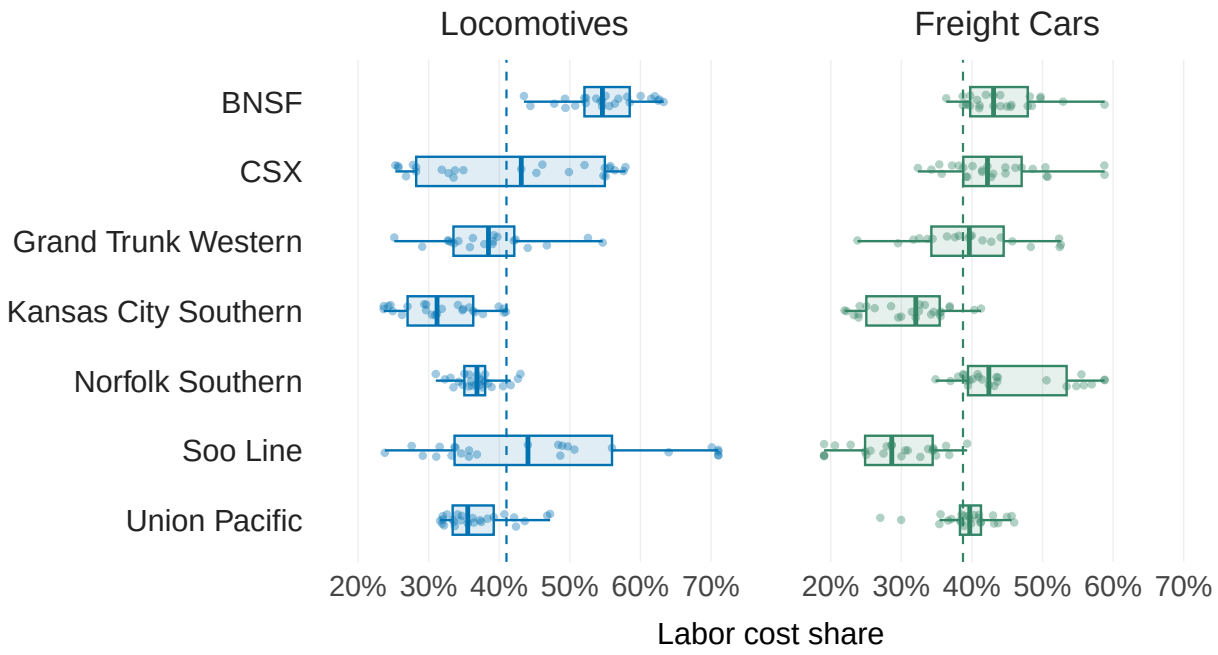
To address this endogeneity, I construct an instrument that isolates exogenous variation in

maintenance costs:

$$Z_{i,j,t} = \frac{\text{Labor}_{i,j,t-2}}{\text{Internal Maintenance}_{i,j,t-2}} \sum_{s=1}^{S} \frac{\text{Rail Miles}_{i,t,s}}{\text{Rail Miles}_{i,t}} W_{s,t}. \tag{22}$$

This is a shift-share design. The first term is each firm's lagged equipment-specific labor share of maintenance expenditure. This captures each firm's internal maintenance technology, and the fact that more labor-intensive firms are more exposed to wage movements. These are the *shares*. Next, the *shifts* are firm-specific state-weighted average wage indices made up of two parts. The first part is a state-level wage index for general maintenance occupations (BLS SOC code 49-0000). The second is the firm's geographic presence in each state, which comes from their share of rail miles in that state. The state-level wages reflect broad labor market conditions in a particular state plausibly outside the control of the railroad.

Figure 6: Internal Labor Cost Share for Rolling Stock




**Note:** The figure plots two-year-lagged internal labor cost shares by railroad and asset type. Shares are winsorized at the 2nd and 98th percentiles within asset type. Boxes show the interquartile range and median within each firm–asset pair; dots are annual observations. Dashed lines are asset-specific means.

Figure 6 plots the distribution of lagged labor cost shares by firm and asset type, showing substantial variation both within and between firms over time. Appendix Table D.1 shows that the the first stage has the correct sign with an effective Montiel-Olea and Pflueger (2013) $F$-statistic of 12.6 in the headline specification. I also show F-stats and first-stages for other instrument types. Since

the instrument is barely above the conventional threshold, I include weak-instrument-robust inference in all specifications.

As supplementary diagnostics for the instrument, I run several balance tests. This is to address the concern that railroads with greater exposure to wage shocks may also have persistently different maintenance needs, so the estimated elasticity could reflect those needs rather than a response to maintenance costs. Appendix Figure D.1 shows that lagged maintenance rates are not significantly predicted by $Z_{i,j,t}$. Similarly, Appendix Figure D.2 shows that three-year-lagged labor cost shares are not significantly related to prior maintenance rates or asset age, although they are positively related to prior relative prices. In all, the diagnostics provide no evidence that the instrument is systematically related to persistent differences in the maintenance requirements of these firms.

For the identification strategy to work, the exclusion restriction must hold. In this case, that means that the wage shock must affect maintenance only through the relative price. This requires that, conditional on fixed effects and controls, state-level wage growth is unrelated to maintenance demand shocks associated with lagged labor cost shares. The instrument has two features which make this plausible. First, state-level wage indices are plausibly exogenous to the railroad's own maintenance choices. Second, lagged labor cost shares prevent current maintenance choices from responding endogenously to the exposure weights. Since wage shocks may nevertheless correlate with broader demand shocks, I include an additional control. In particular, I use a track-mileage-weighted average of state GDP growth to account for differential exposure to local business cycles that could otherwise move wages and maintenance together.[20]

**Results and Robustness.** Table 2 presents estimates of equation (21) for all maintenance and, separately, for its internal and external components, which I examine below. For all maintenance, the baseline OLS estimate yields a point estimate of 1.74. Instrumenting for relative prices produces a substantially larger elasticity: the TSLS point estimate is approximately 4.8, above the economy-wide elasticity from the SOI. The larger IV estimate is consistent with measurement error in relative prices biasing the OLS estimate toward zero. Because the instrument is only moderately strong, I follow Lee et al. (2022) and Lal et al. (2024) and compute 95% confidence intervals by inverting the Anderson and Rubin (1949) and tF statistics. While wider than analytic intervals, both reject $\omega = 0$ and $\omega = 1$. These intervals support the restriction $\omega > 1$ used in the depreciation technology in Section 3.7.

20. Across all specifications, I also control for the age of the equipment, since older equipment may require more maintenance. I proxy for capital age by taking the ratio of gross to net book capital.

Table 2: The Railroad Maintenance Elasticity

| | All | | Internal | | External | |
|---|---|---|---|---|---|---|
| | OLS | IV | OLS | IV | OLS | IV |
| $\hat{\omega}$ | $1.74^{***}$ | $4.76^{***}$ | $2.81^{***}$ | $8.43^{***}$ | $-0.52$ | 0.12 |
| | (0.26) | (1.07) | (0.74) | (1.76) | (0.59) | (3.74) |
| $F$-statistic | 12.63 | | 12.63 | | 12.63 | |
| $t_F$ 95% CI | [1.48, 8.04] | | [3.02, 13.84] | | [-11.35, 11.58] | |
| AR 95% CI | [3.03, 8.37] | | [3.84, 12.13] | | [-6.99, 10.96] | |
| $N$ | 340 | | 340 | | 340 | |
| Twoway FE | ✓ | | ✓ | | ✓ | |
| Controls | ✓ | | ✓ | | ✓ | |

$^{*}$ p < 0.1, $^{**}$ p < 0.05, $^{***}$ p < 0.01

**Notes:** Estimates of (21) for log maintenance expenditure per unit of beginning capital. The 2SLS columns instrument the log internal-maintenance price relative to investment with (22). All columns include age and GDP controls, firm-asset and year fixed effects, and firm-clustered standard errors. $\hat{\omega}$ is minus the price coefficient; $t_F$ and AR are weak-IV-robust 95% intervals.

I now present two types of evidence supporting the baseline results: first, the internal-external maintenance split; second, a summary of additional robustness checks detailed in Appendix D.1.

*Internal vs. External Maintenance.* The preceding results aggregate all maintenance into a single input. In practice, maintenance may be done internally or externally, with the latter often done through outside contracts. That may imply some difference in the degree of flexibility firms have in their maintenance operations. For instance, firms can presumably flexibly change their internal maintenance, while external maintenance may be contracted in advance through service agreements negotiated at equipment purchase, which may limit short-run adjustments. In that case, we may expect maintenance demand to be more flexible through an internal margin.

To examine this mechanism, the remaining columns of Table 2 re-estimate equation (21) separately for internal and external maintenance. While the internal maintenance elasticity is large and significant—nearly double the pooled estimate—external maintenance has point estimates near zero. Its wide confidence interval, however, does not rule out economically meaningful external adjustment. Altogether, the evidence suggests that maintenance price shocks lead firms to

reduce their internal maintenance while holding external maintenance fixed, so that total maintenance declines. This serves as a mechanism check because the instrument itself is based on the internal maintenance technology.

*Additional Robustness Checks.* Appendix D.1 provides extensive additional validation. I show that the first stage is positive across multiple instrument specifications (Table D.1). The appendix also re-estimates the elasticity using physical capital stocks rather than book values, a national wage shifter in place of the state-mix instrument, and a three-year-lagged rather than the two-year-lagged labor share (Table D.4). Across these specifications the elasticity remains positive and large, and the headline estimates are larger still without any controls (Table D.3). The large elasticity persists across these measurement and instrument choices.

I also examine the dynamic response. Local projections in Figure D.3 indicate that the response of maintenance is immediate but also displays persistence. As in the SOI, firms reduce maintenance on impact, but the effect grows for two to three years before attenuating. In contrast to the SOI results, this pattern is consistent with some adjustment frictions.

*Identification at the frontier.* Because the instrument is a shift-share, built from state wage shocks and firm exposure weights, I subject it to additional inference procedures, reported for both the headline specification and the no-controls 2SLS (Table D.2). Since the instrument is bordering on weak, I show that weak-instrument-robust intervals keep the elasticity bounded away from zero. Next, one concern is that there is dependence across firms that clustering does not capture. When allowing for Adão, Kolesár, and Morales (2019)-robust inference, the confidence intervals become very wide and cannot exclude zero, and AKM0 confidence intervals are unbounded. This follows from the fact that there are a small number of effective shifts.

Taken together, the SOI and R-1 designs find a quantitatively similar demand elasticity using different sources of variation, providing complementary evidence that maintenance is sensitive to changes in its relative price. In the next section, I show that this leads to economically meaningful changes in the transmission of tax policy.

# 7 Quantitative Implications of Maintenance Demand

The two mechanisms identified in Section 3 change how investment and capital respond to tax incentives, and each carries an empirical consequence. The capital preservation effect—only the non-maintenance share of user cost responds to tax incentives—attenuates steady-state responses and, in standard regressions, biases the estimated investment coefficient downward. The investment-capital gap—tax cuts endogenously raise depreciation, so observed investment overstates capital accumulation—drives investment and capital elasticities apart. The empirical esti-

mates from Sections 5 and 6 provide the parameters needed to quantify both. In this section, I first embed the estimated maintenance demand function into a general equilibrium model to assess the 2017 Tax Cuts and Jobs Act, comparing steady-state outcomes with and without maintenance. I then apply the OVB correction to estimates from the literature.[21]

## 7.1 Capital Maintenance and the 2017 Tax Cuts and Jobs Act

To assess the aggregate importance of the maintenance channel, I embed it in a full model to assess the 2017 Tax Cuts and Jobs Act. The reform cut the corporate tax rate from 35% to 21% and granted full bonus depreciation to equipment investment, which effectively eliminates the wedge between maintenance and investment for that type. Standard estimates, including Chodorow-Reich et al. (2025), estimate that this substantially increased the corporate capital stock in general equilibrium. To assess how the maintenance channel affects those conclusions, I embed it in an otherwise identical model with matched initial steady states, and compare post-reform steady states for key aggregate variables. Since policy analyses, particularly those from the Congressional Budget Office, often rely on dynamic ten-year scores, I present those along with transition dynamics in Appendix G.

### Model and Calibration

I focus on several different variants of the neoclassical model. The main benchmark is a three-sector model with a perfectly elastic supply of capital and maintenance with an inelastic supply of labor shared across sectors. This is also the primary model in the Chodorow-Reich et al. (2025) analysis of domestic effects of the reform.

In the model, there are three sectors: a corporate sector (denoted as $c$), a non-corporate ($nc$), and an exogenous government sector which does nothing. A representative firm in each sector produces output with identical decreasing-returns-to scale Cobb-Douglas technology and discounts the future at rate $r^k$. Capital and labor shares are $\alpha_K$ and $\alpha_L$ with $\alpha_K + \alpha_L \leq 1$. One unit of labor is supplied inelastically and allocated across the two private sectors to equalize wages. The corporate and non-corporate sectors differ in two ways. First, the corporate sector receives a tax cut, while the non-corporate sector does not. Second, productivity may differ between the two, in a way that I describe in the calibration section below.

Firms adjust capital through investment and maintenance as in Section 3. I use the constant-elasticity maintenance demand function from Section 3.7, so steady-state investment satisfies $I_i = \delta(m_i)K_i$.

21. Appendix G studies transition dynamics and dynamic scoring.

**NGMM Calibration.** The maintenance demand estimates from Section 6 pin down the maintenance demand parameters $(\gamma, \omega)$. I rely entirely on those from the Statistics of Income. This is conservative because the estimated level of maintenance is almost surely an underestimate, as I discussed in the data description. To calibrate baseline depreciation $\delta_0$ and non-corporate productivity $A_{nc}$, I take 5,000 draws from the statistical distributions of the demand parameters to jointly match two targets: the ratio of corporate to total capital ($K_c/K = 0.7$, from Chodorow-Reich et al. 2025) and the pre-TCJA investment-output ratio (0.109, from NIPA).[22] The resulting calibration implies a maintenance share of user cost $s_m \approx 0.18$, consistent with the SOI data. To facilitate comparison with Chodorow-Reich et al. (2025), I adopt their values for the discount rate, capital share, and labor share. Table E.1 reports the calibration in full.

**NGM Calibration.** As a benchmark, I calibrate an otherwise identical model with zero maintenance. I set the constant depreciation rates in each private sector to match the same initial steady state as the NGMM.

**Policy Reform.** I set pre- and post-TCJA tax rates following Chodorow-Reich et al. (2025), assuming permanent bonus depreciation and no change in non-corporate marginal rates.[23] Corporate tax rates come from a capital-weighted average of their domestic estimates. Government output is held fixed, so the percentage change in aggregate output equals 0.81 times the percentage change in private GDP.

I report results under four model closures, which differ in how prices and resources adjust to the reform. In principle, there are four relevant prices that may adjust: wages, interest rates, maintenance prices, and investment prices. I vary all except investment prices, since there is little outside evidence since Goolsbee (1998a) that investment good prices respond to tax policy; to the contrary, House, Mocanu, and Shapiro (2017) and Basu, Kim, and Singh (2024) provide evidence that the supply of investment goods is, on average, perfectly elastic. With that in mind, the four closures are:

1. **Partial Equilibrium:** In partial equilibrium, wages, interest rates, and maintenance prices are held at their pre-reform values, and labor supply is perfectly elastic.

2. **General Equilibrium (Baseline):** Total labor supply is fixed, $L_c + L_{nc} = 1$, and wages adjust to clear the labor market, while maintenance prices and interest rates are held fixed.

3. **General Equilibrium with Crowding Out:** In this closure, wages adjust to clear the labor market and interest rates respond to the deficit effects of the reform. As in Chodorow-Reich

22. I divide the sum of non-residential investment in equipment and structures by GDP less government expenditures in NIPA Table 1.1.5.

23. This is not what actually happened: bonus was temporary and non-corporate taxes fell. The goal is to isolate the maintenance channel rather than score TCJA comprehensively.

et al. (2025) and Barro and Furman (2018), the effects of crowding out are modeled with an exogenously calibrated supply curve. In particular, I compute the change in the 2027 debt-to-GDP ratio implied by TCJA's fiscal cost and each model's revenue feedback, then use Plante, Richter, and Zubairy (2026) to calculate the resulting increase in interest rates.

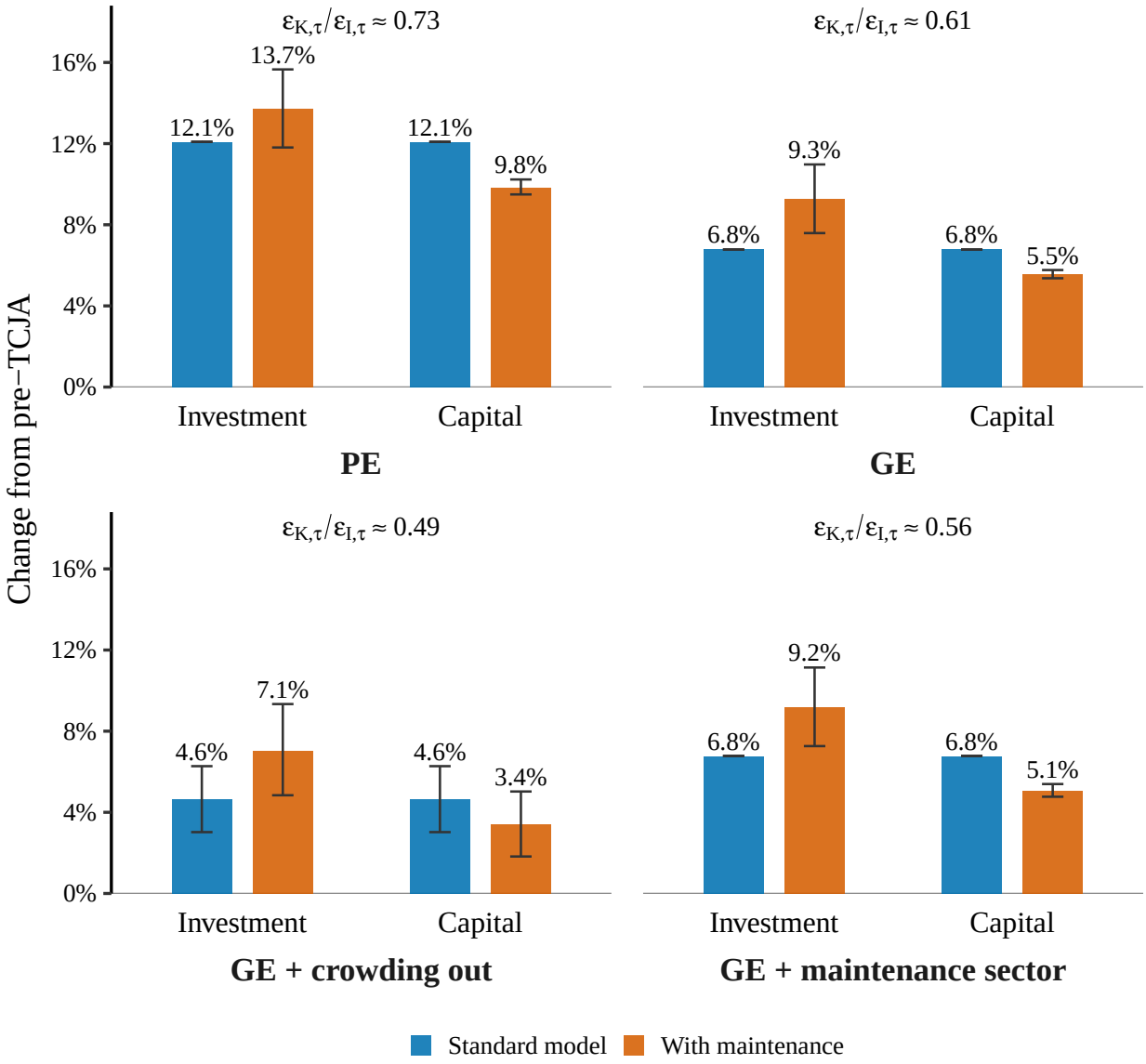
4. **General Equilibrium with Endogenous Maintenance:** Here, I add an intermediate sector which produces maintenance services using materials and labor. Appendix F.3 provides the details on this model. Here, it is important to note that private GDP nets out materials used in maintenance, and that the labor market clears across the corporate sector, the non-corporate sector, and the maintenance sector, with maintenance prices responding endogenously to wages. Interest rates do not respond to tax policy.

I propagate uncertainty from the maintenance demand function via its estimated sampling distribution from Section 6. In the crowding-out closure, I also propagate uncertainty from the investment elasticity and the debt–interest rate coefficient used to compute the fiscal feedback. I hold all other assigned parameters fixed. This differs from Chodorow-Reich et al. (2025), which propagates uncertainty across all structural parameters. I adopt the narrower approach to isolate uncertainty attributable to the maintenance channel and fiscal feedback. The trade-off is that reported confidence intervals understate total uncertainty and should be interpreted as conditional on the broader calibration.

### Aggregate Effects of TCJA

The Tax Cuts and Jobs Act cut the effective tax on corporate capital by around 3.4% on average. In Figure 7, I plot the steady-state responses of investment and capital to that tax cut across all four model closures. In partial equilibrium, investment and capital each rise by 12.1% in the NGM, closely matching the estimate from Chodorow-Reich et al. (2025). In the baseline NGMM, investment rises by 13.7% while capital rises by 9.8%. The implied tax elasticity of capital is approximately 73% of the investment elasticity.

Figure 7: The Effect of TCJA on Investment and Capital

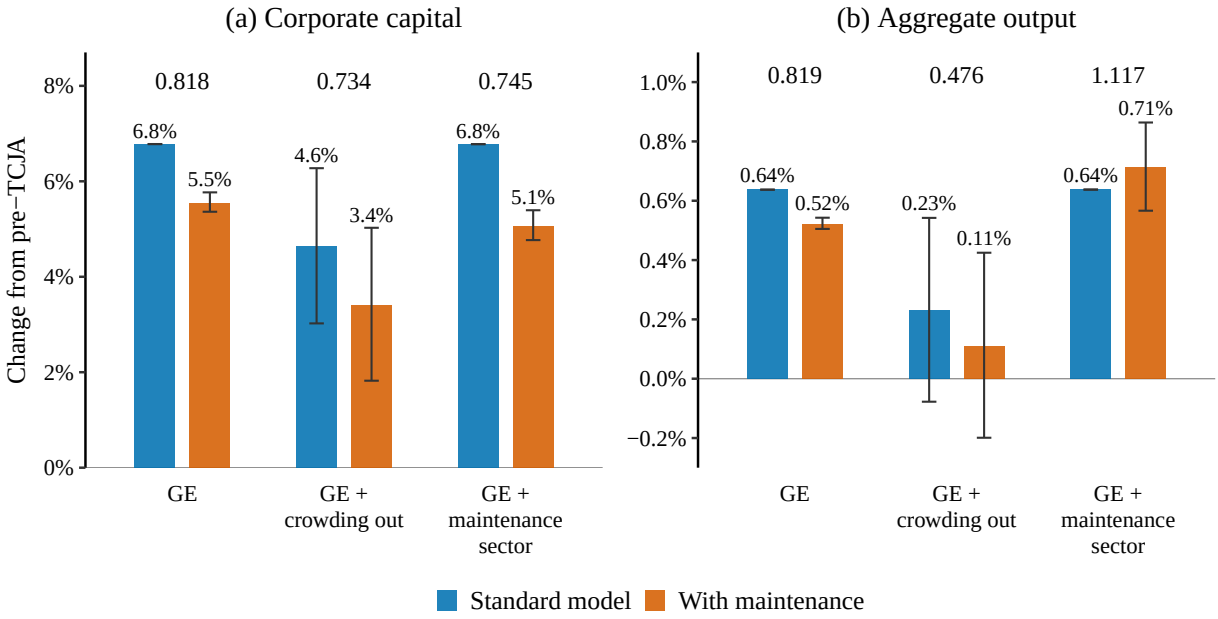



**Notes:** Bars show mean percentage changes in domestic corporate investment and capital between the initial and post-TCJA steady states in the NGM (blue) and NGMM (orange). Error bars are 95% confidence intervals from 5,000 parameter draws. The panels show partial equilibrium (top left), baseline general equilibrium (top right), general equilibrium with fiscal crowding out (bottom left), and general equilibrium with a competitive maintenance sector (bottom right). Ratios above each panel compare the log capital and investment responses in the NGMM.

In the top right panel, which features an inelastic supply of labor, firms bid up wages as they compete for a fixed supply of labor. Here, the maintenance response is unchanged, so the extra replacement investment accompanies a dampened response of capital. In this case, the implied tax elasticity of capital is approximately 61% of the investment elasticity. Crowding out further limits capital expansion through higher interest rates without changing the depreciation response,

lowering the ratio to 49%. With a competitive maintenance sector, higher wages also raise maintenance prices, leading firms to reduce maintenance further and increasing depreciation. The ratio is approximately 56% in this closure.

Figure 8: Steady-State Capital and Output Effects of TCJA across Model Closures




**Notes:** Bars show mean percentage changes in domestic corporate capital (left) and aggregate output (right) between the initial and post-TCJA steady states in the NGM (blue) and NGMM (orange). Error bars are 95% confidence intervals from 5,000 parameter draws. Numbers above each pair of bars divide the mean NGMM response by the mean NGM response. The three closures are general equilibrium, general equilibrium with fiscal crowding out, and general equilibrium with a competitive maintenance sector. Government output is held fixed. In the maintenance-sector closure, private GDP nets out materials used in maintenance.

Next, Figure 8 plots the steady-state responses of corporate capital and aggregate output across the three general equilibrium closures. In baseline general equilibrium, capital and output responses are roughly 82% as large in the NGMM as in the NGM, in line with the $(1 - s_m) \approx 0.82$ markdown. Aggregate output rises by 0.52% in the NGMM, compared with 0.64% in the NGM. Proportionally, this is no different from the response of capital.

The equal proportional responses of capital and output do not hold when accounting for crowding out or for endogenous maintenance. With crowd-out, corporate capital rises by roughly 74% as much in the NGMM as in the NGM, while output rises by less than half as much. This reflects the fact that crowd-out also reduces capital accumulation in the non-corporate sector, so the response of aggregate output need not be proportional to the response of corporate capital. Finally, the closure with competitive maintenance sectors features a similar attenuation as the baseline and crowd-out closures, with capital rising by approximately 75% as much in the NGMM

as the NGM. However, this model produces a starkly different effect on aggregate output. Since the reform induces reallocation of labor from intermediate to final goods production, output rises by more in the NGMM than in the NGM. Thus, accounting for maintenance may mean a smaller response of capital while inducing an even larger response of output. Appendix G estimates the dynamic effects of TCJA using the same model closures with capital adjustment costs.

In sum, the maintenance channel is quantitatively large. It produces a sizable investment-capital elasticity gap, implying that investment is far from a sufficient statistic for capital. Moreover, the maintenance channel clearly attenuates the response of capital, but the corresponding effect on output is not robust across model closures.

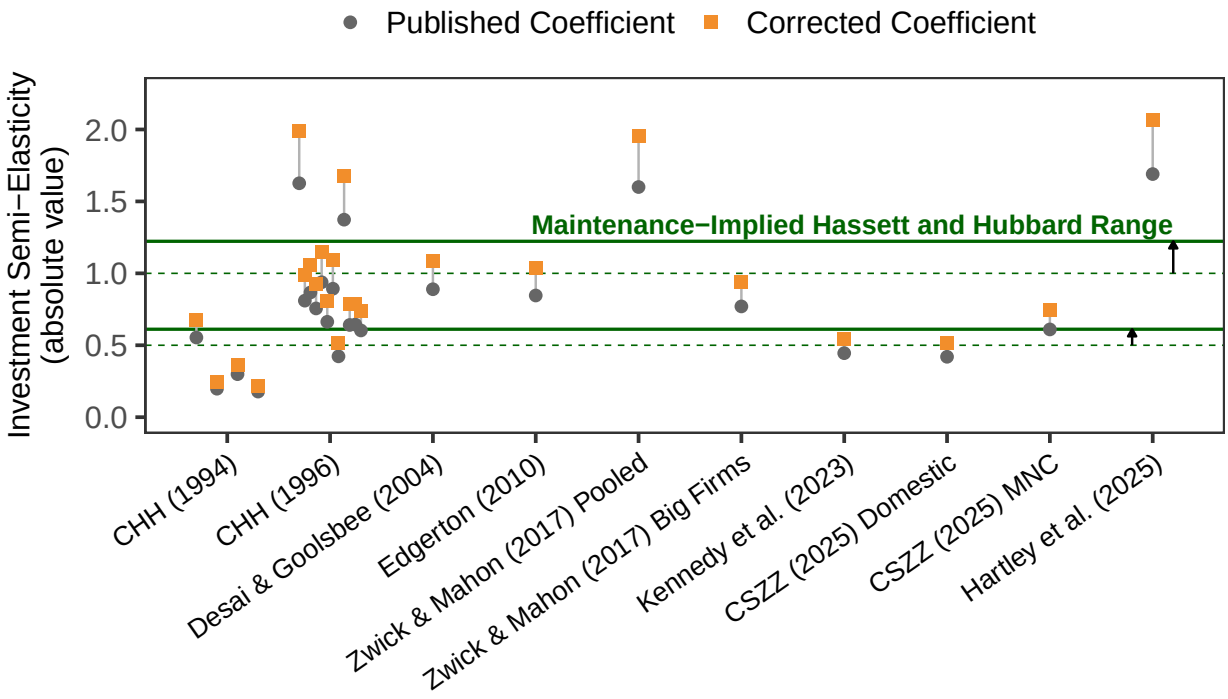
## 7.2 Correcting Existing Estimates

Proposition 3 shows that standard investment regressions understate the coefficient on full user cost because measured user cost overstates the true price change. The correction factor is $1/(1 - s_m)$: if maintenance accounts for share $s_m$ of the user cost, the true coefficient is $1/(1 - s_m)$ times the estimated coefficient. This matters because there is a large literature in public finance which assesses the success of reforms based on the investment response.

Under the estimates from Section 5, the correction factor is approximately 1.22. This adjustment is first-order for interpreting the empirical investment literature. Figure 9 applies Proposition 3 to a broad swath of studies, all of which have the investment rate as the left-hand side variable. The corrected estimates are uniformly larger than published estimates. The green lines plot what Hassett and Hubbard (2002) refer to as the "consensus range" for estimates in the literature, and the correction lifts it from roughly 0.5–1.0 to about 0.61–1.22.[24]

24. As Chodorow-Reich (2025) argues, the coefficients in this figure are not strictly investment elasticities. The correction nonetheless applies because the bias operates through the same channel regardless of how the dependent variable is specified.

Figure 9: Corrected Investment Coefficients from the Literature




**Notes:** This figure applies Proposition 3 to studies from the literature using the maintenance share of user cost measured in Section 5. Most estimates are taken from Chodorow-Reich, Zidar, and Zwick (2024). CHH1994 refers to Cummins, Hassett, and Hubbard (1994). CSZZ2025 refers to Chodorow-Reich et al. (2025). The correction factor is approximately 1.22.

Thus, firms' investment choices are substantially more sensitive to changes in the user cost of capital than published coefficients suggest, but these coefficients are also less informative about the response of capital than commonly understood. That is because the extra investment is actually replacing a more rapidly depreciating capital stock.

# 8 Concluding Remarks

In this paper, I establish that capital maintenance is a first-order margin for understanding how tax incentives transmit to aggregate outcomes. This is driven by two separate effects. Theoretically, the larger maintenance is as a share of the user cost of capital, the less tax-elastic the capital stock will be. Moreover, as long as maintenance demand is elastic, then it is insufficient to look solely at investment as a path to inferring the effect of certain shocks on the capital stock. In the context of tax policy, this implies that accounting for maintenance reduces steady-state capital effects by roughly one-fifth to one-quarter relative to standard models, while the capital-investment elasticity gap may be even larger. That follows from applying my own estimates using new data

on maintenance to standard models augmented with maintenance.

More broadly, my findings imply that any analysis of investment is incomplete without accounting for maintenance expenditures. Both types of flows matter for how capital accumulates, and it is insufficient to look at only one in isolation. While I focused solely on tax policy, the same would hold with any analysis that uses the user cost of capital to analyze the effects of shocks—temporary or otherwise—on capital accumulation. For example, interest rate shocks (Winberry 2021) or depreciation shocks (Bilal and Känzig 2026) would face the same issues with attenuation. Moreover, it implies that steadily declining prices of investment relative to maintenance should produce a secularly rising depreciation rate independently of the composition of the capital stock. Similar effects should be observed when wages rise secularly and raise maintenance costs relative to investment prices.

My findings raise broader questions about the fundamental nature of capital. My analysis focused on a particular law of motion for the capital stock, but it is not obvious that a stock of assets is the relevant economic object, even if it is a useful accounting approximation. The productive services available from those assets over their lifetimes depend on their condition and on firms' choices about maintenance, use, replacement, and retirement. Given the groundwork laid here and in prior work by Feldstein and Rothschild (1974), McGrattan and Schmitz Jr. (1999), and Goolsbee (2004), the case for developing further work along this line, and understanding more deeply how policy influences the lifecycle of capital, is too strong to ignore in public finance and macroeconomics.

# References


Adão, Rodrigo, Michal Kolesár, and Eduardo Morales. 2019. "Shift-Share Designs: Theory and Inference*." *The Quarterly Journal of Economics* 134, no. 4 (November): 1949–2010. ISSN: 0033-5533. https://doi.org/10.1093/qje/qjz025.

Albonico, Alice, Sarantis Kalyvitis, and Evi Pappa. 2014. "Capital maintenance and depreciation over the business cycle." *Journal of Economic Dynamics and Control* 39 (February): 273–286. ISSN: 01651889. https://doi.org/10.1016/j.jedc.2013.12.008.

Anderson, T. W., and Herman Rubin. 1949. "Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations." *The Annals of Mathematical Statistics,* 46–63.

Angelopoulou, Eleni, and Sarantis Kalyvitis. 2012. "Estimating the Euler Equation for Aggregate Investment with Endogenous Capital Depreciation." *Southern Economic Journal* 78, no. 3 (January): 1057–1078. ISSN: 0038-4038. https://doi.org/10.4284/0038-4038-78.3.1057.

Bachmann, Rüdiger, Ricardo J. Caballero, and Eduardo M. R. A. Engel. 2013. "Aggregate Implications of Lumpy Investment: New Evidence and a DSGE Model." *American Economic Journal: Macroeconomics* 5, no. 4 (October): 29–67. ISSN: 1945-7707. https://doi.org/10.1257/mac.5.4.29.

Barro, Robert J., and Jason Furman. 2018. "The macroeconomic effects of the 2017 tax reform."

Basu, Riddha, Doyeon Kim, and Manpreet Singh. 2024. "Spillover Effects of Accelerated Depreciation on Small Business Investment."

Bilal, Adrien, and Diego R Känzig. 2026. "The Macroeconomic Impact of Climate Change: Global Versus Local Temperature." *The Quarterly Journal of Economics* 141, no. 2 (April): 889–944. ISSN: 0033-5533. https://doi.org/10.1093/qje/qjag011.

Bitros, George C. 1976. "A Statistical Theory of Expenditures in Capital Maintenance and Repair." *Journal of Political Economy* 84, no. 5 (October): 917–936. ISSN: 0022-3808. https://doi.org/10.1086/260490.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2024. *A Practical Guide to Shift-Share Instruments.* Technical report.

Boucekkine, R., G. Fabbri, and F. Gozzi. 2010. "Maintenance and investment: Complements or substitutes? A reappraisal." *Journal of Economic Dynamics and Control* 34, no. 12 (December): 2420–2439. ISSN: 01651889. https://doi.org/10.1016/j.jedc.2010.06.007.

Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H C Sant'anna. 2025. *Difference-in-Differences with a Continuous Treatment.* Technical report.

Chaisemartin, Clément de, Diego Ciccia, Xavier D'Haultfœuille, and Felix Knau. 2026. "Difference-in-Differences Estimators When No Unit Remains Untreated" (April). http://arxiv.org/abs/2405.04465.

Chaisemartin, Clément de, and Xavier D'Haultfœuille. 2026. "Difference-in-Differences Estimators of Intertemporal Treatment Effects." *Review of Economics and Statistics* (May): 1–18. ISSN: 0034-6535. https://doi.org/10.1162/rest{_}a{_}01414.

Chodorow-Reich, Gabriel. 2025. "The Neoclassical Theory of Firm Investment and Taxes: A Reassessment."

Chodorow-Reich, Gabriel, Matthew Smith, Owen Zidar, and Eric Zwick. 2025. "Tax Policy and Investment in a Global Economy."

Chodorow-Reich, Gabriel, Owen Zidar, and Eric Zwick. 2024. "Lessons from the Biggest Business Tax Cut in US History." *Journal of Economic Perspectives* 38, no. 3 (August): 61–88. ISSN: 0895-3309. https://doi.org/10.1257/jep.38.3.61.

Cooley, Thomas F., Jeremy Greenwood, and Mehmet Yorukoglu. 1997. "The replacement problem." *Journal of Monetary Economics* 40, no. 3 (December): 457–499. ISSN: 03043932. https://doi.org/10.1016/S0304-3932(97)00055-X.

Cummins, Jason G., Kevin A. Hassett, and R. Glenn Hubbard. 1994. "A Reconsideration of Investment Behavior Using Tax Reforms as Natural Experiments." *Brookings Papers on Economic Activity* 2:1–74.

———. 1996. "Tax reforms and investment: A cross-country comparison." *Journal of Public Economics* 62, nos. 1-2 (October): 237–273. ISSN: 00472727. https://doi.org/10.1016/0047-2727(96)01580-0.

Cunningham, Christopher R., and Gary V. Engelhardt. 2008. "Housing capital-gains taxation and homeowner mobility: Evidence from the Taxpayer Relief Act of 1997." *Journal of Urban Economics* 63, no. 3 (May): 803–815. ISSN: 00941190. https://doi.org/10.1016/j.jue.2007.05.002.

Curtis, E. Mark, Daniel Garrett, Eric Ohrn, Kevin Roberts, and Juan Carlos Suárez Serrato. 2021. *Capital Investment and Labor Demand.* Technical report. Cambridge, MA: National Bureau of Economic Research, November. https://doi.org/10.3386/w29485.

Desai, Mihir A., and Austan Goolsbee. 2004. "Investment, Overhang, and Tax Policy." *Brookings Papers on Economic Activity* 35 (2): 285–355.

Edgerton, Jesse. 2010. "Investment incentives and corporate tax asymmetries." *Journal of Public Economics* 94, nos. 11-12 (December): 936–952. ISSN: 00472727. https://doi.org/10.1016/j.jpubeco.2010.08.010.

Engen, Eric M., and R. Glenn Hubbard. 2004. "Federal Government Debt and Interest Rates." *NBER Macroeconomics Annual* 19 (January): 83–138. ISSN: 0889-3365. https://doi.org/10.1086/ma.19.3585331.

Feldstein, Martin S., and Michael Rothschild. 1974. "Towards an Economic Theory of Replacement Investment." *Econometrica* 42 (3): 393–424.

Fuest, Clemens, Andreas Peichl, and Sebastian Siegloch. 2018. "Do Higher Corporate Taxes Reduce Wages? Micro Evidence from Germany." *American Economic Review* 108, no. 2 (February): 393–418. ISSN: 0002-8282. https://doi.org/10.1257/aer.20130570.

Garrett, Daniel G., Eric Ohrn, and Juan Carlos Suárez Serrato. 2020. "Tax Policy and Local Labor Market Behavior." *American Economic Review: Insights* 2, no. 1 (March): 83–100. ISSN: 2640-205X. https://doi.org/10.1257/aeri.20190041.

Goolsbee, Austan. 1998a. "Investment Tax Incentives, Prices, and the Supply of Capital Goods." *The Quarterly Journal of Economics* 113, no. 1 (February): 121–148. ISSN: 0033-5533. https://doi.org/10.1162/003355398555540.

———. 1998b. "The Business Cycle, Financial Performance, and the Retirement of Capital Goods." *Review of Economic Dynamics* 1, no. 2 (April): 474–496. ISSN: 10942025. https://doi.org/10.1006/redy.1998.0012.

———. 2004. "Taxes and the quality of capital." *Journal of Public Economics* 88, nos. 3-4 (March): 519–543. ISSN: 00472727. https://doi.org/10.1016/S0047-2727(02)00190-1.

Gormsen, Niels, and Kilian Huber. 2022. "Discount Rates: Measurement and Implications for Investment."

Hall, Robert E., and Dale Jorgenson. 1967. "Tax Policy and Investment Behavior." *American Economic Review* 57:391–414.

Harding, John P., Stuart S. Rosenthal, and C.F. Sirmans. 2007. "Depreciation of housing capital, maintenance, and house price inflation: Estimates from a repeat sales model." *Journal of Urban Economics* 61, no. 2 (March): 193–217. ISSN: 00941190. https://doi.org/10.1016/j.jue.2006.07.007.

Hartley, Jonathan S., Kevin A. Hassett, and Joshua Rauh. 2025. "Firm Investment and the User Cost of Capital: New U.S. Corporate Tax Reform."

Hassett, Kevin A., and R. Glenn Hubbard. 2002. "Tax Policy and Business Investment," 1293–1343. https://doi.org/10.1016/S1573-4420(02)80024-6.

Hernandez, Manuel A., and Danilo R. Trupkin. 2021. "Asset maintenance as hidden investment among the poor and rich: Application to housing." *Review of Economic Dynamics* 40 (April): 128–145. ISSN: 10942025. https://doi.org/10.1016/j.red.2020.09.004.

House, Christopher, Ana-Maria Mocanu, and Matthew Shapiro. 2017. *Stimulus Effects of Investment Tax Incentives: Production versus Purchases.* Technical report. Cambridge, MA: National Bureau of Economic Research, May. https://doi.org/10.3386/w23391.

House, Christopher L, and Matthew D Shapiro. 2008. "Temporary Investment Tax Incentives: Theory with Evidence from Bonus Depreciation." *American Economic Review* 98, no. 3 (May): 737–768. ISSN: 0002-8282. https://doi.org/10.1257/aer.98.3.737. https://pubs.aeaweb.org/doi/10.1257/aer.98.3.737.

House, Christopher L. 2014. "Fixed costs and long-lived investments." *Journal of Monetary Economics* 68 (November): 86–100. ISSN: 03043932. https://doi.org/10.1016/j.jmoneco.2014.07.011.

Kabir, Poorya, and Eugene Tan. 2024. *Maintenance Volatility, Firm Productivity, and the User Cost of Capital *.* Technical report.

Kabir, Poorya, Eugene Tan, and Ia Vardishvili. 2024. *Quantifying the Allocative Efficiency of Capital: The Role of Capital Utilization *.* Technical report.

Kennedy, Patrick J, Christine Dobridge, Paul Landefeld, and Jacob Mortenson. 2023. *The Efficiency-Equity Tradeoff of the Corporate Income Tax: Evidence from the Tax Cuts and Jobs Act.* Technical report.

Kitchen, John, and Matthew Knittel. 2011. "Business Use of Special Provisions for Accelerated Depreciation: Section 179 Expensing and Bonus Depreciation, 2002-2009." *SSRN Electronic Journal,* ISSN: 1556-5068. https://doi.org/10.2139/ssrn.2789660.

Knight, John R., and C.F. Sirmans. 1996. "Depreciation, Maintenance, and Housing Prices." *Journal of Housing Economics* 5, no. 4 (December): 369–389. ISSN: 10511377. https://doi.org/10.1006/jhec.1996.0019.

Koby, Yann, and Christian K. Wolf. 2020. "Aggregation in Heterogeneous-Firm Models: Theory and Measurement."

Lal, Apoorva, Mackenzie Lockhart, Yiqing Xu, and Ziwen Zu. 2024. "How Much Should We Trust Instrumental Variable Estimates in Political Science? Practical Advice Based on 67 Replicated Studies." *Political Analysis* 32, no. 4 (October): 521–540. ISSN: 1047-1987. https://doi.org/10.1017/pan.2024.2.

Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. "Valid t-ratio Inference for IV." *American Economic Review* 112, no. 10 (October): 3260–3290. ISSN: 0002-8282. https://doi.org/10.1257/aer.20211063.

Lian, Chen, and Yueran Ma. 2020. "Anatomy of Corporate Borrowing Constraints*." *The Quarterly Journal of Economics* 136, no. 1 (December): 229–291. ISSN: 0033-5533. https://doi.org/10.1093/qje/qjaa030.

McGrattan, Ellen R., and James A. Schmitz Jr. 1999. "Maintenance and Repair: Too Big to Ignore." *Federal Reserve Bank of Minneapolis Quarterly Review,* no. Fall, 213.

Montiel-Olea, José Luis, and Carolin Pflueger. 2013. "A Robust Test for Weak Instruments." *Journal of Business & Economic Statistics* 31, no. 3 (July): 358–369. ISSN: 0735-0015. https://doi.org/10.1080/00401706.2013.806694.

Neveu, Andre R, and Jeffrey Schafer. 2024. "Revisiting the Relationship Between Debt and Long-Term Interest Rates." May.

Occhino, Filippo. 2023. "The macroeconomic effects of the tax cuts and jobs act." *Macroeconomic Dynamics* 27, no. 6 (September): 1495–1527. ISSN: 1365-1005. https://doi.org/10.1017/S1365100522000311.

Pfeifer, Sylvia. 2023. *Boom time for the $110bn a year industry keeping airlines flying.* London, October.

Plante, Michael D., Alexander W. Richter, and Sarah Zubairy. 2026. "Revisiting the Interest Rate Effects of Federal Debt." *Review of Economics and Statistics* (March): 1–24. ISSN: 0034-6535. https://doi.org/10.1162/REST.a.1746.

PWBM. 2019. *Penn Wharton Budget Model: Dynamic OLG Model.* Technical report. Penn Wharton Budget Model. www.budgetmodel.wharton.upenn.edu.

Romer, Christina D, and David H Romer. 2010. "The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks." *American Economic Review* 100 (3): 763–801.

Saunders, Richard. 2003. *Main lines: rebirth of the North American railroads, 1970-2002.* 436. Northern Illinois University Press. ISBN: 0875803164.

Sedlacek, Petr, and Vincent Sterk. 2019. "Reviving american entrepreneurship? tax reform and business dynamism." *Journal of Monetary Economics* 105 (August): 94–108. ISSN: 03043932. https://doi.org/10.1016/j.jmoneco.2019.04.009.

Shan, Hui. 2011. "The effect of capital gains taxation on home sales: Evidence from the Taxpayer Relief Act of 1997." *Journal of Public Economics* 95, nos. 1-2 (February): 177–188. ISSN: 00472727. https://doi.org/10.1016/j.jpubeco.2010.10.006.

Suárez Serrato, Juan Carlos, and Owen Zidar. 2016. "Who Benefits from State Corporate Tax Cuts? A Local Labor Markets Approach with Heterogeneous Firms." *American Economic Review* 106, no. 9 (September): 2582–2624. ISSN: 0002-8282. https://doi.org/10.1257/aer.20141702.

Summers, Lawrence H. 1981. "Taxation and Corporate Investment: A q-Theory Approach." *Brookings Papers on Economic Activity* 1:67–127.

Winberry, Thomas. 2021. "Lumpy Investment, Business Cycles, and Stimulus Policy." *American Economic Review* 111, no. 1 (January): 364–396. ISSN: 0002-8282. https://doi.org/10.1257/aer.20161723.

Yatchew, A. 1997. "An elementary estimator of the partial linear model." *Economics Letters* 57, no. 2 (December): 135–143. ISSN: 01651765. https://doi.org/10.1016/S0165-1765(97)00218-8.

Zeida, Teegawende H. 2022. "The Tax Cuts and Jobs Act (TCJA): A quantitative evaluation of key provisions." *Review of Economic Dynamics* 46 (October): 74–97. ISSN: 10942025. https://doi.org/10.1016/j.red.2021.08.003.

Zwick, Eric, and James Mahon. 2017. "Tax Policy and Heterogeneous Investment Behavior." *American Economic Review* 107, no. 1 (January): 217–248. ISSN: 0002-8282. https://doi.org/10.1257/aer.20140855. https://pubs.aeaweb.org/doi/10.1257/aer.20140855.

# Appendices

**A Institutional Background** **46**

**B Model Derivations** **48**

B.1 Proofs . . . . . 48

B.2 General Equilibrium . . . . . 50

B.3 The Tax Elasticity of Investment . . . . . 56

**C Data** **58**

C.1 R-1 Data Construction . . . . . 58

C.2 SOI Data Construction . . . . . 66

**D Empirical Robustness and Identification Details** **76**

D.1 R-1 Estimates . . . . . 76

D.2 SOI Robustness Checks . . . . . 81

**E Quantification** **86**

**F Model Extensions** **89**

F.1 Capital Reallocation . . . . . 89

F.2 Extension to Multiple Maintenance Inputs . . . . . 93

F.3 A Competitive Maintenance Sector . . . . . 95

**G Adjustment Costs and Dynamic Scoring** **98**

G.1 Proof of Proposition 9 . . . . . 99

G.2 Example with Quadratic Adjustment Costs . . . . . 100

G.3 The Dynamic Effects of the Tax Cuts and Jobs Act . . . . . 102

**H Additional Figures and Tables** **106**

# A Institutional Background

Once classified as an expenditure that must be capitalized, assets are slotted into one of eight lives under rules governed by the Modified Accelerated Cost Recovery System (MACRS)—three, five, seven, ten, fifteen, twenty, 27.5, or 39 years—which govern how quickly they may be depreciated; shorter class lives yield faster deductions and larger present-value tax benefits. Given a class life of $T$ years and a discount rate $r^k$, the net present value of a dollar of deductions for a capital investment is

$$z = \sum_{t=0}^{T} \left( \frac{1}{1 + r^k} \right)^t d_t,$$

where $d_t$ is the allowable deduction by the IRS in period $t$. Table A.1 shows typical assets and associated present value of deductions $z$ for each class category, and Table A.2 works out the year-by-year tax consequences of spending a marginal dollar investing in a new seven-year asset versus maintaining an existing one.

Table A.1: MACRS Asset Lives and NPV of Depreciation Allowances $z$ at a 6% discount rate

| **MACRS Asset Life (Years)** | **Representative Examples** | $z$ |
|---|---|---|
| 3 | Racehorses; special tools | 0.9467 |
| 5 | Automobiles; computers; office machinery | 0.9038 |
| 7 | Furniture; fixtures; general-purpose equipment; locomotives[25] | 0.8645 |
| 10 | Appliances; vessels; barges | 0.8108 |
| 15 | Land improvements; sewage treatment facilities; telephone poles | 0.6942 |
| 20 | Farm buildings; municipal sewers | 0.6219 |
| 27.5 | Residential rental property | 0.5001 |
| 39 | Nonresidential real property (commercial buildings) | 0.3958 |

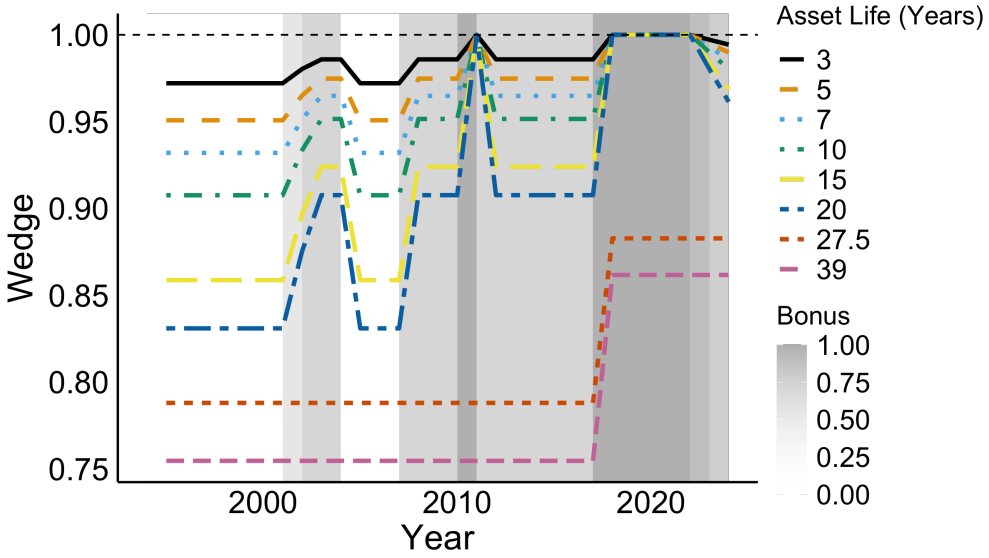
Table A.2: The Tax Treatment of Investment and Maintenance for a Seven-Year Asset

| Year: | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Total | $z$ |
|---|---|---|---|---|---|---|---|---|---|---|
| *Investment* | | | | | | | | | | |
| Deductions (000s) | 142.9 | 244.9 | 174.9 | 124.9 | 89.3 | 89.2 | 89.3 | 44.6 | 1,000 | |
| Tax benefit ($\tau = 35\%$) | 50.0 | 85.7 | 61.2 | 43.7 | 31.3 | 31.2 | 31.3 | 15.6 | 350 | 0.86 |
| Tax Benefit (PV) | 50.0 | 80.9 | 54.5 | 36.7 | 24.8 | 23.3 | 22.0 | 10.4 | 302.6 | |
| *Maintenance* | | | | | | | | | | |
| Deductions (000s) | 1,000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1,000 | |
| Tax benefit ($\tau = 35\%$) | 350 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 350 | 1.00 |
| Tax Benefit (PV) | 350 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 350 | |

**Notes:** This table adapts Table 1 from Zwick and Mahon (2017) to a seven-year MACRS schedule (8-year life with half-year convention and 200% declining balance until straightline becomes optimal). It includes year-by-year deductions, tax benefits, and present values using a 6% discount rate, and reports $z$, the PV factor per dollar of tax benefit.

Figure A.1 plots the wedge since 1995 for all MACRS assets. The wedge is largest for long-lived assets and with 100% bonus, the wedge vanishes for qualifying assets. Of course, some firms may elect not to claim bonus if they are not taxable because the deductions are worthless to them (Kitchen and Knittel 2011). Regardless, the wedge has varied considerably between asset types over time.

Figure A.1: The Maintenance-Investment Wedge for MACRS Assets




**Notes:** The wedge is defined in the main text as the ratio $(1-\tau)/(1-\tau\tilde{z})$, where $\tilde{z}$ is the net present value of depreciation allowances after accounting for bonus depreciation. I set the discount rate as $r = 0.06$.

# B Model Derivations

This appendix contains proofs and additional results and discussion for the theoretical results in Section 3. Throughout, the NGM is the $s_m \to 0$ limit of the NGMM, with $\Psi^{NGM} = \lim_{s_m \to 0} \Psi^{NGMM}$.

## B.1 Proofs

### Proof of Proposition 2

In steady state, $K_{t+1} = K_t = K$. From the accumulation equation, $K = (1-\delta(m^*))K + I$, which implies $I = \delta(m^*)K$. Taking logs and differentiating with respect to $\log(1-\tau)$ yields the decomposition $\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau}$.

For the signs, investment incentives reduce maintenance: $\partial m^*/\partial(1-\tau) < 0$. Since $\delta'(m) < 0$ by Assumption 1(i), lower $m^*$ raises $\delta(m^*)$. A tax cut raises $(1-\tau)$, so $\delta$ increases, meaning $\varepsilon_{\delta,\tau} > 0$. Lower user cost also raises capital, so $\varepsilon_{K,\tau} > 0$. Since both elasticities are positive, $\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau} > \varepsilon_{K,\tau}$. □

**Proof of Proposition 1**

The proof relies on the envelope theorem. Denote the user cost at the optimum as $\Psi^{NGMM} = \Psi(m^*(x), x)$ where $x$ represents any parameter and $m^*(x)$ is optimally chosen maintenance. By the chain rule:

$$\frac{d\Psi^{NGMM}}{dx} = \left.\frac{\partial \Psi}{\partial m}\right|_{m=m^*} \cdot \frac{\partial m^*}{\partial x} + \left.\frac{\partial \Psi}{\partial x}\right|_{m=m^*}$$

At the optimum, $\frac{\partial \Psi}{\partial m}|_{m=m^*} = 0$, so $\frac{d\Psi^{NGMM}}{dx} = \frac{\partial \Psi}{\partial x}|_{m=m^*}$.

Computing partial derivatives with respect to $\tau$ yields

$$\frac{\partial \Psi}{\partial \tau}|_{m=m^*} = \frac{p^I(r + \delta(m^*))}{(1-\tau)^2},$$

which is equivalent to $\frac{\partial \Psi^{NGM}}{\partial \tau}$ when $\bar{\delta} = \delta(m^*)$. The same equality holds for derivatives with respect to $r$ and $p^I$. For $p^M$, $\frac{\partial \Psi}{\partial p^M}|_{m=m^*} = m^*$.

To convert to elasticities, note that for any $x \in \{\tau, r, p^I\}$, $\frac{\partial \Psi^{NGMM}}{\partial x} = \frac{\partial \Psi^{NGM}}{\partial x}$. Since $\varepsilon_{\Psi,x} = \frac{\partial \Psi}{\partial x} \cdot \frac{x}{\Psi}$:

$$\frac{\varepsilon_{\Psi^{NGMM},x}}{\varepsilon_{\Psi^{NGM},x}} = \frac{\Psi^{NGM}}{\Psi^{NGMM}} = 1 - s_m$$

Therefore $\varepsilon_{\Psi^{NGMM},x} = (1 - s_m)\varepsilon_{\Psi^{NGM},x}$. For the maintenance price elasticity: $\varepsilon_{\Psi^{NGMM},p^M} = m^* \cdot \frac{p^M}{\Psi^{NGMM}} = \frac{p^M m^*}{\Psi^{NGMM}} = s_m$. □

**Proof of Proposition 3**

The correctly specified regression uses the full user cost $\Psi = \Psi^I + \Psi^M$:

$$f(I_{i,t}, K_{i,t-1}) = \beta \times \log\left(\frac{r^k + \delta(m_{i,t})}{1 - \tau_{i,t}} + m_{i,t}\right) + X_{i,t} + u_{i,t}. \tag{A.1}$$

Standard regressions use only $\Psi^I = \frac{r^k+\delta}{1-\tau}$, omitting the maintenance component. The coefficient $\beta$ represents the short-run investment response to a change in user cost; the omission biases this short-run estimate toward zero.

Under the constant-elasticity functional form, optimal maintenance is $m^* = \gamma(1-\tau)^{-\omega}$ and depreciation is $\delta(m^*) = \delta_0 + \frac{\gamma\omega}{1-\omega}(1-\tau)^{1-\omega}$. The omitted term is the difference between the true

and standard log user costs. Taking a first-order approximation around $\tau \approx 0$:

$$\begin{aligned}
\text{Omitted Term} &= \log\left(\frac{r^k + \delta(m^*)}{1-\tau} + m^*\right) - \log\left(\frac{r^k+\delta}{1-\tau}\right) \\
&= \log\left(\frac{r^k + \delta_0 + \frac{\gamma}{1-\omega}(1-\tau)^{1-\omega}}{1-\tau}\right) - \log\left(\frac{r^k+\delta}{1-\tau}\right) \\
&\approx \log\left(\frac{r^k + \delta_0 + \frac{\gamma}{1-\omega}(1-(1-\omega)\tau)}{r^k+\delta}\right) \\
&= \log\left(\frac{r^k + \delta_0 + \frac{\gamma}{1-\omega}}{r^k+\delta}\left(1 - \frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}}\right)\right) \\
&\approx -\frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}},
\end{aligned}$$

where constant terms are absorbed by fixed effects.

Applying the omitted variable bias formula:

$$\begin{aligned}
\text{Bias} &= \beta \cdot \frac{\text{Cov}\left(\log\left(\frac{r^k+\delta}{1-\tau}\right), -\frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}}\right)}{\text{Var}\left(\log\left(\frac{r^k+\delta}{1-\tau}\right)\right)} \\
&\approx -\beta \cdot \frac{\text{Cov}\left(\tau, \frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}}\right)}{\text{Var}(\tau)} \\
&= -\beta \cdot \frac{\gamma}{r^k+\delta_0+\frac{\gamma}{1-\omega}}.
\end{aligned}$$

Since $\hat{\beta} = \beta + \text{Bias}$, rearranging yields:

$$\beta = \frac{\hat{\beta}}{1 - \frac{\gamma}{r^k+\delta_0+\frac{\gamma}{1-\omega}}}. \tag{A.2}$$

Under the functional form assumptions, the denominator equals $(1 - s_m)$ evaluated at the steady state, yielding the expression in Proposition 3. The correction factor depends only on the relationship between the included and omitted terms on the right-hand side, not on the left-hand side variable $f(\cdot)$. □

## B.2 General Equilibrium

To analyze general equilibrium, I extend the partial equilibrium model to include labor. Production is $F(K_t, L_t)$, where $L_t$ is labor.

**Assumption 2** (Production Technology)**.** *The production function $F : \mathbb{R}^2_+ \to \mathbb{R}_+$ is twice continuously differentiable with $F_K > 0$, $F_L > 0$, $F_{KK} < 0$, and $F_{LL} < 0$.*

The firm's after-tax dividends become

$$d_t = (1 - \tau^c)\left[F(K_t, L_t) - p_t^M M_t - w_t L_t\right] - (1 - \tau^c z - c)p_t^I I_t, \tag{A.3}$$

where $w_t$ is the wage. The firm's first-order conditions now include labor demand:

$$F_L(K_t, L_t) = w_t, \tag{A.4}$$

$$(1 - \tau_t^c z_t - c_t)p_t^I = \frac{1}{1 + r_t}\Big\{(1 - \tau_{t+1}^c)F_K(K_{t+1}, L_{t+1}) + (1 - \tau_{t+1}^c z_{t+1} - c_{t+1})p_{t+1}^I\left[1 - \delta(m_{t+1}) + \delta'(m_{t+1})m_{t+1}\right]\Big\}, \tag{A.5}$$

$$(1 - \tau_t^c)p_t^M = (1 - \tau_t^c z_t - c_t)p_t^I[-\delta'(m_t)]. \tag{A.6}$$

The maintenance and capital conditions are unchanged from the partial equilibrium analysis; labor demand (A.4) governs the wage response to capital accumulation.

Throughout this appendix, I compare the maintenance model (NGMM) to a standard neoclassical model without maintenance (NGM). To match their initial steady states, I calibrate the NGM's constant depreciation rate $\bar{\delta}$ so that initial user costs are equal:

$$\frac{p^I(r + \bar{\delta})}{1 - \tau} = \frac{p^I[r + \delta(m^*)]}{1 - \tau} + p^M m^*.$$

Equivalently,

$$\bar{\delta} = \delta(m^*) + (1 - \tau)\frac{p^M}{p^I}m^*.$$

All quantities in this calibration are evaluated at the initial steady state. I hold $\bar{\delta}$ fixed following the reform. Given the same marginal-product schedule, matching initial user costs matches the initial capital stock.

The capital preservation effect is a partial equilibrium result: it characterizes how user cost responds to tax policy holding prices fixed. But the effects that matter for policy occur in general equilibrium, where capital accumulation feeds back through factor markets. When capital demand rises, interest rates, wages, and input prices may change. This appendix separates the direct $(1 - s_m)$ attenuation from the effects of equilibrium price responses.

**Assumption 3** (GE Closure)**.** *The general equilibrium is characterized by:*

*(i) A GE-adjusted marginal product $\tilde{F}_K(K)$ with elasticity*

$$\tilde{\varepsilon}_{F_K} \equiv \frac{d \log \tilde{F}_K}{d \log K} < 0,$$

*incorporating diminishing returns and labor-market clearing.*

*(ii) Common reduced-form price-capital elasticities $\varepsilon_{r,K}$ and $\varepsilon_{p^I,K}$ across models, with $\varepsilon_{p^M,K}$ governing the maintenance-price response in the NGMM.*

*(iii) The function $\tilde{F}_K$ is identical across models for this comparison.*

*(iv) Stability in both models:*

$$\tilde{\varepsilon}_{F_K} - \Lambda_0 < 0,$$

$$\tilde{\varepsilon}_{F_K} - \Lambda_0 - s_m \left( \varepsilon_{p^M,K} - \varepsilon_{p^I,K} \right) < 0,$$

*where $\Lambda_0$ is defined below.*

To state the results, I first derive the user-cost elasticities with respect to prices. Matching initial user costs implies

$$\varepsilon_{\Psi^{NGM},r} = \varepsilon_{\Psi^{NGMM},r} = \frac{r}{r + \bar{\delta}}.$$

The investment-price elasticities are

$$\varepsilon_{\Psi^{NGM},p^I} = 1, \qquad \varepsilon_{\Psi^{NGMM},p^I} = 1 - s_m.$$

Define the GE price feedback in the standard model as the effect of capital accumulation on user cost through prices:

$$\Lambda_0 \equiv \frac{r}{r + \bar{\delta}} \varepsilon_{r,K} + \varepsilon_{p^I,K}.$$

The first term captures the interest-rate response to capital accumulation. The second term captures investment-goods price effects.

The maintenance model has the same interest-rate elasticity at the matched initial steady state, but only share $1 - s_m$ of its user cost responds directly to investment prices. Its user-cost elasticity with respect to the maintenance price is $s_m$. I compare steady states before and after a permanent tax change, with hats denoting log-deviations around the matched initial steady state. The tax shock is $\widehat{1-\tau}$.

**Proposition 4** (GE Response Ratio). *Under Assumptions 2–3, the first-order steady-state capital responses to a tax shock $\widehat{1-\tau}$ are*

$$\hat{K}^{NGM} = \frac{\varepsilon_{\Psi^{NGM},\tau}}{\tilde{\varepsilon}_{F_K} - \Lambda_0} \widehat{1-\tau},$$

$$\hat{K}^{NGMM} = \frac{(1-s_m)\varepsilon_{\Psi^{NGM},\tau}}{\tilde{\varepsilon}_{F_K} - \Lambda_0 - s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})}\widehat{1-\tau}.$$

*The ratio is*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = (1-s_m)\frac{\tilde{\varepsilon}_{F_K} - \Lambda_0}{\tilde{\varepsilon}_{F_K} - \Lambda_0 - s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})}. \tag{A.7}$$

*Proof.* The standard user cost is

$$\Psi^{NGM} = \frac{p^I(r+\bar{\delta})}{1-\tau}.$$

Taking logs and differentiating,

$$\hat{\Psi}^{NGM} = -\widehat{1-\tau} + \frac{r}{r+\bar{\delta}}\hat{r} + \hat{p}^I.$$

Thus, $\varepsilon_{\Psi^{NGM},\tau} = -1$, holding prices fixed. Substituting the price responses gives

$$\hat{\Psi}^{NGM} = \varepsilon_{\Psi^{NGM},\tau}\widehat{1-\tau} + \Lambda_0\hat{K}^{NGM}.$$

The equilibrium condition $\tilde{F}_K(K) = \Psi$ log-linearizes to

$$\tilde{\varepsilon}_{F_K}\hat{K} = \hat{\Psi}.$$

Substituting and solving yields the NGM capital response.

For the maintenance model, the envelope theorem gives

$$\hat{\Psi}^{NGMM} = -(1-s_m)\widehat{1-\tau} + \frac{r}{r+\bar{\delta}}\hat{r} + (1-s_m)\hat{p}^I + s_m\hat{p}^M.$$

The interest-rate term is the same as in the NGM because both models have derivative $p^I/(1-\tau)$ with respect to $r$ and the same initial user cost. Substituting the price responses,

$$\hat{\Psi}^{NGMM} = (1-s_m)\varepsilon_{\Psi^{NGM},\tau}\widehat{1-\tau} + \left[\Lambda_0 + s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})\right]\hat{K}^{NGMM}.$$

Using $\tilde{\varepsilon}_{F_K}\hat{K}^{NGMM} = \hat{\Psi}^{NGMM}$ and solving yields the NGMM capital response. Taking the ratio establishes the result. □

The numerator of the maintenance model is $(1-s_m)$ times the standard model—the capital preservation effect carries through to general equilibrium. The denominators reflect equilibrium adjustment. The common term $\Lambda_0$ contains interest-rate and investment-price feedbacks. The additional term in the maintenance model reflects its different exposure to maintenance and investment prices.

The common $\tilde{F}_K$ schedule is imposed in Assumption 3(iii). This holds the production-side response fixed while allowing the models to differ in their exposure to input prices. When maintenance production draws labor away from other production, labor allocation can differ between models even at a given capital stock. Appendix F.3 considers this case.

**Proposition 5** (GE Attenuation)**.** *Under Assumptions 2–3,*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = 1 - s_m \quad \Longleftrightarrow \quad \varepsilon_{p^M,K} = \varepsilon_{p^I,K}. \tag{A.8}$$

*Proof.* From (A.7), the ratio equals $1 - s_m$ if and only if

$$\tilde{\varepsilon}_{F_K} - \Lambda_0 = \tilde{\varepsilon}_{F_K} - \Lambda_0 - s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K}).$$

Since $s_m > 0$, this is equivalent to

$$\varepsilon_{p^M,K} = \varepsilon_{p^I,K}.$$

□

When maintenance and investment prices respond proportionately to capital accumulation, the ratio equals $1 - s_m$ exactly.

Two economic forces shape the response ratio. The first is interest-rate feedback. A change in $r$ raises each model's user cost by $p^I/(1-\tau)$ times the change in $r$. Because initial user costs are matched, interest rates produce identical proportional cost changes in both models. The same interest-rate term consequently appears in both denominators. When maintenance and investment prices are fixed, common interest-rate feedback leaves the ratio equal to $1 - s_m$. This conclusion holds for the common price-capital feedback assumed here; the quantitative crowding-out exercise also allows the fiscal cost of the reform to change with each model's revenue response.

The second force is wage sensitivity. Maintenance is labor-intensive—repairs require technicians, upkeep requires workers. When capital accumulates and wages rise, maintenance prices rise more than investment-goods prices if maintenance is more labor-intensive. The maintenance model is then more exposed to wage-driven cost increases than the standard model, strengthening the attenuation.

The condition $\varepsilon_{p^M,K} = \varepsilon_{p^I,K}$ holds when maintenance and investment prices respond equally to capital accumulation. In a small open economy where interest rates and input prices are pinned globally, neither force operates and the condition holds trivially. The empirically relevant case is likely an open economy where investment goods are traded globally but maintenance services are local. House and Shapiro (2008) and House, Mocanu, and Shapiro (2017) find that investment-goods prices did not respond to tax policy changes during the 2000s, largely due to

foreign competition—suggesting $\varepsilon_{p^I,K} \approx 0$. Interest-rate crowding out depends on how tax cuts are financed: deficit-financed cuts may raise interest rates (Engen and Hubbard 2004; Neveu and Schafer 2024; Plante, Richter, and Zubairy 2026), though the magnitude is debated. But maintenance requires domestic labor and cannot be imported. Fuest, Peichl, and Siegloch (2018) and Kennedy et al. (2023) document that wages rise following corporate tax cuts. When these wage increases raise maintenance prices more than investment prices, $\varepsilon_{p^M,K} > \varepsilon_{p^I,K}$, and the ratio falls below $1 - s_m$: general equilibrium amplifies the attenuation.

With labor fixed in final-good production and the same Cobb-Douglas technology $Y = K^\alpha L^{1-\alpha}$ in both models, output and wages satisfy

$$d \log Y = \alpha \, d \log K, \qquad d \log w = \alpha \, d \log K.$$

Their responses consequently inherit the ratio of the capital responses:

$$\frac{\hat{Y}^{NGMM}}{\hat{Y}^{NGM}} = \frac{\hat{w}^{NGMM}}{\hat{w}^{NGM}} = \frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}}.$$

This ratio equals $1 - s_m$ when $\varepsilon_{p^M,K} = \varepsilon_{p^I,K}$; otherwise, it is given by (A.7). With competitive maintenance production, labor allocation and materials use also change. Appendix F.3 shows how the resources released from maintenance can generate a larger GDP response even when capital accumulation is smaller.

### Additional Results

The following results characterize how the response ratio deviates from $1 - s_m$ when maintenance and investment prices respond differently to capital accumulation.

**Corollary 1** (Special Cases)**.** *Under Assumptions 2–3:*

*(a) If $\varepsilon_{r,K} = \varepsilon_{p^I,K} = \varepsilon_{p^M,K} = 0$, the ratio equals $1 - s_m$.*

*(b) If $\varepsilon_{p^I,K} = \varepsilon_{p^M,K} = 0$, the ratio remains $1 - s_m$ for any common interest-rate feedback satisfying stability.*

*(c) If $\varepsilon_{r,K} = \varepsilon_{p^I,K} = 0$, then*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = (1 - s_m) \frac{\tilde{\varepsilon}_{F_K}}{\tilde{\varepsilon}_{F_K} - s_m \varepsilon_{p^M,K}}.$$

*Proof.* Parts (a) and (b) follow from Proposition 5. Part (c) follows by substitution into (A.7). □

**Proposition 6** (Deviation from $(1 - s_m)$)**.** *Under Assumptions 2–3:*

*(a) If* $\varepsilon_{p^M,K} > \varepsilon_{p^I,K}$*, then*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} < 1 - s_m.$$

*(b) If* $\varepsilon_{p^M,K} < \varepsilon_{p^I,K}$*, then*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} > 1 - s_m.$$

*(c) If* $\varepsilon_{p^M,K} = \varepsilon_{p^I,K}$*, then*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = 1 - s_m.$$

*Proof.* Subtracting $1 - s_m$ from (A.7),

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} - (1 - s_m) = \frac{(1 - s_m)s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})}{\tilde{\varepsilon}_{F_K} - \Lambda_0 - s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})}.$$

The denominator is negative by stability. The difference therefore has the opposite sign to $\varepsilon_{p^M,K} - \varepsilon_{p^I,K}$, establishing all three cases. □

**Proposition 7** (First-Order Approximation)**.** *For*

$$|\Lambda_0|, \quad |\varepsilon_{p^M,K} - \varepsilon_{p^I,K}| \ll |\tilde{\varepsilon}_{F_K}|,$$

*the response ratio satisfies*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} \approx (1 - s_m)\left[1 + \frac{s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})}{\tilde{\varepsilon}_{F_K}}\right].$$

*Proof.* The exact response ratio can be written as

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = (1 - s_m)\left[1 + \frac{s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})}{\tilde{\varepsilon}_{F_K} - \Lambda_0 - s_m(\varepsilon_{p^M,K} - \varepsilon_{p^I,K})}\right].$$

The numerator of the correction is first order in the price feedbacks. Replacing its denominator by $\tilde{\varepsilon}_{F_K}$ changes only second-order terms, yielding the approximation. □

## B.3 The Tax Elasticity of Investment

In steady state, investment is given by $I = \delta(m)K$. Defining investment explicitly in terms of the tax wedge $(1 - \tau)$:

$$I(1 - \tau) = \delta(m(1 - \tau)) \cdot K(1 - \tau, m(1 - \tau), \delta(m(1 - \tau))) \tag{A.9}$$

Taking logs and differentiating:

$$\begin{aligned}\varepsilon_{I,\tau} &= \frac{d\ln I}{d\ln(1-\tau)} = \frac{d\ln\delta}{d\ln(1-\tau)} + \frac{d\ln K}{d\ln(1-\tau)} \\ &= \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau}\end{aligned} \tag{A.10}$$

I now derive explicit expressions for each component using the constant-elasticity depreciation technology:

$$\delta(m) = \delta_0 - \frac{\gamma^{1/\omega}}{1-1/\omega} m^{1-1/\omega}$$

and Cobb-Douglas production $F(K) = K^{\alpha}$.

**Depreciation elasticity.** Substituting optimal maintenance $m^* = \gamma(1-\tau)^{-\omega}$ into the depreciation technology yields:

$$\delta(m^*) = \delta_0 + \frac{\gamma\omega}{1-\omega}(1-\tau)^{1-\omega}$$

Taking the derivative with respect to $(1-\tau)$:

$$\frac{\partial\delta}{\partial(1-\tau)} = \gamma\omega(1-\tau)^{-\omega}$$

Evaluating at $\tau \approx 0$:

$$\varepsilon_{\delta,\tau} = \frac{\partial\ln\delta}{\partial\ln(1-\tau)} = \frac{\gamma\omega}{\delta_0 + \frac{\gamma\omega}{1-\omega}} \tag{A.11}$$

**Capital elasticity.** With Cobb-Douglas production, the capital stock satisfies:

$$K = \left(\frac{\alpha}{\Psi}\right)^{\frac{1}{1-\alpha}}$$

where $\Psi = \frac{r^k+\delta(m^*)}{1-\tau} + m^*$ is the user cost. Taking logs and differentiating:

$$\varepsilon_{K,\tau} = \frac{d\ln K}{d\ln(1-\tau)} = \frac{-1}{1-\alpha}\cdot\varepsilon_{\Psi}$$

where $\varepsilon_{\Psi} = \frac{d\ln\Psi}{d\ln(1-\tau)}$ is the user cost elasticity. By Proposition 1, $\varepsilon_{\Psi} = -(1-s_m)$, so:

$$\varepsilon_{K,\tau} = \frac{1-s_m}{1-\alpha} \tag{A.12}$$

where $s_m = \frac{m^*}{\Psi}$ is the maintenance share of user cost.

**Investment elasticity.** Combining (A.11) and (A.12):

$$\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau} = \frac{\gamma\omega}{\delta_0 + \frac{\gamma\omega}{1-\omega}} + \frac{1 - s_m}{1 - \alpha} \tag{A.13}$$

Both terms are positive: investment increases in response to higher $(1 - \tau)$ both because capital expands ($\varepsilon_{K,\tau} > 0$) and because depreciation rises as firms substitute away from maintenance ($\varepsilon_{\delta,\tau} > 0$). The investment-capital gap from Proposition 2 is precisely $\varepsilon_{\delta,\tau}$.

### Balanced Growth

Note that with balanced growth, the steady state investment condition is

$$I = (\delta(m) + n + g)K,$$

where $n$ and $g$ are exogenous rates of population and productivity growth. In that case, the gap would simply become

$$\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau}\left(\frac{\delta(m)}{\delta(m) + n + g}\right) + \varepsilon_{K,\tau}$$

# C Data

## C.1 R-1 Data Construction

### Data Sources and Digitization

Class I freight railroads must file annual R-1 reports with the Surface Transportation Board (STB) under 49 CFR §1241. These reports contain information about firm financials, expenditures, assets, and more. While they are similar in some respects to 10-K statements filed by publicly traded corporations, the reports follow the Uniform System of Accounts for Railroad Companies, which may differ meaningfully from Generally Accepted Accounting Principles (GAAP) in some respects. For example, railroads often use composite depreciation rates which must be approved ex ante by the STB. Each report is independently must be independently audited by an accounting firms and then reviewed by the STB before publication.

I hand-collected and digitized R-1 reports for all Class I railroads from 1999-2023. From 2012 onward, reports are available on the STB's website. For earlier years, I used Amazon Textract to translate all information into usable data.

The reports contain dozens of detailed schedules. For this paper, I extract data from the following:

- **Schedule 410:** All components of maintenance expenditures come from Line 202 (Locomotives) and Line 221 (Freight Cars). This schedule also breaks down maintenance costs by materials, labor, and purchased services.
- **Schedules 330 and 335:** Investment expenditures and capital stocks for locomotives and freight cars
- **Schedule 702:** Miles of track by state (used for constructing geographic exposure weights)
- **Schedule 710:** Detailed capital inventories by asset type
- **Wage Form A&B:** Hourly wage rates for maintenance workers by occupation and firm

**Sample Construction**

My sample includes seven Class I railroads:

1. Burlington Northern & Santa Fe Railway (BNSF)
2. CSX Transportation
3. Norfolk Southern Railway (NS)
4. Union Pacific Railroad (UP)
5. Kansas City Southern Railway (KCS)
6. Soo Line Railroad (SOO)
7. Grand Trunk Western Railroad (GT)

These firms are geographically dispersed. Burlington Northern and Union Pacific together are largely in the western United States, with extensive networks covering the Great Plains and West Coast. By contrast, CSX and Norfolk Southern operate primarily on the eastern seaboard and in the South, while the Soo Line, Kansas City Southern, and Grand Trunk (operated by Canadian National Railway) have networks concentrated in the Midwest, with Kansas City Southern also arts of the serving the Southwest and Mexico.

The industry was highly fragmented prior to the Staggers Rail Act of 1980, which deregulated much of the industry. Throughout the 1980s and 1990s, extensive consolidation occurred through mergers and acquisitions. By the late 1990s, the industry had largely stabilized. As a result, I begin my sample in the late 1990s to avoid maintenance behavior being influenced by anticipated merger activity. I end the sample in 2023 because Canadian Pacific Railway formally took control

of both the Soo Line and Kansas City Southern by 2024, fundamentally altering the competitive structure. Including post-merger years would conflate firm-level maintenance decisions with merger-related reorganization.

**Institutional Background: The R-1 Reporting Requirement.** The R-1 reporting requirement originates from the Interstate Commerce Commission (ICC), the predecessor to the STB. Prior to the 1990s, the ICC extensively used R-1 reports to regulate rate-setting through cost-of-service regulation. The detailed reporting requirements remain in place primarily for regulatory monitoring and dispute resolution.

### Maintenance Rate Construction

The primary maintenance rate is:

$$m_{i,j,t} = \frac{\text{Maintenance Expenditures}_{i,j,t}}{\text{Beginning Book Capital}_{i,j,t}}, \tag{A.14}$$

where $i$ indexes firms, $j$ indexes asset types (locomotives or freight cars), and $t$ indexes years.

Maintenance expenditures come directly from Schedule 410, Line 202 for locomotives and Line 221 for freight cars. These lines capture all repair and maintenance activities and are broken down into whether the expenditures were for labor, materials, or external services. Beginning book capital is the prior year's ending value from Schedules 330 and 335. This timing ensures maintenance in year $t$ responds to the capital stock at the start of year $t$, consistent with the theoretical model. I construct three alternative measures to check robustness:

**Physical Maintenance Rate.** To control for inflation and asset quality changes, I construct:

$$m^{\text{phys}}_{i,j,t} = \begin{cases} \dfrac{\text{Maintenance Expenditures}_{i,j,t}}{\text{Total Horsepower}_{i,j,t}} & \text{if } j = \text{locomotives} \\ \dfrac{\text{Maintenance Expenditures}_{i,j,t}}{\text{Total Freight Ton Capacity}_{i,j,t}} & \text{if } j = \text{freight cars} \end{cases} \tag{A.15}$$

Horsepower and freight ton capacity come from Schedule 710, which reports quantities and physical characteristics of equipment.

**Internal vs. External Maintenance.** Schedule 410 breaks down maintenance costs into three components:

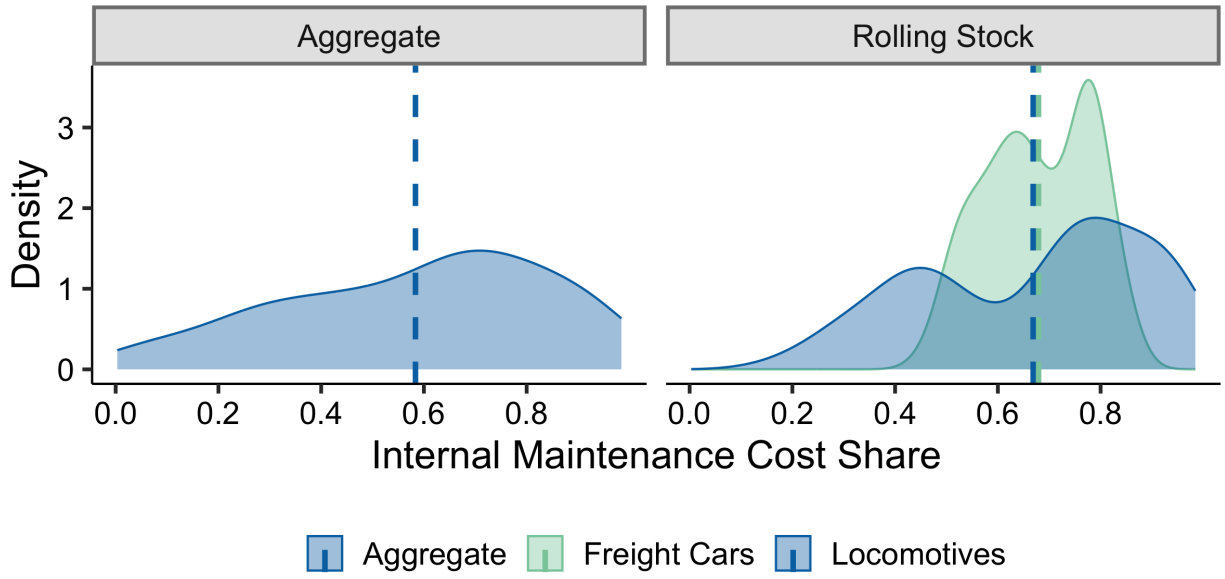
- Materials costs (e.g., replacement parts, lubricants, consumables)
- Labor costs (wages and benefits for railroad employees performing maintenance)

- Purchased services (maintenance contracted to outside vendors)

I define **internal maintenance** as the sum of materials and labor costs, and **external maintenance** as purchased services. This distinction is crucial for identification. Internal maintenance costs vary with local labor market conditions and firm-specific wage policies, while external maintenance costs reflect national market prices for specialized services. If maintenance demand is elastic, the adjustment will occur primarily through the internal margin in the short run, as external maintenance often involves predetermined contracts.

There is significant variability across firms in their internal maintenance practices. On average, approximately 65% of total maintenance is performed internally. However, this share varies considerably: some firms like Norfolk Southern perform nearly all locomotive maintenance in-house, while similarly sized competitors like CSX outsource around 70% of locomotive maintenance. Freight car maintenance tends to have less variability, perhaps because locomotives are more technically complex and require specialized expertise.

Figure C.1: Ratio of Internal Maintenance to Total Maintenance




**Notes:** The distribution of aggregate internal maintenance rates is constructed using Table 1 from the Statistics of Income, which encompasses all firms regardless of legal type, together with input-output tables from the BEA on purchases of equipment repairs from NAICS code 811 except housing services. For 2007, 2012, and 2017, I subtract payments to NAICS 811 from total maintenance expenditures after applying a labor cost correction. The distribution of the share of internal railroad equipment maintenance comes from dividing internal maintenance expenditures by total maintenance expenditures for all Class I freight railroads for locomotives and freight cars.

**Relative Price Construction**

The after-tax relative price is:

$$P_{i,j,t} = \frac{p^M_{i,j,t}}{p^I_{j,t}} \frac{1-\tau_{i,t}}{1-\tau_{i,t}z_t}. \tag{A.16}$$

I construct each component as follows:

**Investment Price ($p^I_{j,t}$).** The investment price varies by asset type but not by firm. I use BLS Producer Price Indices:

**Freight Cars:** Through 2016, I use BLS series PCU3365103365103Z ("Producer Price Index by Industry: Railroad Rolling Stock Manufacturing: Passenger and Freight Train Cars, New (Excluding Parts)"). For subsequent years, I splice this series with WPU14440102 ("Wholesale Price Index: Passenger and Freight Train Cars, New (Excluding Parts)"). The two series have a correlation of 0.98 in the overlapping period.

**Locomotives:** Through 2018, I use BLS series WDU1441 ("Wholesale Price Index: Locomotives"). The BLS discontinued this series in 2019 without a direct replacement. For 2019-2023, I construct a locomotive-specific price index by decomposing the broader rolling stock price index (WPU144) using investment shares from the R-1 data. Specifically, I calculate the locomotive share of total rolling stock investment for each year and construct a Tornqvist index that separates locomotive price movements from freight car price movements.

**Maintenance Price ($p^M_{i,j,t}$).** The pre-tax maintenance price is a firm-asset-time specific Törnqvist index of the two internal input prices, labor and materials:

$$\begin{aligned}\Delta \log p^M_{i,j,t} = &\frac{\alpha^{\text{labor}}_{i,j,t} + \alpha^{\text{labor}}_{i,j,t-1}}{2} \Delta \log w_{i,t} \\ &+ \frac{\alpha^{\text{materials}}_{i,j,t} + \alpha^{\text{materials}}_{i,j,t-1}}{2} \Delta \log p^{\text{mat}}_t,\end{aligned} \tag{A.17}$$

where:

- $\alpha^{\text{labor}}_{i,j,t}$ and $\alpha^{\text{materials}}_{i,j,t}$ are labor and materials cost shares from Schedule 410, normalized to sum to one. I winsorize the labor share at the 5th and 95th percentiles within asset type and compute the materials share as its complement.
- $w_{i,t}$ is the firm-specific average hourly wage for rolling stock maintenance workers from Wage Form A&B filed with the STB. This form reports detailed wage information by occupation and skill level.

- $p_t^{\text{mat}}$ is BLS series PCU33651033651054 ("Producer Price Index by Industry: Railroad Rolling Stock Manufacturing: Railway Maintenance of Way Equipment and Parts, Parts for All Railcars, and Other Railway Vehicles").

I chain the annual price changes, normalizing each firm-asset index to one in its first year. The headline regressions use this internal-maintenance price index.

**Tax Parameters.** I construct firm-specific statutory tax rates ($\tau_{i,t}$) as:

$$\tau_{i,t} = \tau_t^{\text{federal}} + \tau_{i,t}^{\text{state}} - \tau_t^{\text{federal}} \cdot \tau_{i,t}^{\text{state}}, \tag{A.18}$$

where $\tau_t^{\text{federal}}$ is the federal corporate tax rate (35% before 2018, 21% thereafter) and $\tau_{i,t}^{\text{state}}$ is a revenue-weighted average state corporate tax rate.

The state component is calculated using the geographic distribution of each railroad's operations. Schedule 702 reports track miles by state for each firm. I use these track miles as weights to construct a firm-specific exposure to state corporate tax rates:

$$\tau_{i,t}^{\text{state}} = \sum_s \left( \frac{\text{Track Miles}_{i,s,t}}{\text{Total Track Miles}_{i,t}} \right) \cdot \tau_{s,t}^{\text{state}}, \tag{A.19}$$

where $\tau_{s,t}^{\text{state}}$ is the statutory corporate tax rate in state $s$ at time $t$. I obtain state tax rates by extending the dataset of Suárez Serrato and Zidar (2016) through 2023.

The present value of depreciation allowances ($z_t$) varies over time due to changes in bonus depreciation and MACRS schedules. Following House and Shapiro (2008), I calculate:

- Both locomotives and freight cars are classified as MACRS 7-year property
- Bonus depreciation rates
- Discount rate: $r^k = 0.0713$ (industry average for NAICS 48 from Gormsen and Huber (2022))

Because the IRS places both asset types in the same depreciation class, $z_t$ does not vary between locomotives and freight cars. However, there is firm-level variation in the effective tax rate through the state tax component.

### Additional Control Variables

**Local GDP Exposure.** I construct a firm-specific measure of exposure to local demand shocks using:

$$\Delta \log Y_{i,t} = \sum_s \left( \frac{\text{Track Miles}_{i,s,t}}{\text{Total Track Miles}_{i,t}} \right) \cdot \Delta \log \text{GDP}_{s,t}, \tag{A.20}$$

where $\Delta \log \text{GDP}_{s,t}$ is the log change in gross state product from the Bureau of Economic Analysis. This measure captures how aggregate demand shocks in a railroad's service territory might affect maintenance decisions.

**Local Wage Index.** I construct a firm-specific maintenance wage index using Bureau of Labor Statistics data on wages by state in the Installation, Maintenance, and Repair Occupation (SOC Code 49-0000):

$$W_{i,t} = \sum_{s} \left( \frac{\text{Track Miles}_{i,s,t}}{\text{Total Track Miles}_{i,t}} \right) \cdot \frac{w_{s,t}}{w_{s,1999}}, \tag{A.21}$$

where $w_{s,t}$ is the average wage in state $s$ at time $t$, normalized to 1999 = 1. This provides an alternative measure of maintenance labor costs that is not firm-specific and can serve as an instrument for firm-level wage variation.

**Maintenance Share of User Cost.** Following the definition in Section 3, I calculate:

$$s_{m,i,j,t} = \frac{e_{i,j,t}}{\frac{1-\tau_{i,t} z_t}{1-\tau_{i,t}} p^I_{j,t} (r^k + \delta_j) + e_{i,j,t}}, \tag{A.22}$$

where $e_{i,j,t}$ is reported maintenance expenditure divided by beginning book capital. It directly measures the maintenance-expenditure component $p^M_{i,j,t} m_{i,j,t}$.

The depreciation rate $\delta_j$ is assumed constant across firms but varies by asset type:

- Locomotives: $\delta = 0.04$ (4% annual depreciation)
- Freight cars: $\delta = 0.03$ (3% annual depreciation)

These rates are industry standards for freight railroads and are consistent with the depreciation rates railroads report to the STB for composite depreciation calculations. The discount rate is $r^k = 0.0713$ as noted above.

### Summary Statistics

Tables C.1 and C.2 present summary statistics for the R-1 sample, separately for locomotives and freight cars.

Table C.1: Summary Statistics: R-1 Data (Locomotives)

| Variable | Mean | 10th Pctl | Median | 90th Pctl | N |
|---|---|---|---|---|---|
| Year | 2011.19 | 2001 | 2011 | 2021 | 172 |
| $m_{i,j,t}$ (Total) | 0.17 | 0.07 | 0.14 | 0.30 | 172 |
| $m_{i,j,t}$ (Internal) | 0.11 | 0.04 | 0.09 | 0.19 | 172 |
| $m_{i,j,t}$ (External) | 0.06 | 0.00 | 0.05 | 0.11 | 172 |
| $m_{i,j,t}$ (Physical) | 0.03 | 0.02 | 0.02 | 0.04 | 172 |
| $\log M_{i,j,t}$ | 11.98 | 10.35 | 12.36 | 13.41 | 172 |
| $\log I_{i,j,t}$ | 11.56 | 9.40 | 11.94 | 13.32 | 172 |
| Investment Rate | 0.15 | 0.02 | 0.10 | 0.28 | 172 |
| $P_{i,j,t}$ | 1.11 | 1.00 | 1.10 | 1.25 | 172 |
| Capital Age | 1.48 | 1.24 | 1.51 | 1.69 | 172 |
| Local GDP Exposure | 1.99 | -0.21 | 2.13 | 4.09 | 172 |
| $z_{i,j,t}$ | 0.56 | 0.36 | 0.50 | 0.83 | 172 |
| Labor Cost Share (lag) | 0.41 | 0.29 | 0.37 | 0.57 | 172 |
| Local Wage Index | 1.33 | 1.08 | 1.32 | 1.64 | 172 |
| Effective Tax Rate | 1.04 | 1.00 | 1.04 | 1.09 | 172 |
| Maintenance Share ($s_m$) | 0.53 | 0.34 | 0.53 | 0.71 | 172 |

**Notes:** Sample covers 7 firms over 25 years (1999-2023), with some missing observations. Maintenance and investment rates are scaled as shares of beginning-of-period book capital. Reported maintenance expenditure enters user cost directly. Capital Age is the ratio of gross to net book capital.

Table C.2: Summary Statistics: R-1 Data (Freight Cars)

| Variable | Mean | 10th Pctl | Median | 90th Pctl | N |
|---|---|---|---|---|---|
| Year | 2011.19 | 2001 | 2011 | 2021 | 172 |
| $m_{i,j,t}$ (Total) | 0.22 | 0.08 | 0.16 | 0.45 | 172 |
| $m_{i,j,t}$ (Internal) | 0.15 | 0.05 | 0.10 | 0.34 | 172 |
| $m_{i,j,t}$ (External) | 0.07 | 0.02 | 0.05 | 0.12 | 172 |
| $m_{i,j,t}$ (Physical) | 0.03 | 0.02 | 0.03 | 0.06 | 172 |
| $\log M_{i,j,t}$ | 11.70 | 10.35 | 11.92 | 13.03 | 172 |
| $\log I_{i,j,t}$ | 10.33 | 8.12 | 10.72 | 12.24 | 172 |
| Investment Rate | 0.08 | 0.00 | 0.05 | 0.19 | 172 |
| $P_{i,j,t}$ | 0.90 | 0.77 | 0.88 | 1.04 | 172 |
| Capital Age | 1.59 | 1.18 | 1.62 | 1.93 | 172 |
| Local GDP Exposure | 1.99 | -0.21 | 2.13 | 4.09 | 172 |
| $z_{i,j,t}$ | 0.52 | 0.36 | 0.53 | 0.67 | 172 |
| Labor Cost Share (lag) | 0.39 | 0.26 | 0.39 | 0.51 | 172 |
| Local Wage Index | 1.33 | 1.08 | 1.32 | 1.64 | 172 |
| Effective Tax Rate | 1.04 | 1.00 | 1.04 | 1.09 | 172 |
| Maintenance Share ($s_m$) | 0.55 | 0.34 | 0.55 | 0.76 | 172 |

**Notes:** Sample covers 7 firms over 25 years (1999-2023), with some missing observations. Maintenance and investment rates are scaled as shares of beginning-of-period book capital. Reported maintenance expenditure enters user cost directly. Capital Age is the ratio of gross to net book capital.

## C.2 SOI Data Construction

### Data Sources and Sample Construction

The Statistics of Income (SOI) is published annually by the Internal Revenue Service and aggregates corporate tax return data into industry-level samples. The SOI uses a stratified sampling procedure that oversamples large corporations and then applies weights to make the sample representative of the full corporate population. Corporations report maintenance expenditures (labeled "repairs and maintenance") and book capital as line items on their tax forms (Form 1120), which the SOI aggregates across firms within industries.

I use the SOI Corporate Sample files from 1999-2021. The data are publicly available at the industry level but not at the firm level. Each observation is an industry-year cell, with all dollar amounts representing weighted aggregates across firms in that industry. The SOI reports data at varying levels of aggregation, roughly corresponding to 2-3 digit NAICS codes. However, the number of SOI industries fluctuates over time as the IRS adjusts its classification scheme. In 2014, for example, the SOI changed from a "major" to a "minor" industry scheme.

To maintain consistency across years, I map SOI industries to Bureau of Economic Analysis (BEA) industry definitions, which are more stable over time. Specifically, I use the 49-industry classification from the BEA's Fixed Asset Tables. This mapping is not one-to-one—some SOI industries must be aggregated—but it ensures definitional consistency across my sample period. I exclude several categories, the financial industry and passthrough entities, from the analysis, focusing only on corporate filings. The resulting sample consists of 49 industries observed annually from 1999-2021, yielding up to 1,127 industry-year observations. Some industry-year cells have missing data, resulting in a final sample of approximately 1,116 observations.

I use the BEA industry definition rather than the raw SOI definition for three reasons. First, I combine SOI data with BEA data in some specifications (e.g., using BEA depreciation rates, capital composition weights, and price deflators). Using a consistent industry definition facilitates this merging. Second, the BEA definition is more aggregated than the SOI definition. This aggregation arguably better captures general equilibrium effects: when national tax policy changes, firms can reallocate capital within broadly defined industries, and the BEA aggregation level implicitly nets out this within-industry reallocation. Third, the BEA classification is stable over time, whereas the SOI classification changes periodically, creating artificial breaks in the panel.

**Taxable vs. Untaxable Firms.** For some analyses, I separate the sample into taxable and untaxable firms. I define **taxable firms** as those with positive net income in a given year, and **untaxable firms** as those without positive net income. The SOI reports both categories separately.

This classification is imperfect. Some firms in the "taxable" category may not actually face positive tax liability because they have accumulated net operating losses (NOLs) from prior years that they can carry forward to offset current income. Conversely, some "untaxable" firms may face some tax liability through alternative minimum tax provisions or other mechanisms. Despite these limitations, the taxable/untaxable split provides a useful approximation: taxable firms face stronger incentives to respond to tax policy changes than untaxable firms.

### Variable Construction

**Maintenance Rate.** The maintenance rate is defined as the ratio of the maintenance and repairs line item to lagged book capital:

$$m_{i,t} = \frac{\text{Repairs and Maintenance}_{i,t}}{\text{Book Capital}_{i,t-1}}, \tag{A.23}$$

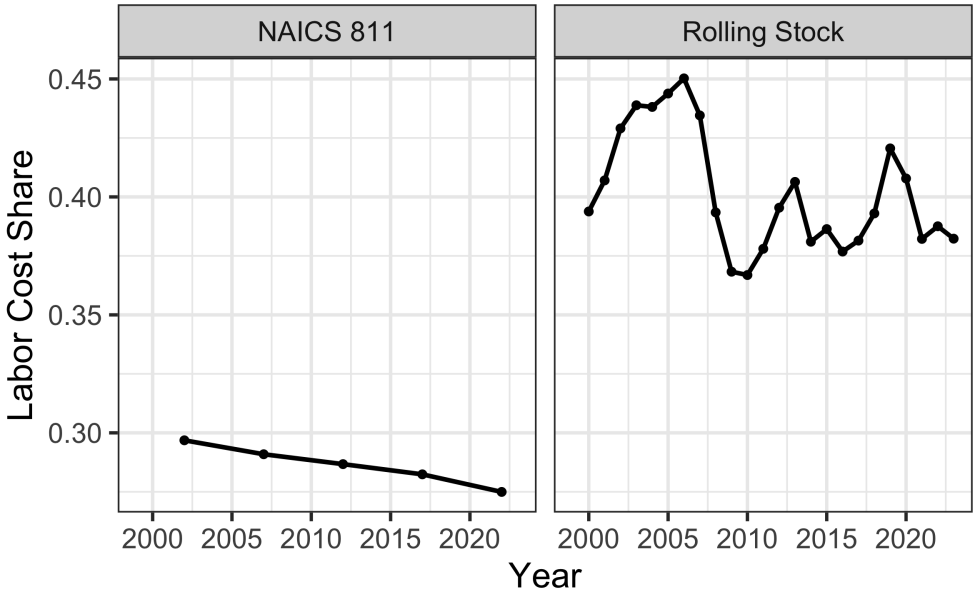
where $i$ indexes industries and $t$ indexes years.

The SOI maintenance measure is noisier than the R-1 measure for two important reasons:

**First, labor costs are likely understated.** On corporate tax forms, labor expenditures are reported as a separate line item (wages and salaries). As a result, the repairs and maintenance line primarily reflects spending on materials and external maintenance services purchased from vendors. Internal maintenance performed by the firm's own employees—which can be substantial—is not fully captured. Additionally, some materials costs may be classified under cost of goods sold rather than the maintenance line item, further understating true maintenance expenditures.

In the R-1 railroad panel, labor accounts for about 40 percent of internal maintenance but 26 percent of total maintenance expenditure (Figure 6). Appendix Figure C.2 plots labor costs relative to total receipts for the maintenance and repair sector using the Economic Census from 2002–2022; that share is about 30%. These benchmarks imply that reported SOI maintenance omits a meaningful, but not 40-percent, labor component.

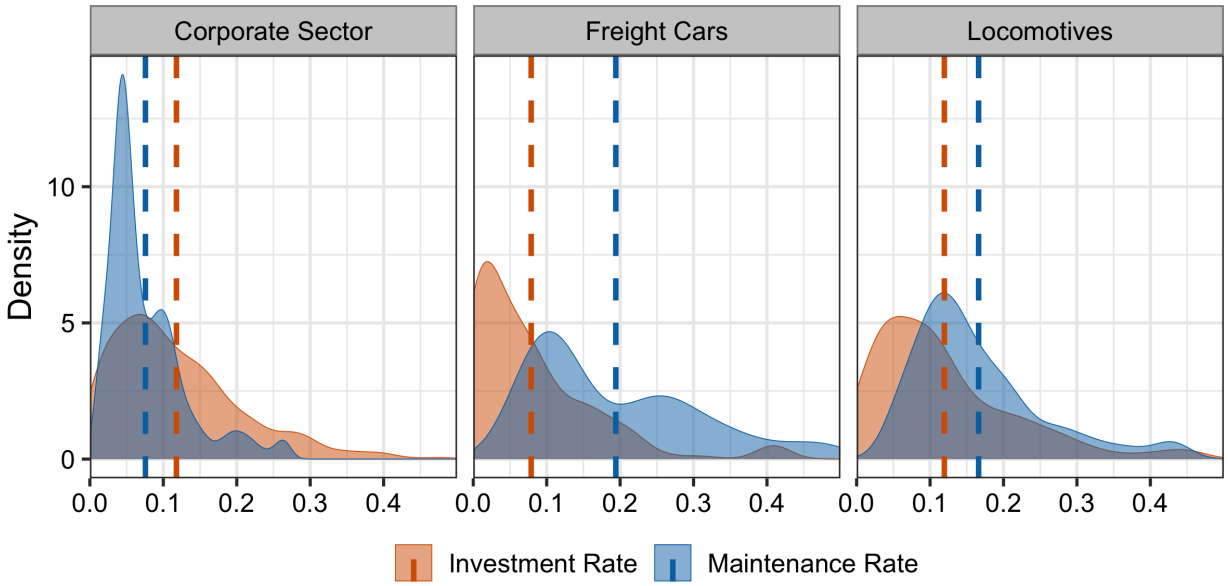
Figure C.2: Aggregate Internal Labor Cost Share




**Notes:** The share is computed by dividing labor costs by total internal maintenance costs. The left panel plots the ratio of labor costs to total receipts for NAICS code 811, which is the maintenance and repair sector, from 2002-2022. Each data point comes from the Economic Census. The right panel is the average for all rolling stock in the R-1 data.

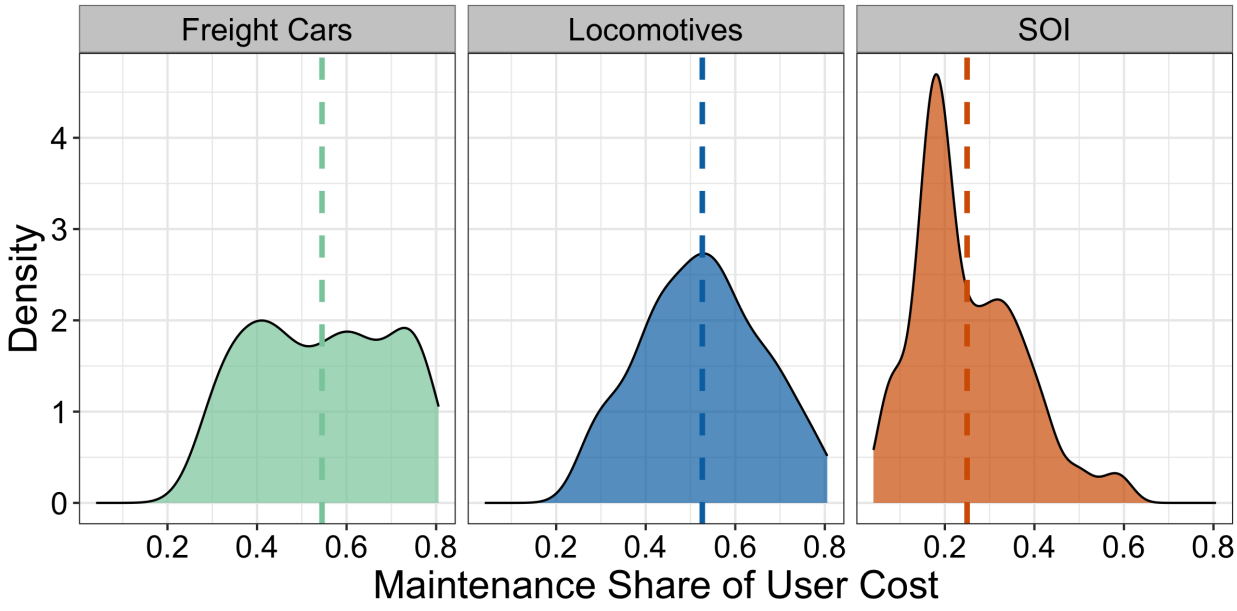
The Equipment Maintenance and Repair sector (NAICS 811) suggests that labor accounts for about 30% of total maintenance costs. As a deliberately generous adjustment, I divide reported SOI maintenance by 0.65 to approximate total maintenance inclusive of omitted labor. The elasticity results are invariant to this rescaling; I treat the unadjusted level as a lower bound and the adjusted level as an upper-bound exercise.

Figure C.3: Density plots for maintenance and gross investment rates (adjusted)




**Notes:** Dashed lines are mean maintenance and investment rates. From left to right, maintenance rates are 7.6%, 19.4%, and 16.6%; investment rates are 11.8%, 7.9%, and 11.9%. The Corporate Sector maintenance rate is divided by 0.65.

Figure C.4: Density plots for the maintenance share of user cost (adjusted)




**Notes:** Dashed lines are means: 25% (Corporate Sector), 55% (Freight Cars), and 53% (Locomotives). The Corporate Sector maintenance rate is divided by 0.65.

**Second, the capital stock denominator uses tax depreciation.** The SOI constructs book

capital using firms' reported accumulated depreciation, which follows tax depreciation schedules rather than economic depreciation. Tax depreciation can be accelerated (especially during bonus depreciation periods), causing book capital to understate true economic capital. This introduces measurement error in the denominator, which may attenuate estimated elasticities. Despite these measurement issues, SOI maintenance rates are comparable to those observed in aggregate Canadian *survey* data, which is constructed by Statistics Canada.

**Investment Rate.** The investment rate is the year-over-year change in gross property, plant, and equipment, scaled by lagged book capital:

$$\frac{I_{i,t}}{K_{i,t-1}} = \frac{\text{Gross PP\&E}_{i,t} - \text{Gross PP\&E}_{i,t-1}}{\text{Book Capital}_{i,t-1}}. \tag{A.24}$$

**Capital Age.** I proxy for capital age using the ratio of gross to net book capital:

$$\text{Capital Age}_{i,t} = \frac{\text{Gross Book Capital}_{i,t}}{\text{Net Book Capital}_{i,t}}. \tag{A.25}$$

Higher values indicate older capital (more accumulated depreciation relative to gross value). This measure is imperfect because it reflects tax depreciation schedules rather than true physical age, but it provides a useful control for capital vintage effects.

I winsorize maintenance rates, investment rates, and capital age at the 2nd and 98th percentiles to reduce the influence of outliers and measurement error.

### Policy Wedge Construction

The key source of identification in the SOI data is variation in the policy wedge:

$$\text{Wedge}_{i,t} = \frac{1 - \tau_t}{1 - \tau_t z_{i,t}}, \tag{A.26}$$

where $\tau_t$ is the statutory federal corporate tax rate (35% before 2018, 21% thereafter) and $z_{i,t}$ is the net present value of depreciation allowances for industry $i$ at time $t$.

I construct $z_{i,t}$ in three steps:

**Step 1: Asset-Level Present Value of Depreciation.** For every asset type $j$ with MACRS class life $T$, I compute the baseline present value of depreciation allowances as:

$$z_{i,j} = \sum_{t=0}^{T} \left( \frac{1}{1 + r_i^k} \right)^t d_t, \tag{A.27}$$

where $d_t$ is the depreciation rate in year $t$ under the applicable MACRS schedule (e.g., 200% declining balance for 5-year property, 150% declining balance for 15-year property, straight-line for structures), and $r_i^k$ is the industry-specific discount rate.

I obtain industry-specific discount rates from Gormsen and Huber (2022). I map their firm-level estimates to BEA industries and take a time-series average within each industry.

**Step 2: Adjusting for Bonus Depreciation.** Time-series variation in $z_{i,j,t}$ comes from changes in bonus depreciation policy. Let $\theta_t$ denote the bonus depreciation rate (the percentage of asset cost that can be immediately expensed). Then:

$$z_{i,j,t} = \theta_t + (1 - \theta_t) z_{i,j}, \tag{A.28}$$

where $\theta_t$ only applies to eligible assets (primarily equipment). Structures are generally ineligible for bonus depreciation. My sample ends before the TCJA bonus provisions began phasing out.

**Step 3: Aggregating to Industry-Level Wedges.** To construct the industry-specific $z_{i,t}$, I aggregate across the 39 detailed asset categories in the BEA's Fixed Asset Tables using capital-weighted shares:

$$z_{i,t} = \sum_{j=1}^{39} \alpha_{i,j} \cdot z_{i,j,t}, \tag{A.29}$$

where $\alpha_{i,j}$ is the share of asset type $j$ in industry $i$'s total capital stock. I map the 39 BEA assets to their corresponding MACRS depreciation classes using the concordance from House and Shapiro (2008).

**Capital Weights and Exogeneity.** A key consideration is which years to use for constructing the capital weights $\alpha_{i,j}$. Using contemporaneous weights would introduce endogeneity: if tax policy affects investment, it also affects capital composition, which would feed back into the construction of $z_{i,t}$.

I use two different weighting periods:

- **1998–2001 weights** for years 1999–2012.
- **2014–2017 weights** for years 2013–2021.

The weights are held fixed within each application period. The later weighting window is applied to both pre- and post-TCJA observations, including observations within and before the weighting window. The 2017 TCJA represented a major structural break (corporate rate cut from

35% to 21% plus 100% bonus), so using 2014–2017 weights better captures the capital composition relevant to that reform.

Industries with more equipment-intensive capital structures (higher shares of short-lived assets eligible for bonus depreciation) experience larger changes in $z_{i,t}$ when bonus depreciation rates change. Figure 3 in the main text shows substantial cross-industry variation in the wedge over time, with equipment-intensive industries like manufacturing experiencing larger policy-driven changes than structures-intensive industries like utilities or real estate.

### Maintenance Share of User Cost

Following the definition in Section 3, I calculate the maintenance share of user cost as:

$$s_{m,i,t} = \frac{e_{i,t}}{\frac{1-\tau_t^c z_{i,t}}{1-\tau_t^c} p_t^I (r_i^k + \delta_i) + e_{i,t}}, \tag{A.30}$$

where:

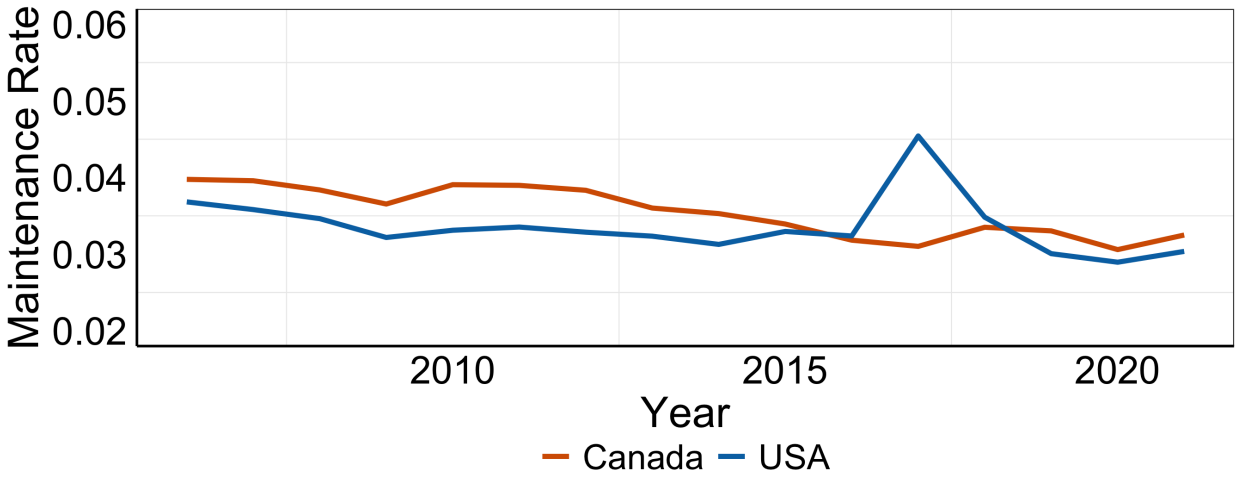
- $e_{i,t}$ is reported repairs and maintenance divided by lagged book capital; it directly measures $p_t^M m_{i,t}$
- $r_i^k$ is the industry-average discount rate from Gormsen and Huber (2022)
- $\delta_i$ is a capital-weighted industry-average depreciation rate constructed from the BEA Fixed Asset Tables. For each industry, I compute $\delta_i = \sum_j \alpha_{i,j} \delta_j$, where $\delta_j$ is the BEA's estimate of economic depreciation for asset type $j$.
- $p_t^I$ is the implicit price deflator for fixed investment from the BEA National Income and Product Accounts (NIPA Table 1.1.4)

Table C.3 also reports an "adjusted" maintenance share that accounts for the labor understatement issue discussed above. It divides reported maintenance by 0.65 before computing the share, treating a 35% omitted labor component as a deliberately generous upper-bound adjustment.

### Comparison to Canadian Data

To validate the SOI maintenance rates, I compare them to aggregate maintenance data from Canada. The Canadian data compiled by Statistics Canada are from survey data on maintenance expenditures combined with an estimated aggregate economic capital stock. Figure C.5 plots the aggregate maintenance rate (total maintenance divided by total book capital) for both the U.S. SOI sample and Canadian data.

Figure C.5: American vs. Canadian Maintenance Rates




**Notes:** This figure plots the maintenance rate (net of government capital and maintenance) from Statistics Canada against the SOI maintenance rate.

The two series track each other closely, with mean maintenance rates of 4.8% (U.S.) and 5.1% (Canada). Both exhibit similar cyclical patterns, rising during recessions (when investment falls but maintenance continues) and declining during booms. This similarity suggests that despite measurement issues, the SOI provides a reasonable proxy for economy-wide maintenance behavior.

### Summary Statistics

Table C.3 presents summary statistics for the SOI sample, separately for all firms, taxable firms, and untaxable firms.

Table C.3: Summary Statistics: SOI Data

| Variable | Mean | 10th Pctl | Median | 90th Pctl | N |
|---|---|---|---|---|---|
| **All Firms** | | | | | |
| Year | 2009.95 | 2001 | 2010 | 2019 | 1116 |
| $m_{i,t}$ | 0.05 | 0.02 | 0.04 | 0.09 | 1116 |
| Investment Rate | 0.06 | -0.14 | 0.06 | 0.23 | 1116 |
| Wedge | 0.83 | 0.76 | 0.82 | 0.92 | 1116 |
| Maintenance Share ($s_m$) | 0.18 | 0.08 | 0.15 | 0.31 | 1116 |
| $s_m$ (Adjusted) | 0.25 | 0.12 | 0.22 | 0.41 | 1116 |
| Capital Age | 2.21 | 1.66 | 2.15 | 2.80 | 1116 |
| **Taxable Firms** | | | | | |
| Year | 2009.04 | 2000 | 2009 | 2019 | 1114 |
| $m_{i,t}$ | 0.05 | 0.02 | 0.04 | 0.10 | 1114 |
| Investment Rate | 0.11 | -0.47 | 0.05 | 0.61 | 1114 |
| Wedge | 0.83 | 0.76 | 0.82 | 0.92 | 1114 |
| Maintenance Share ($s_m$) | 0.19 | 0.07 | 0.17 | 0.33 | 1114 |
| $s_m$ (Adjusted) | 0.26 | 0.11 | 0.23 | 0.43 | 1114 |
| Capital Age | 2.25 | 1.69 | 2.19 | 2.93 | 1114 |
| **Untaxable Firms** | | | | | |
| Year | 2009.05 | 2000 | 2009 | 2019 | 1113 |
| $m_{i,t}$ | 0.05 | 0.01 | 0.04 | 0.09 | 1113 |
| Investment Rate | 0.13 | -0.40 | 0.02 | 0.83 | 1113 |
| Wedge | 0.83 | 0.76 | 0.82 | 0.92 | 1113 |
| Maintenance Share ($s_m$) | 0.19 | 0.06 | 0.16 | 0.36 | 1113 |
| $s_m$ (Adjusted) | 0.26 | 0.08 | 0.23 | 0.47 | 1113 |
| Capital Age | 2.12 | 1.52 | 2.06 | 2.75 | 1113 |

**Notes:** Sample covers 49 industries over 23 years (1999-2021). Maintenance and investment rates are winsorized at 2nd and 98th percentiles. Reported maintenance expenditure enters user cost directly; "Adjusted" divides reported maintenance by 0.65. Wedge is $\frac{1-\tau_t}{1-\tau_t z_{i,t}}$. Capital Age is the ratio of gross to net book capital.

# D Empirical Robustness and Identification Details

This appendix provides detailed validation of identification strategies, robustness checks, and supplementary analyses for the maintenance demand estimates in Section 6.

## D.1 R-1 Estimates

**Identifying assumption.** Identification requires that the instrument affect maintenance only through the relative price—the exclusion restriction. The state wage indices $W_{s,t}$ (BLS Installation, Maintenance, and Repair occupations, SOC 49-0000) reflect broad labor-market conditions: tightness, cost of living, unionization, and prevailing-wage rules. Railroad maintenance facilities are geographically dispersed by network structure and historical infrastructure. Union Pacific's major facilities sit in Omaha (its historical headquarters), North Platte (the system's geographic midpoint), and division points set by crew-change and fueling needs—locations established, in many cases, in the nineteenth century and not chosen to minimize current labor costs. The exposure-weighted wage is therefore plausibly orthogonal to firm-specific maintenance demand, and the internal-versus-external split in the main text provides evidence consistent with a maintenance cost shock.

**First stage and balance.** The first stage is strong and correctly signed: across instruments the effective $F$ runs from about 10 to 15, and the instrument raises the relative price as predicted (Table D.1). I also examine pre-shock outcomes. Figure D.1 regresses lagged maintenance rates on the instrument—no lag is significant, providing no evidence that the instrument predicts prior maintenance. Figure D.2 regresses three-year-lagged maintenance, relative prices, and capital age on the three-year-lagged labor share. The share is not significantly related to past maintenance or capital age, although it is positively related to past relative prices. This association does not by itself establish a relationship with maintenance demand shocks, but the figure does not support balance across all three outcomes.

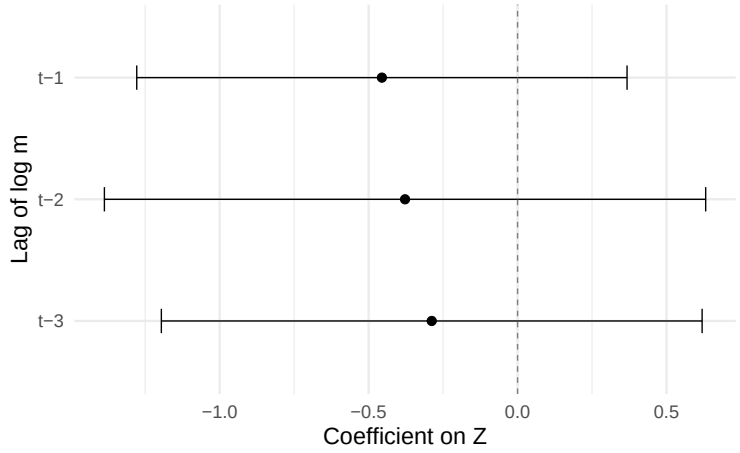
Table D.1: First-stage regressions

| Instrument | Coef. on $Z_{i,j,t}$ | Cluster SE | Effective $F$ | $N$ |
|---|---|---|---|---|
| Baseline (no controls) | 0.165*** | (0.049) | 11.54 | 340 |
| Baseline (all controls) | 0.168*** | (0.047) | 12.63 | 340 |
| Three-year-lagged shares | 0.156*** | (0.041) | 14.68 | 326 |
| National wage | 0.001*** | (0.000) | 10.68 | 314 |

* p < 0.1, ** p < 0.05, *** p < 0.01

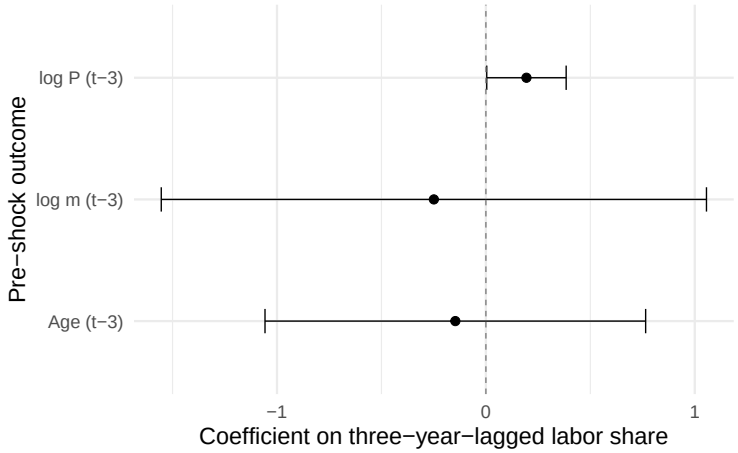
**Notes:** First stage for the log internal-maintenance price relative to investment. All rows include firm-asset and year fixed effects and firm-clustered standard errors; controlled rows add age and GDP exposure. Effective $F$ is from Montiel-Olea and Pflueger (2013).

Figure D.1: Lead tests of the log maintenance rate on $Z_{i,j,t}$




**Notes:** Coefficients from regressing the log maintenance rate one, two, and three years earlier on the instrument $Z_{i,j,t}$ of (22), controlling for firm trends, year fixed effects, and firm-asset fixed effects. Standard errors clustered by firm. Bars show normal-based 95% confidence intervals. The respective samples contain 328, 316, and 302 observations.

Figure D.2: Balance tests on pre-shock outcomes




**Notes:** Coefficients from regressing each three-year-lagged outcome on the three-year-lagged labor share, controlling for firm-asset and year fixed effects and firm trends. Standard errors clustered by firm. Bars show normal-based 95% confidence intervals. Each regression contains 302 observations.

**Inference at the shift-share frontier.** Because the instrument aggregates state wages with firm exposure weights, it has a shift-share structure. Table D.2 reports the inference appropriate to such designs: weak-IV-robust intervals, exposure-robust standard errors, and the effective number of independent wage shocks.

Table D.2: Railroad: exposure-robust and weak-IV inference

| Inference | With controls ($\hat{\omega} = 4.76$) | No controls ($\hat{\omega} = 5.41$) |
|---|---|---|
| Firm-clustered (analytic) | [2.66, 6.85] | [3.34, 7.48] |
| weak-IV-robust ($t_F$) | [1.48, 8.04] | [2.04, 8.78] |
| weak-IV-robust (AR) | [3.03, 8.37] | [2.50, 7.63] |
| Shift-level, clustered by state | [2.09, 7.43] | [2.24, 8.58] |
| Shift-level, clustered by year | [2.52, 7.00] | [2.93, 7.89] |
| AKM exposure-robust | [-384.95, 394.47] | [-204.36, 215.18] |
| AKM0 (weak-IV-robust) | $(-\infty, \infty)$ | $(-\infty, \infty)$ |
| Effective number of shifts (raw) | 627 | |
| Effective number of shifts (state-cluster) | 26 | |

**Notes:** Inference on the 2SLS of (21) with the instrument (22), for the headline with-controls specification and the no-controls specification. Firm-clustered rows give the conventional and weak-IV-robust ($t_F$, AR) intervals (Lee et al. 2022; Anderson and Rubin 1949). Shift-level rows aggregate the instrument to the state×year shock and cluster by state and by year (Borusyak, Hull, and Jaravel 2024), and AKM and AKM0 are exposure-robust intervals (Adão, Kolesár, and Morales 2019). The effective number of shifts is $1/\sum_s \bar{s}_s^2$, with $\bar{s}_s$ the mean exposure share on shift $s$, a property of the route-mileage shares and the lagged labor share that is identical across specifications. The with-controls column adds capital age and GDP exposure (the Table 2 specification), the no-controls column the twoway fixed effects only.

**Robustness.** Table D.3 re-estimates the main results without controls. Table D.4 re-estimates the elasticity across instruments—three-year-lagged versus two-year-lagged shares, and a national versus the state-mix wage shifter—and across outcomes (physical versus book maintenance). Table D.5 reports the reduced form and the implied elasticity.

Table D.3: The Railroad Maintenance Elasticity (No Controls)

| | All | | Internal | | External | |
|---|---|---|---|---|---|---|
| | OLS | IV | OLS | IV | OLS | IV |
| $\hat{\omega}$ | $1.84^{***}$ | $5.41^{***}$ | $2.91^{**}$ | $9.21^{***}$ | $-0.41$ | 0.85 |
| | (0.55) | (1.05) | (1.15) | (2.55) | (0.47) | (3.40) |
| $F$-statistic | 11.54 | | 11.54 | | 11.54 | |
| $t_F$ 95% CI | [2.04, 8.78] | | [1.07, 17.35] | | [-10.01, 11.70] | |
| AR 95% CI | [2.50, 7.63] | | [0.77, 13.60] | | [-6.70, 9.69] | |
| $N$ | 340 | | 340 | | 340 | |
| Twoway FE | ✓ | | ✓ | | ✓ | |
| Controls | | | | | | |

* p < 0.1, ** p < 0.05, *** p < 0.01

**Notes:** As in Table 2, without age and GDP controls. Firm-asset and year fixed effects and firm-clustered standard errors are included throughout.

Table D.4: Railroad: robustness of $\hat{\omega}$ across specifications

| Specification | $\hat{\omega}$ | Cluster SE | First-stage $F$ |
|---|---|---|---|
| Headline (all controls) | $4.76^{***}$ | (1.07) | 12.63 |
| No controls | $5.41^{***}$ | (1.05) | 11.54 |
| Three-year-lagged-share instrument | $6.06^{***}$ | (1.07) | 14.68 |
| National-wage instrument | $6.77^{***}$ | (0.91) | 10.68 |
| Physical maintenance outcome | $3.70^{**}$ | (1.67) | 12.63 |

* p < 0.1, ** p < 0.05, *** p < 0.01

**Notes:** The 2SLS elasticity of (21), re-estimated under alternative instruments (22) and outcomes. Except for the no-controls row, all add capital-age and GDP-exposure controls; all include firm-asset and year fixed effects, with standard errors clustered on the seven firms. Rows vary the labor-share weighting (three-year vs. two-year lag), the wage shifter (national vs. state-mix), and the maintenance measure (physical vs. book). The first-stage $F$ is the effective $F$ of Montiel-Olea and Pflueger (2013).

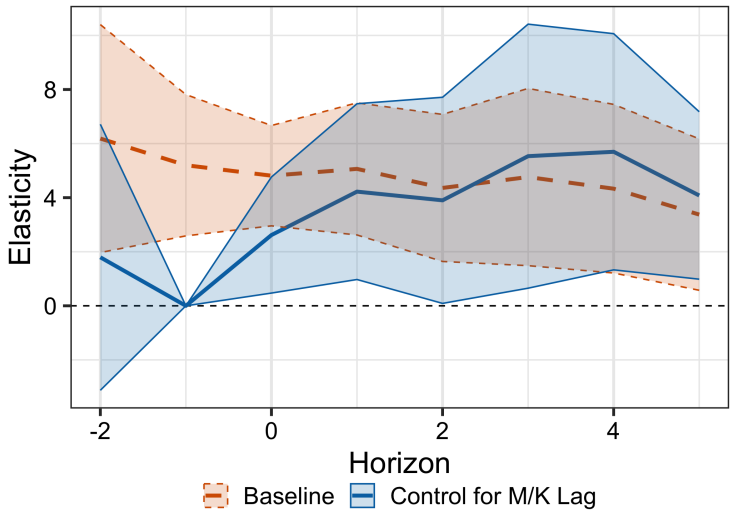
Table D.5: Railroad: reduced form and implied elasticity

| | Main | Internal | External |
|---|---|---|---|
| Reduced form ($M/K$ on exposure) | -0.801*** | -1.418*** | -0.020 |
| | (0.194) | (0.534) | (0.627) |
| Implied $\hat{\omega}$ (= − RF/FS) | 4.76 | 8.43 | 0.12 |

**Notes:** Each outcome regressed directly on the intensity-weighted wage exposure of (22) (capital age, GDP exposure, firmtype and year fixed effects; standard errors clustered on the seven firms). The first stage (relative price on exposure) is 0.168 (0.047), $F = 12.63$. The implied elasticity is $\hat{\omega} = -\text{RF/FS}$. */**/***: 10/5/1%.

**Dynamics.** Figure D.3 traces the local-projection response of maintenance to the relative price by horizon.

Figure D.3: Local Projection of Relative Price Shocks on the Maintenance Rate (R-1)




**Notes:** IV coefficients $\omega_h$ from the local projection $\log m_{i,j,t+h} = \alpha_{ij} + \lambda_t - \omega_h \log P_{i,j,t} + \text{Controls} + u_{i,j,t+h}$. The solid line controls for the lagged maintenance rate, and the dashed line does not.

## D.2 SOI Robustness Checks

**Specification band.** Table D.6 collects the headline elasticity across alternative specifications: the trend ladder varies the industry time trends; the pre-2014 window ends the sample before the 2013 change in SOI industry sampling, which made some industries too small to report maintenance separately; the alternative capital stock replaces SOI book capital, built from tax depreciation, with BEA capital built from economic depreciation; and the leave-one-out row deletes each

industry in turn. The elasticity stays positive and between 1.3 and 3.2 throughout.

Table D.6: Robustness of the SOI Maintenance Elasticity

| Specification (All sample) | $\hat{\omega}$ | Cluster SE |
|---|---|---|
| FE only (no trends) | 1.83* | (1.03) |
| Linear NAICS trends | 2.45** | (0.94) |
| Quadratic NAICS trends *(headline)* | 2.61** | (1.01) |
| Industry-specific linear trends | 1.33* | (0.79) |
| Pre-2014 window | 1.43** | (0.65) |
| Alternative capital stock | 1.50 | (1.11) |
| Leave-one-industry-out (range) | [2.18, 3.17] | — |

* p < 0.1, ** p < 0.05, *** p < 0.01

**Notes:** The headline maintenance elasticity (−1× the coefficient on $\log(\text{wedge}_k)$ in (20), All sample, clustered on the 49 BEA industries) across trend specifications, the pre-2014 (bonus-era) window, an alternative BEA capital stock, and the 49 single-industry deletions. The elasticity is positive in every specification and economically large (the 1.3–3.2 band, which spans the leave-one-out interval), and significant at conventional levels in five of the six point specifications—the lone exception being the alternative capital-stock measure. The wide standard errors reflect the limited within-industry variation in the wedge, so the SOI alone pins down $\omega$ imprecisely.

**Design diagnostics.** The specification is a continuous-intensity difference-in-differences: each industry's asset-composition exposure supplies the cross-sectional dose and the national bonus-depreciation schedule supplies the timing, following Zwick and Mahon (2017). Because the schedule is national, all industries are treated simultaneously and no not-yet-treated comparison group exists, so the heterogeneity-robust difference-in-differences estimators of Callaway, Goodman-Bacon, and Sant'anna (2025) and Chaisemartin and D'Haultfœuille (2026) are not identified in this design. I therefore assess the two-way fixed-effects slope directly. A heterogeneous-adoption consistency test (*did_had*, Chaisemartin et al. 2026), implemented as a Yatchew (1997) nonparametric test of mean-independence of the effect from the dose, does not reject ($p$ between 0.40 and 0.85). This is supportive of the linear slope under the test's assumptions, but it is not a second heterogeneity-robust estimate. An exposure event study also finds flat pre-trends across exposure terciles (joint $F = 0.66$, $p = 0.62$), and a balance test finds no differential pre-2002 trend in exposure ($-0.07$, $p = 0.68$). Finally, a within-fixed-effects dose-response check is monotone in the residualized dose ($p < 0.001$).

Table D.7: SOI design: identification diagnostics

| Diagnostic | Result | Reading |
|---|---|---|
| Heterogeneity-robust consistency (*did_had*) | not rej., $p = 0.40$–$0.85$ | TWFE consistent for the weighted-average slope |
| Pre-trends across exposure (event study) | $F = 0.66$ ($p = 0.62$) | not rejected |
| Selection on exposure (balance) | $-0.07$ ($p = 0.68$) | no differential pre-2002 trend |
| Dose-response monotonicity (within-FE bins) | increasing, $p < 0.001$ | linear form not masking heterogeneity |

**Notes:** The SOI specification is a continuous-intensity difference-in-differences, with each industry's asset-composition exposure as the dose and the national bonus-depreciation schedule as the timing, following Zwick and Mahon (2017). Because adoption is synchronized across industries (no not-yet-treated control), the heterogeneity-robust difference-in-differences estimators of Callaway, Goodman-Bacon, and Sant'anna (2025) and Chaisemartin and D'Haultfœuille (2026) are not identified here, so the two-way fixed-effects slope is validated directly. *did_had* (Chaisemartin et al. 2026) tests mean-independence of the effect from the dose with a Yatchew (1997) nonparametric test, the consistency condition for the slope. The event study tests parallel pre-trends across exposure terciles, the balance test checks for a differential pre-2002 trend in exposure, and the dose-response check confirms the response is monotone in the dose.

**Investment Weights.** Table D.8 re-estimates equation (20) with $z_i$ built from pre-reform investment flows rather than the capital stock. The pooled estimate is wrong-signed at about $-2.3$ and significant at the 10% level, while the tax-paying estimate is statistically insignificant. The sign change shows that the result is sensitive to the weighting scheme; I use capital weights because maintenance applies to the installed stock rather than only to recent investment flows.

Table D.8: The SOI Maintenance Elasticity: Investment-Weighted Wedge

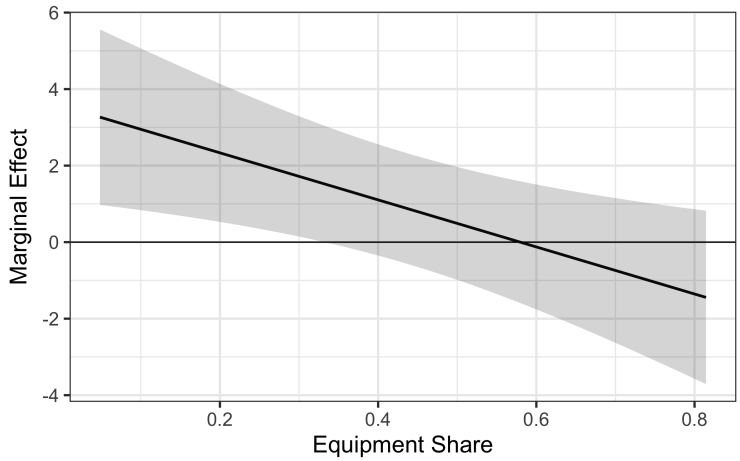
| | All | | Tax-paying | | Untaxed | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $\hat{\omega}$ | $-2.29^{*}$ | $-2.24^{*}$ | $-0.33$ | $-0.15$ | 0.45 | 0.19 |
| | (1.29) | (1.23) | (1.32) | (1.30) | (1.72) | (1.70) |
| $N$ (industry-years) | 1,116 | 1,113 | 1,051 | 1,051 | 1,049 | 1,049 |
| Industry and year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Capital age control | | ✓ | | ✓ | | ✓ |
| Two-digit NAICS trends | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

$^{*}$ $p < 0.1$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$

**Notes:** As in Table 1, but the industry exposure $z_i$ in (20) is built from pre-reform investment flows rather than the capital stock. All columns include two-way (industry and year) fixed effects and linear and quadratic two-digit NAICS trends, clustered by industry. Columns (2), (4), and (6) add a capital-age control; (1)–(2) use all firms, (3)–(6) split into tax-paying and untaxed firms.

**No Selection by Equipment Intensity.** Koby and Wolf (2020) raise concerns that bonus depreciation identification may be confounded by selection: equipment-intensive industries may have inherently different elasticities than structures-intensive industries. If true, differential exposure to bonus would be correlated with pre-existing elasticity heterogeneity, biasing estimates. To test this, I add an interaction between the equipment-capital ratio and the wedge term to equation (20). If the elasticity varies by equipment intensity, the interaction should be significant. Figure D.4 shows the interaction is small and statistically insignificant, indicating the elasticity is homogeneous across industries. This rules out selection effects.

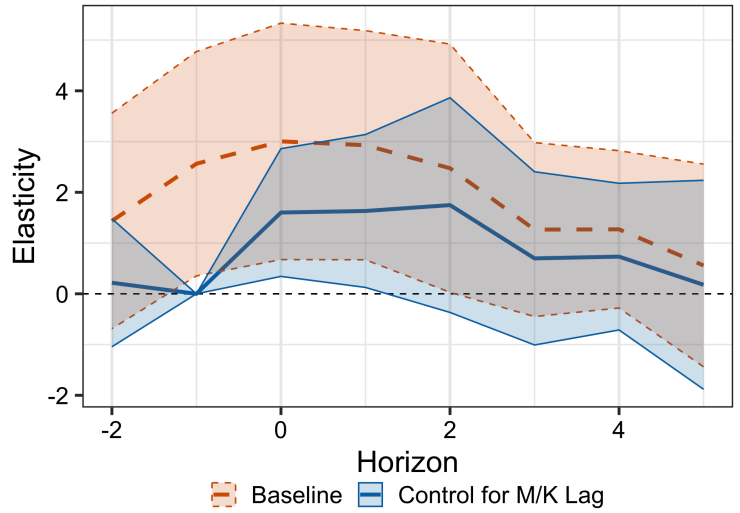
Figure D.4: Marginal Effect of Wedge on Maintenance Rate by Equipment Share




**Notes:** Marginal maintenance-wedge elasticity by equipment-capital ratio. The black line plots $\partial \log m / \partial \log(\text{wedge})$ against each industry's average equipment-capital ratio. Shaded bands are 95% confidence intervals.

**Dynamics.** Figure D.5 traces the response of maintenance to the wedge by horizon.

Figure D.5: Local Projection of Relative Price Shocks on the Maintenance Rate (SOI)




**Notes:** This figure plots the coefficients $\omega_h$ from the regression

$$\log m_{i,t+h} = \alpha_i + \lambda_t - \omega_h \log\left(\frac{1-\tau_t^c}{1-\tau_t^c z_{i,t}}\right) + \text{Controls} + \eta_{i,t+h}.$$

The solid path controls for the lagged maintenance rate. Both specifications control for industry trends and correspond to the SOI sample with all types of firms.

# E Quantification

This appendix collects the full calibration of the quantitative model and the variants referenced in Section 7.1: partial equilibrium, general equilibrium, fiscal crowding out, and a competitive maintenance sector. Table E.1 lists the assigned, estimated, and calibrated parameters.

Table E.1: Calibration of the Quantitative Model

| Description | Symbol | Value | Source / target |
|---|---|---|---|
| *A. Technology and aggregation* | | | |
| Capital elasticity | $\alpha_K$ | 0.245 | Chodorow-Reich et al. (2025) |
| Labor elasticity | $\alpha_L$ | 0.650 | Chodorow-Reich et al. (2025) |
| Discount rate | $r^k$ | 0.060 | Chodorow-Reich et al. (2025) |
| Sectoral output weights (c / nc) | $c_c, c_{nc}$ | 0.70 / 0.30 | Chodorow-Reich et al. (2025) |
| Private share of output | — | 0.81 | NIPA |
| Maintenance-sector labor share | $\eta$ | 0.30 | competitive maintenance sector |
| *B. Tax rates* ($\tau^c$, $\tau^x$) | | | |
| Corporate, pre-TCJA | — | 0.249, 0.210 | Chodorow-Reich et al. (2025) |
| Corporate, post-TCJA | — | 0.151, 0.138 | Chodorow-Reich et al. (2025) |
| Non-corporate (held fixed) | — | 0.280, 0.230 | Chodorow-Reich et al. (2025) |
| *C. Estimated parameters (standard error)* | | | |
| Maintenance demand level | $\gamma$ | 0.0305 (0.0056) | §6 |
| Maintenance demand elasticity | $\omega$ | 2.608 (1.012) | §6 |
| Investment coefficient (raw) | $\widehat{\varepsilon_I}$ | 4.27 (0.51) | Chodorow-Reich et al. (2025) |
| Crowding-out coefficient | — | 3.20 (1.22) | Plante, Richter, and Zubairy (2026) |
| *D. Calibrated to match targets* | | | |
| Non-corporate productivity | $A_{nc}$ | 0.919 | $K_c/K = 0.70$ |
| Baseline depreciation | $\delta_0$ | 0.137 | $I/Y = 0.109$ |
| *E. Implied at the calibration (corporate sector)* | | | |
| Steady-state depreciation | $\delta(m^*)$ | 0.083 | |
| Maintenance rate | $m^*$ | 0.035 | |
| Maintenance share of user cost | $s_m$ | 0.187 | |
| *F. First-order transition calibration* | | | |
| Maintenance share in the first-order calibration | $s_m$ (approx.) | 0.171 | zero-wedge expression |

*continued on next page*

*Table E.1, continued*

| Description | Symbol | Value | Source / target |
|---|---|---|---|
| Depreciation elasticity | $\varepsilon_\delta$ | 0.91 | |
| Corrected investment coefficient | $\widetilde{\varepsilon}_I$ | 5.15 | $= \widehat{\varepsilon}_I/(1-s_m)$ |
| Adjustment cost, NGMM | $\phi_{\text{NGMM}}$ | 2.83 | |
| Adjustment cost, NGM | $\phi_{\text{NGM}}$ | 2.01 | |
| *G. Competitive maintenance-sector calibration* | | | |
| Baseline depreciation, NGMM | $\delta_0$ | 0.132 | $I/G = 0.109$ |
| Non-corporate productivity, NGMM | $A_{nc}$ | 0.892 | $K_c/K = 0.70$ |
| Production labor elasticity, NGMM | $\alpha_L$ | 0.614 | aggregate labor share 0.65 |
| Maintenance productivity | $A_m$ | 1.675 | $p_0^M = 1$ |
| Production labor elasticity, NGM | $\alpha_L$ | 0.650 | matched initial steady state |
| Corporate productivity, NGM | $A_c$ | 0.966 | matched initial steady state |
| Non-corporate productivity, NGM | $A_{nc}$ | 0.887 | matched initial steady state |
| Corporate depreciation, NGM | $\delta_c$ | 0.106 | matched initial steady state |
| Non-corporate depreciation, NGM | $\delta_{nc}$ | 0.105 | matched initial steady state |

**Notes:** SOI main-spec estimates. In the baseline calibration, $\delta_0$ and $A_{nc}$ are jointly calibrated so the initial steady state matches $K_c/K = 0.70$ (Chodorow-Reich et al. 2025) and gross $I/Y = 0.109$ (NIPA). Panel E reports initial pre-reform steady-state quantities. Panel F reports the maintenance-share and depreciation-elasticity expressions used in the first-order transition calibration and the resulting investment coefficient and adjustment-cost parameters. The transition calibration uses the zero-wedge expressions, so its maintenance share is reported separately from the initial pre-reform share. Panel G reports the additional calibration for competitive maintenance with $\eta = 0.30$. Private GDP is $G = Y_c + Y_{nc} - X_m$. Its NGM benchmark matches initial corporate and non-corporate capital, private GDP, wages, and total labor supply.

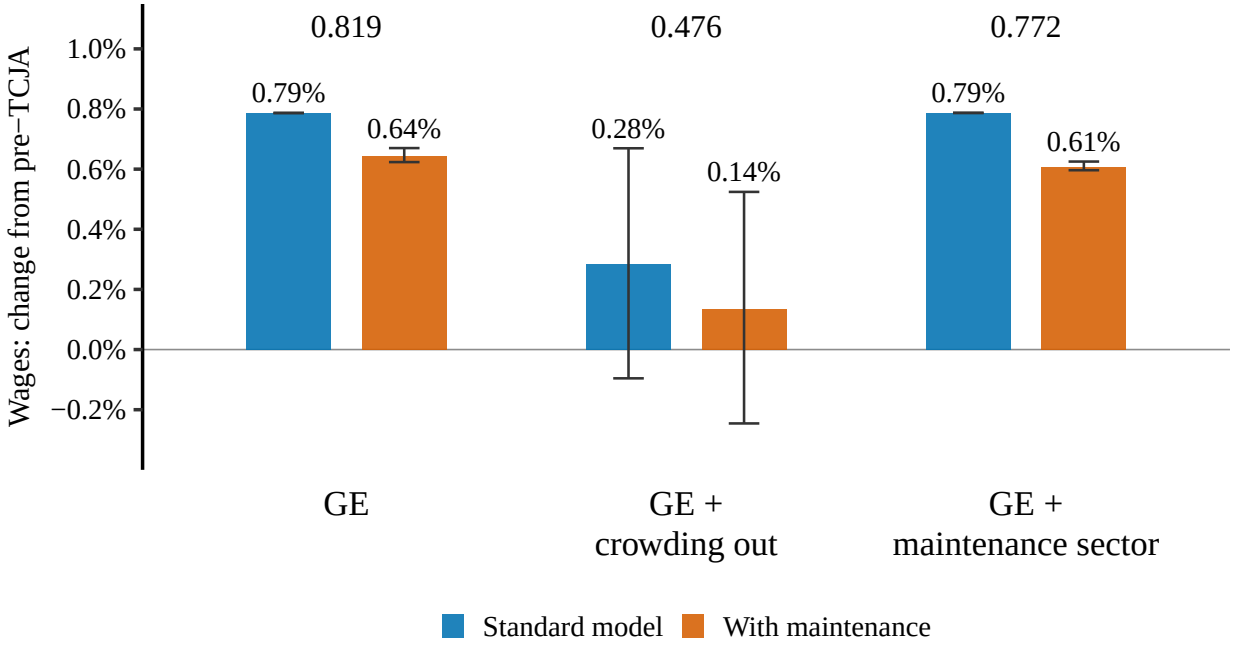
The baseline depreciation rate $\delta_0$ is calibrated to match the pre-TCJA investment-output ratio. Table E.2 sweeps it across a range around its calibrated value, re-calibrating non-corporate productivity at each node so the corporate capital share still hits 0.70. This relaxes the investment-output target; the resulting $I/Y$ is reported at each node. The maintenance markdown holds throughout, with the NGMM capital response smaller than the NGM response at every node. Its magnitude does move with $\delta_0$: the NGMM capital response ranges from approximately 70% to 87% of the NGM response, compared with 81% at the calibrated value.

Table E.2: Sensitivity of Steady-State Capital Responses to the Baseline Depreciation Rate $\delta_0$

| $\delta_0$ | $I/Y$ | $s_m$ | NGM (%) | NGMM (%) | NGMM/NGM |
|---|---|---|---|---|---|
| 0.068 | 0.031 | 0.306 | 6.78 | 4.73 | 0.698 |
| 0.080 | 0.050 | 0.277 | 6.78 | 4.93 | 0.727 |
| 0.103 | 0.079 | 0.233 | 6.78 | 5.22 | 0.770 |
| **0.137** | **0.109** | **0.187** | **6.78** | **5.52** | **0.815** |
| 0.171 | 0.129 | 0.157 | 6.78 | 5.73 | 0.845 |
| 0.205 | 0.143 | 0.135 | 6.78 | 5.87 | 0.866 |
| 0.210 | 0.145 | 0.133 | 6.78 | 5.89 | 0.869 |

**Notes:** Steady-state changes in domestic corporate capital under the GE closure with fixed interest rates and input prices. Each row holds $\gamma$, $\omega$, and the assigned production parameters at their central values, varies $\delta_0$, and recalibrates $A_{nc}$ to maintain $K_c/K = 0.70$. The NGM benchmark matches the initial capital stocks and labor allocation at every row. The induced gross $I/Y$ is reported; only the calibrated row (bold) matches the 0.109 target. $s_m$ is evaluated at the initial pre-reform steady state. Ratios divide the NGMM capital response by the NGM response.

Figure E.1: Steady-State Wage Effects of TCJA across Model Closures




**Notes:** Bars show mean percentage changes in wages between the initial and post-TCJA steady states in the NGM (blue) and NGMM (orange). Error bars are 95% confidence intervals from 5,000 parameter draws. Numbers above each pair of bars divide the mean NGMM response by the mean NGM response. The three closures are general equilibrium, general equilibrium with fiscal crowding out, and general equilibrium with a competitive maintenance sector.

# F Model Extensions

## F.1 Capital Reallocation

This subsection extends the base model to heterogeneous firms which only differ in their tax status. That leads to reallocation through variation in the marginal product of capital. A fraction $\eta \in (0,1)$ of firms is taxable (Type $T$) and the remaining $1-\eta$ are untaxed (Type $U$). Variation in taxability comes from realization of an i.i.d. fixed cost $\bar{F}$, which is assumed to be sufficiently large that it exceeds profits. Untaxed firms also pay a higher cost of investment $p^I + b$, which is meant to account (in a reduced-form way) for the fact that untaxable firms often face some kind of financial constraint that hinders their ability to access new investment (Lian and Ma 2020). This formulation allows us to model the fact that investment is typically more expensive for unprofitable firms without explicitly modeling the borrowing constraint.

Each period, a firm observes its tax type $\theta \in \{T, U\}$:

- Type $T$ (taxable) firms face corporate tax $\tau^c$.

- Type $U$ (untaxed) firms pay no corporate tax because they have a fixed cost $\bar{F}$. They also have a higher cost of investment $p^I + b$.

After observing its type, a firm chooses:

1. Maintenance expenditure $M$, with $m = \frac{M}{K}$ being the maintenance intensity.
2. Investment $I$ (at new capital price $p^I$).
3. Net used–capital sales $s$, where $s > 0$ indicates selling a fraction of the capital and $s < 0$ indicates buying used capital. Firms pay a convex adjustment cost $G(s)$ for participating in the used capital market, which can be thought of as accounting for information frictions in a reduced–form way. For example, a firm selling a used car must furnish information about the vehicle, which is costly to do. The equilibrium used–capital price is denoted by $q$.

Putting together maintenance expenditures $m = M/K$, investment $I$, and capital sales $s$, the law of motion for capital (independently of tax status) is

$$K' = \Big[1 - \delta(m)\Big](1 - s)K + I. \tag{A.31}$$

Because firms vary in their tax status, they also vary in their cash flows. Profitable firms have cash flows

$$\pi^T = (1 - \tau^c)\Big[F(K) - p^m M + q\, s\, K - G(s)K\Big] - \big(1 - c - \tau^c z\big)p^I I,$$

where $z$ incorporates the possibility that the firm may be untaxable in the future and hence will not always be able to take advantage of a tax depreciation allowance. Note that capital sales are taxed at rate $\tau^c$, reflecting current policy practice. Given those cash flows, we get the following recursive formulation for a firm choosing $K', s, M$:

$$\begin{aligned} V^T(K) = \max_{m,s,K'} \Bigg\{ & (1 - \tau^c)\Big[F(K) - p^m\, mK + q\, s\, K - G(s)K\Big] \\ & - \big(1 - c - \tau^c z\big)p^I\Big(K' - [1 - \delta(m)](1 - s)K\Big) + \frac{V^*(K')}{1 + r^k}\Bigg\}, \end{aligned} \tag{A.32}$$

where

$$V^*(K') = \eta\, V^T(K') + (1 - \eta)\, V^U(K')$$

is the expected continuation value. Taxable firms therefore have the following FOCs:

$$\textbf{Investment:} \quad \frac{V^{*'}(K')}{1+r^k} = \left(1-c-\tau^c z\right)p^I, \tag{A.33}$$

$$\textbf{Maintenance:} \quad -\delta'(m) = \frac{1-\tau^c}{1-c-\tau^c z}\frac{p^m}{p^I\,(1-s)}, \tag{A.34}$$

$$\textbf{Sales:} \quad (1-\tau^c)\Big[q - G'(s)\Big] = \left(1-c-\tau^c z\right)p^I\,[1-\delta(m)]. \tag{A.35}$$

On the other hand, untaxed type $U$ firms do not face any tax, so their cash flows are

$$\pi^U = F(K) - p^m M + q\,s\,K - G(s)K - (1-\tau^c\tilde{z})(p^I+b)I - \bar{F}.$$

Here $\tilde{z}$ is the net present value of depreciation allowances for a currently untaxed firm, the counterpart to $z$ for taxable firms. Given those cash flows, we get the following recursive formulation for a firm choosing $K', s, M$:

$$\begin{aligned} V^U(K) = \max_{m,s,K'}\Bigg\{ & F(K) - p^m\,mK + q\,s\,K - G(s)K \\ & - (1-\tau^c\tilde{z})(p^I+b)\Big(K' - [1-\delta(m)](1-s)K\Big) - \bar{F} + \frac{V^*(K')}{1+r^k}\Bigg\}. \end{aligned} \tag{A.36}$$

The type $U$ FOCs are:

$$\textbf{Investment:} \quad \frac{V^{*'}(K')}{1+r^k} = (1-\tau^c\tilde{z})(p^I+b), \tag{A.37}$$

$$\textbf{Maintenance:} \quad -\delta'(m) = \frac{p^m}{(1-\tau^c\tilde{z})(p^I+b)(1-s)}, \tag{A.38}$$

$$\textbf{Sales:} \quad q - G'(s) = (1-\tau^c\tilde{z})(p^I+b)\,[1-\delta(m)]. \tag{A.39}$$

The envelope condition enables us to define the user cost of capital in this economy. For taxed firms,

$$V^{T\prime}(K) = (1-\tau^c)\Big[F'(K) - p^m\,m^T + q\,s^T - G(s^T)\Big] + \left(1-c-\tau^c z\right)p^I\,(1-\delta(m^T))(1-s^T),$$

while for untaxed firms we have

$$V^{U\prime}(K) = F'(K) - p^m\,m^U + q\,s^U - G(s^U) + (1-\tau^c\tilde{z})(p^I+b)\,(1-\delta(m^U))(1-s^U).$$

The combined envelope condition is

$$V^{*'}(K) = \eta\, V^{T\prime}(K) + (1-\eta)\, V^{U\prime}(K).$$

**Equilibrium**

An equilibrium in this economy is a collection of policies $\{V^T,\ V^U\}$, prices $\{q, p^I, p^M\}$ (the latter two exogenous) and allocations $\{\, m^T,\ s^T,\ K'^T,\ m^U,\ s^U,\ K'^U\}$ such that:

1. For all $K$, the Bellman equations for $V^T(K)$ and $V^U(K)$ are satisfied with the corresponding optimal policies.

2. The chosen policy functions satisfy the FOCs for investment, maintenance, and used–capital sales (with type–specific controls $m^\iota$ and $s^\iota$) and the envelope conditions hold.

3. The law of motion for capital,

$$K' = \Big[1 - \delta(m^\iota)\Big](1-s)K + I,$$

   is satisfied for each firm.

4. The used–capital market clears at the equilibrium price $q$; that is, the total net sales of used capital by all firms (taxable and untaxed) sum to zero:

$$\int s^T(K)\cdot K\, d\mu(K) + \int s^U(K)\cdot K\, d\nu(K) = 0,$$

   where $\mu(K)$ and $\nu(K)$ are the distributions of capital among taxable and untaxed firms.

5. The expected continuation value is given by

$$V^*(K') = \eta\, V^T(K') + (1-\eta) V^U(K'),$$

   and firms form rational expectations consistent with the aggregate outcomes.

**Trading Conditions**

Trading of used capital will only arise under certain conditions. In particular, an active trading market is characterized by:

- Taxable firms choosing $s^T > 0$ (selling used capital) because their subsidized cost $\frac{1-c-\tau^c z}{1-\tau^c}\, p^I(1-\delta(m^T))$ is lower than that faced by untaxed firms.

- Untaxed firms choosing $s^U < 0$ (buying used capital) because the equilibrium used capital price $q$ (adjusted by the marginal cost $G'(s^U)$) falls below the cost of acquiring new capital $p^I(1-\delta(m^U))$.

- An equilibrium price $q$ that satisfies

$$q = \frac{1-c-\tau^c z}{1-\tau^c}\, p^I(1-\delta(m^T)) + G'(s^T) = (1-\tau^c\tilde{z})(p^I+b)(1-\delta(m^U)) + G'(s^U),$$

  along with the market clearing condition

$$\eta\, s^T K^T + (1-\eta)\, s^U K^U = 0.$$

Essentially, we require that the gap between the pre–tax price and after–tax price of new capital is sufficiently large that it is worth it for firms to sell used capital and for untaxed firms to buy it, i.e.,

$$\frac{(1-c-\tau^c z)p^I\,[1-\delta(m^T)]}{1-\tau^c} < q < (1-\tau^c\tilde{z})(p^I+b)[1-\delta(m^U)].$$

I focus on this equilibrium because we observe active used capital trading in practice.

**Aggregate User Cost of Capital**

Let $\Psi^\iota \equiv F'(K^\iota)$ denote the steady-state user cost of capital for a type-$\theta$ firm, the marginal product of capital obtained by combining its investment FOC with its envelope condition. The aggregate user cost is the capital-weighted average of the two type-specific user costs:

$$\boxed{\Psi_{\text{agg}} = \frac{\eta\, K^T}{K}\,\Psi^T + \frac{(1-\eta)\, K^U}{K}\,\Psi^U, \quad \text{with } K = \eta\, K^T + (1-\eta)\, K^U.}$$

Indeed, one can observe that the proportional change in user cost will be strictly smaller to first order in this economy than in the representative firm economy in the main model because the share of untaxable firms will not react very much. Moreover, capital sales will prop up the maintenance rate (since the maintenance optimality condition implies a higher maintenance rate for untaxable firms), so maintenance and depreciation change less in this economy in the aggregate.

## F.2 Extension to Multiple Maintenance Inputs

In this section, I extend the model to consider multiple inputs to maintenance production as well as a choice between using internal or external maintenance services. Production is carried out

entirely using capital:

$$Y_t = F(K_t).$$

Capital evolves according to

$$K_{t+1} = \Big[1 - \delta(m_{I,t}, m_{E,t})\Big]K_t + I_t,$$

where $m_{I,t}$ is internal maintenance intensity and $m_{E,t}$ is external maintenance intensity. I assume that internal maintenance is the sum of labor and materials purchases

$$M_{I,t} = g(L_{m,t}, I_{m,t}).$$

The firm purchases internal maintenance labor at price $w_t$ and materials at price $p_t^X$. External maintenance $m_{E,t} = M_{E,t}/K_t$ is purchased at price $p_t^m$ and faces some convex cost of adjustment $C(M_{E,t}, M_{E,t-1})$, with $C(M_E, M_E) = 0$ in steady state. This is to reflect the fact that external maintenance contracts are typically quite sticky. Since internal maintenance also includes labor, then internal maintenance is also sticky, but presumably less so than external maintenance because internal resources can be reallocated more quickly. I could capture that by having an outside option for labor or introducing a separate adjustment cost for internal maintenance, but it would be unnecessarily complicated. Note that with multiple maintenance types, the price of maintenance is a weighted average of each.

Taking $K_0$ as given, the firm chooses the sequence $\{K_{t+1},\, L_{m,t},\, I_{m,t},\, M_{E,t}\}_{t\geq 0}$ to maximize

$$\mathcal{L} = \sum_{t=0}^{\infty}\left(\frac{1}{1+r^k}\right)^t \Bigg\{(1-\tau_t^c)\Big[F(K_t) - w_t\, L_{m,t} - p_t^X\, I_{m,t} - p_t^m M_{E,t} - C\big(M_{E,t}, M_{E,t-1}\big)\Big] \\ -\ (1-\tau_t^c z_t) p_t^I\Big[K_{t+1} - \Big(1 - \delta\Big(m_{I,t}, m_{E,t}\Big)\Big)K_t\Big]\Bigg\},$$

subject to $m_{I,t} = \frac{g(L_{m,t}, I_{m,t})}{K_t}$. The first-order conditions are not substantially different from the baseline model. The capital Euler equation is:

$$(1-\tau_t^c z_t)p_t^I = \frac{1}{1+r^k}\Bigg\{(1-\tau_{t+1}^c)\, F_K(K_{t+1}) + (1-\tau_{t+1}^c z_{t+1})\, p_{t+1}^I\Big[1 - \delta\big(m_{I,t+1}, m_{E,t+1}\big) \\ + \delta_1(t+1)\, m_{I,t+1} + \delta_2(t+1)\, m_{E,t+1}\Big]\Bigg\}, \tag{A.40}$$

which in steady state simplifies to

$$F_K = p^I(1-\tau^c z)\left(\frac{r^k + \delta(m_I, m_E)}{1-\tau^c}\right) + p^{M,I} m_I + p^m m_E, \tag{A.41}$$

where $p^{M,I}$ is the price of internal maintenance. If internal maintenance is a CES aggregator of labor and materials, then $p^{M,I}$ would just be the usual CES price index of $p^X$ and $w$.

There are three maintenance choices. Throughout, $\delta_1 = \partial\delta/\partial m_I$ and $\delta_2 = \partial\delta/\partial m_E$ denote the partials of depreciation with respect to internal and external maintenance intensity, $g_L = \partial g/\partial L_m$ and $g_M = \partial g/\partial I_m$ the marginal products of labor and materials in internal maintenance, and $C_1, C_2$ the partials of the adjustment cost $C(M_{E,t}, M_{E,t-1})$ with respect to its current and lagged arguments. Beginning with the choice of internal labor $L_{m,t}$, the firm's optimal choice is satisfied when

$$-\delta_1(t)\, g_L(t) = \frac{(1-\tau_t^c) w_t}{(1-\tau_t^c z_t) p_t^I}. \tag{A.42}$$

Similarly, optimal materials is given by

$$-\delta_1(t)\, g_M(t) = \frac{(1-\tau_t^c) p_t^X}{(1-\tau_t^c z_t) p_t^I}, \tag{A.43}$$

which implies the marginal rate of substitution between materials and labor is

$$\frac{g_L(t)}{g_M(t)} = \frac{w_t}{p_t^X}. \tag{A.44}$$

On the other hand, external maintenance choice is given by

$$-\delta_2(t) = \frac{(1-\tau_t^c)\left[p_t^m + C_1(t)\right] + \frac{1}{1+r^k}(1-\tau_{t+1}^c)\, C_2(t+1)}{(1-\tau_t^c z_t) p_t^I}. \tag{A.45}$$

## F.3 A Competitive Maintenance Sector

In this section, I extend the model to allow maintenance to be supplied by a competitive intermediate sector. Corporate and non-corporate firms, indexed by $s \in \{c, nc\}$, produce output $Y_s = F_s(K_s, L_s)$ and purchase maintenance services at price $p^M$. Maintenance producers use labor $L_m$ and materials $X_m$ according to

$$Q = A_m L_m^{\eta} X_m^{1-\eta} = M_c + M_{nc}, \qquad 0 < \eta < 1,$$

where the second equality imposes maintenance-market clearing.

Normalizing the materials price to one, competition implies that the maintenance price equals unit cost. Optimal input choices satisfy

$$p^M = \frac{1}{A_m}\left(\frac{w}{\eta}\right)^{\eta}\left(\frac{1}{1-\eta}\right)^{1-\eta},$$
$$wL_m = \eta p^M Q, \qquad X_m = (1-\eta)p^M Q.$$

Labor moves freely across the three sectors. With total labor supply normalized to one, equilibrium requires

$$F_{L,c}(K_c, L_c) = F_{L,nc}(K_{nc}, L_{nc}) = w, \qquad L_c + L_{nc} + L_m = 1.$$

Conditional on prices, the investment and maintenance conditions take the same form as in the baseline model. Let $\tau_s$ denote the marginal tax rate in sector $s$, defined as in the baseline model. Each production sector's user cost satisfies

$$\begin{aligned} F_{K,s}(K_s, L_s) &= \frac{p^I[r + \delta(m_s)]}{1-\tau_s} + p^M m_s \\ &= \mathrm{UCC}_s^{\mathrm{HJ}} + p^M m_s \equiv \mathrm{UCC}_s, \end{aligned}$$

where $m_s = M_s/K_s$ and the Hall and Jorgenson component is evaluated at $\delta(m_s)$.

Let $s_{m,s} = p^M m_s/\mathrm{UCC}_s$ denote the maintenance share of user cost in sector $s$. Holding interest rates and investment prices fixed, the envelope theorem gives

$$d\log \mathrm{UCC}_s = -(1-s_{m,s})\, d\log(1-\tau_s) + s_{m,s}\, d\log p^M.$$

Holding maintenance productivity and the materials price fixed, the wage elasticity of $p^M$ is $\eta$. Hence,

$$d\log \mathrm{UCC}_s = -(1-s_{m,s})\, d\log(1-\tau_s) + s_{m,s}\eta\, d\log w.$$

The first term is the capital preservation effect: the maintenance share attenuates the direct response of user cost to tax policy. The second term captures the equilibrium maintenance-price response. If wages rise following a tax cut, maintenance becomes more expensive, offsetting part of the decline in user cost.

The investment-capital relationship also continues to hold: $I_s = \delta(m_s)K_s$. Maintenance intensity now responds to both the tax wedge and the equilibrium maintenance price, changing the depreciation response and hence the investment-capital elasticity gap. Moreover, reduced maintenance demand releases labor for production. The equilibrium jointly determines these price and labor-allocation effects, which modify the aggregate response while preserving the firm-level optimality conditions.

Maintenance is an intermediate input for the production sectors, and materials are an intermediate input for maintenance producers. Maintenance receipts therefore cancel when value added is aggregated. Private GDP is

$$G = (Y_c + Y_{nc} - p^M Q) + (p^M Q - X_m) = Y_c + Y_{nc} - X_m.$$

This is the output measure used for calibration and aggregate outcomes in the quantitative model with a maintenance sector.

**Proposition 8** (Output and Maintenance)**.** *Holding aggregate labor supply, maintenance productivity, and the materials price fixed, the first-order change in private GDP between steady states is*

$$dG = \sum_{s\in\{c,nc\}} \mathrm{UCC}_s^{\mathrm{HJ}}\, dK_s - p^M \sum_{s\in\{c,nc\}} K_s\, dm_s.$$

*For a given change in capital, lower maintenance intensity raises the output response.*

*Proof.* Differentiating private GDP,

$$dG = \sum_s F_{K,s}\, dK_s + \sum_s F_{L,s}\, dL_s - dX_m.$$

Since firms equate the marginal product of labor to the wage and total labor supply is fixed,

$$dG = \sum_s F_{K,s}\, dK_s - w\, dL_m - dX_m.$$

The maintenance production function and the producer's input conditions imply

$$\begin{aligned} p^M\, dQ &= p^M \frac{\eta Q}{L_m}\, dL_m + p^M \frac{(1-\eta)Q}{X_m}\, dX_m \\ &= w\, dL_m + dX_m. \end{aligned}$$

Substituting gives

$$dG = \sum_s F_{K,s}\, dK_s - p^M\, dQ.$$

Maintenance-market clearing implies

$$dQ = \sum_s \left(m_s\, dK_s + K_s\, dm_s\right),$$

so

$$dG = \sum_s \left(F_{K,s} - p^M m_s\right) dK_s - p^M \sum_s K_s\, dm_s.$$

The capital condition gives $F_{K,s} - p^M m_s = \text{UCC}_s^{\text{HJ}}$, yielding the result. □

The intuition is straightforward. More capital raises production, but it also requires maintenance. The first term measures the extra output net of that maintenance. The second term comes from changing how intensively firms maintain the capital they already have. Cutting maintenance releases labor for production and reduces materials expenditures. Thus, accounting for maintenance can imply less capital accumulation but more output.

The timing matters as well. Following an unanticipated reform, firms can change maintenance immediately, while capital is initially fixed. At the initial wage, a decline in maintenance reduces labor demand. Since labor supply is fixed, wages fall and the corporate and non-corporate sectors hire the workers released from maintenance. The price of maintenance falls with wages. Later, as capital accumulates, greater labor demand in production can raise wages and maintenance prices.

Using the maintenance demand function in the quantitative model, and holding investment prices fixed,

$$d \log m_s = -\omega \left[ d \log(1 - \tau_s) + \eta \, d \log w \right].$$

Falling wages initially make maintenance cheaper, moderating its decline. If wages subsequently rise, they reinforce the reduction in maintenance and the increase in depreciation. These price responses therefore change over the transition. Moreover, the labor and materials released from maintenance can raise output before capital has adjusted, increasing revenue feedback over the ten-year scoring window even when capital accumulation is smaller.

# G Adjustment Costs and Dynamic Scoring

The gap in elasticities shapes both the magnitude of the long-run response of capital as well as the convergence rate. In practice, researchers often estimate short-run investment responses to tax changes and use these to infer long-run effects on capital. This requires a model of transition dynamics. With adjustment costs, capital converges gradually to its new steady state. Suppose the transition path for investment following a permanent tax change takes the form:

$$\varepsilon_{I,\tau}(t) = \varepsilon_{I,\tau}^{LR} \left[ 1 - (1 - \theta) e^{-\kappa t} \right]. \tag{A.46}$$

*Definition* 1 (Transition Parameters). The *impact effect* $\theta \in (0, 1)$ is the fraction of the long-run response achieved on impact: $\theta = \varepsilon_{I,\tau}(0)/\varepsilon_{I,\tau}^{LR}$. The *convergence rate* $\kappa > 0$ governs the speed of subsequent adjustment.

Different adjustment cost specifications generate different feasible $(\theta, \kappa)$ pairs. The following proposition assumes that both models share an increasing relationship between impact shares

and convergence rates: higher $\theta$ is associated with higher $\kappa$.

**Proposition 9** (Slower Convergence)**.** *Suppose both models have transition paths of the form* (A.46) *with* $\theta \in (0,1)$*, both models share an increasing relationship between* $\theta$ *and* $\kappa$*, and both models match the same short-run investment elasticity at horizon* $T$*, measured in periods after reform. If*

$$\left|\varepsilon_{I,\tau}^{LR,NGMM}\right| > \left|\varepsilon_{I,\tau}^{LR,NGM}\right|,$$

*then the maintenance model has a lower convergence rate.*

## G.1 Proof of Proposition 9

The proof proceeds in three steps. First, I state the condition under which the long-run investment elasticity is higher in the NGMM. Second, I characterize the assumed transition paths. Third, I show that matching the same short-run elasticity then implies slower convergence.

**Step 1: Long-run investment elasticities.** In steady state, investment replaces depreciated capital: $I = \delta(m)K$. Taking logs and differentiating with respect to the tax term yields

$$\begin{aligned}\varepsilon_{I,\tau}^{LR,NGM} &= \varepsilon_{K,\tau}^{NGM},\\ \varepsilon_{I,\tau}^{LR,NGMM} &= \varepsilon_{\delta,\tau}^{NGMM} + \varepsilon_{K,\tau}^{NGMM}.\end{aligned}$$

The depreciation rate is fixed in the NGM. Because capital elasticities differ across models, a nonzero depreciation response alone is not sufficient to establish a larger investment response in absolute value. The required condition is

$$\left|\varepsilon_{\delta,\tau}^{NGMM}\right| > \left|\varepsilon_{K,\tau}^{NGM}\right| - \left|\varepsilon_{K,\tau}^{NGMM}\right|.$$

In the Cobb–Douglas special case, holding input prices fixed, the right-hand side is

$$\frac{s_m}{1-\alpha}\left|\frac{\tau}{1-\tau}\right|.$$

**Step 2: Transition paths.** The proposition restricts attention to transition paths of the form

$$\varepsilon_{I,\tau}(t) = \varepsilon_{I,\tau}^{LR}\left[1-(1-\theta)e^{-\kappa t}\right], \tag{A.47}$$

where $\kappa > 0$ determines the convergence rate and $\theta \in (0,1)$ is the impact response as a share of the long-run response. The proposition assumes that feasible $\theta$ and $\kappa$ are co-monotonic along the common relationship: larger impact shares also imply faster convergence.

**Step 3: Calibration and convergence rates.** Suppose both models are calibrated to match the same observed short-run investment elasticity in absolute value, $\widehat{\varepsilon}_I^{SR}$, at horizon $T$:

$$\widehat{\varepsilon}_I^{SR} = \left|\varepsilon_{I,\tau}^{LR,NGM}\right| \left[1 - (1-\theta^{NGM})e^{-\kappa^{NGM}T}\right],$$
$$\widehat{\varepsilon}_I^{SR} = \left|\varepsilon_{I,\tau}^{LR,NGMM}\right| \left[1 - (1-\theta^{NGMM})e^{-\kappa^{NGMM}T}\right].$$

Rearranging each expression:

$$(1-\theta)e^{-\kappa T} = 1 - \frac{\widehat{\varepsilon}_I^{SR}}{\left|\varepsilon_{I,\tau}^{LR}\right|}. \tag{A.48}$$

Since $\left|\varepsilon_{I,\tau}^{LR,NGMM}\right| > \left|\varepsilon_{I,\tau}^{LR,NGM}\right|$, the right-hand side is larger for the NGMM:

$$(1-\theta^{NGMM})e^{-\kappa^{NGMM}T} > (1-\theta^{NGM})e^{-\kappa^{NGM}T}. \tag{A.49}$$

The left-hand side $(1-\theta)e^{-\kappa T}$ is monotonically decreasing in both $\theta$ and $\kappa$. By co-monotonicity, a higher value of this expression implies lower $\theta$ and lower $\kappa$. Therefore $\kappa^{NGMM} < \kappa^{NGM}$: the NGMM converges more slowly. □

The restriction $\theta \in (0,1)$ is substantive: it excludes impact overshooting, under which the monotonicity argument need not hold.

## G.2 Example with Quadratic Adjustment Costs

The intuition is fairly straightforward. Under the conditions of Proposition 9, if both models start from the same initial steady state and display the same short-run investment response, then the NGMM converges more slowly. That is because its long-run investment response is higher, which implies that the NGMM must be traveling more slowly along the adjustment path.

To give an example of the difference in convergence rates, consider the standard quadratic adjustment cost specification from Summers (1981) and Koby and Wolf (2020). A firm chooses sequences of investment, maintenance, labor, and capital to maximize the present value of after-tax profits:

$$\max_{\{I_t,M_t,L_t,K_{t+1}\}} \sum_{t=0}^{\infty} \left(\frac{1}{1+r^k}\right)^t \left\{(1-\tau_t^c)\left[F(K_t,L_t) - w_t L_t - \frac{\phi}{2}\left(\frac{I_t}{K_t} - \delta(m_t)\right)^2 K_t - p_t^M M_t\right] - (1-c_t - z_t\tau_t^c)p_t^I I_t\right\}, \tag{A.50}$$

subject to

$$K_{t+1} = (1 - \delta(m_t))K_t + I_t.$$

The parameter $\phi$ governs how costly it is to deviate from replacement investment $\delta(m_t)K_t$. Higher $\phi$ raises the cost of adjustment. Note that this specification implies maintenance adjusts instantaneously—I rely on capital adjustment costs because it allows for easier comparison with Chodorow-Reich et al. (2025), who use the same specification.

Letting $q_t$ denote the shadow value of capital, the first-order conditions for investment and maintenance are:

$$(1 - \tau_t^c)\phi\left(\frac{I_t}{K_t} - \delta(m_t)\right) = q_t - (1 - c_t - z_t\tau_t^c)p_t^I, \tag{A.51}$$

$$-\delta'(m_t) = \frac{1 - \tau_t^c}{1 - c_t - z_t\tau_t^c}\frac{p_t^M}{p_t^I}. \tag{A.52}$$

The maintenance condition is static and identical to the model without adjustment costs. The investment condition shows that the gap between the investment rate and the depreciation rate is proportional to the gap between the shadow value $q_t$ and the after-tax purchase price.

Define $i \equiv I/K$. Rearranging the investment FOC:

$$i = \frac{q - (1 - c - z\tau^c)p^I}{(1 - \tau^c)\phi} + \delta(m). \tag{A.53}$$

In steady state, $i = \delta(m)$, so the investment rate equals the depreciation rate. Taking a first-order approximation around the initial steady state gives

$$di = \frac{dq - d\left[(1 - c - z\tau^c)p^I\right]}{(1 - \tau^c)\phi} + d\delta(m).$$

The change in the denominator contributes no first-order term because $q = (1 - c - z\tau^c)p^I$ at the initial steady state. Using the marginal tax rate defined in the main text, this is equivalently

$$di = \frac{1}{\phi}\left[d\left(\frac{q}{1 - \tau^c}\right) - d\left(\frac{p^I}{1 - \tau}\right)\right] + d\delta(m).$$

At a zero-wedge steady state with $p^I = 1$, $q/(1 - \tau^c) = 1$. Holding the investment price fixed and dividing by $i = \delta(m)$, the first-order approximation becomes

$$d\log i \approx \frac{1}{\delta(m)\phi}\left[d\log\left(\frac{q}{1 - \tau^c}\right) + d\log(1 - \tau)\right] + d\log\delta(m).$$

Expressing the response relative to the net-of-tax term gives

$$\frac{d\log i}{d\log(1-\tau)} \approx \frac{1}{\delta(m)\phi}\left[1+\frac{d\log(q/(1-\tau^c))}{d\log(1-\tau)}\right]+\frac{d\log\delta(m)}{d\log(1-\tau)}.$$

The key insight is that the investment-rate response has two components: the response through the shadow-value gap and the indirect effect through endogenous depreciation. Only the first component contains the adjustment-cost parameter; the second is a consequence of elastic maintenance demand. On impact, capital is predetermined, so the investment-rate response is also the response of investment.

## G.3 The Dynamic Effects of the Tax Cuts and Jobs Act

The remainder of this appendix compares the dynamic responses of investment, maintenance, capital, and output across the NGM and NGMM models across different closures. I start by discussing how I calibrate the model, then wrap up by comparing the 10-year budgetary effects.

### Calibration

I retain the steady-state calibration from the main text. The SOI estimates pin down the maintenance demand parameters $(\gamma, \omega)$, while $\delta_0$ and non-corporate productivity are calibrated to match the pre-TCJA investment-output ratio and the corporate share of capital. Within each closure, I calibrate the NGM benchmark so that both models start from the same capital stocks, private output, and wages. This allows the comparison to isolate how maintenance changes adjustment following the reform.

The additional parameters needed to characterize the transition are the adjustment costs. Let $|\widehat{\varepsilon_I}|$ denote the magnitude of the empirical investment coefficient used in the main text, with point estimate 4.27. In the NGM, I use this coefficient directly. In the NGMM, I apply the maintenance correction,

$$|\widetilde{\varepsilon_I}| = \frac{|\widehat{\varepsilon_I}|}{1-s_m}.$$

For the reported exercise, I use the following calibration rules:

$$\phi^{NGM} = \frac{1}{\delta|\widehat{\varepsilon_I}|}, \qquad \phi^{NGMM} = \frac{1}{\delta(m)\left(|\widetilde{\varepsilon_I}| - |\widehat{\varepsilon_\delta}|\right)}.$$

Here, $|\widehat{\varepsilon_\delta}|$ is the local depreciation elasticity with respect to $1-\tau$, evaluated at the zero-wedge benchmark holding input prices fixed. I apply these rules separately in each private sector.

These expressions specify how the empirical coefficients enter the adjustment-cost calibration. They do not impose that the two models deliver the same investment response to the reform.

Given the calibrated parameters, I solve jointly for investment and the shadow value of capital along the transition, allowing $q/(1-\tau^c)$ to respond endogenously. The preceding first-order expression describes that response but does not, by itself, identify $\phi$.

I then apply the same TCJA experiment as in the main text. The corporate tax changes and bonus depreciation are permanent, while non-corporate tax provisions remain unchanged. Starting from the initial steady state, I solve the perfect-foresight transition following an unanticipated reform. I consider the same four closures: partial equilibrium, general equilibrium with fixed aggregate labor supply, general equilibrium with fiscal crowding out, and general equilibrium with a competitive maintenance sector.

To propagate uncertainty, I conduct 5,000 paired parameter draws. For each draw, I draw the maintenance elasticity and recalibrate $\gamma$, draw the empirical investment coefficient, and recalibrate the initial steady states and adjustment costs. For the crowding-out closure, I also draw the coefficient relating government debt to interest rates. I use the same draws across models and closures, holding the remaining calibration targets fixed. The resulting transition paths determine capital and output at the ten-year horizon and the cumulative revenue feedback over the scoring window.

**Scoring TCJA**

Transition dynamics matter for how we score tax reform. Government bodies like the Congressional Budget Office and the Joint Committee on Taxation publish budgetary and macroeconomic effects over a ten-year horizon. Static scores hold aggregate output fixed. Dynamic scores account for the extra output, and hence the extra revenue, from growth effects. Because convergence typically takes decades, the transition path matters for revenue over the scoring window.

To score the reform, I follow Barro and Furman (2018). For both models, I adjust the CBO's static output projections using the model-implied growth path. The extra output generates additional tax revenue given the CBO's projected corporate revenue share. I express results as the percent of the bill's static cost offset by this additional revenue. If the offset is 100%, the bill pays for itself.

Table G.1 presents ten-year outcomes for corporate capital accumulation, aggregate output growth, and the static-score offset across a variety of models.

Table G.1: Ten-Year Corporate Capital, Aggregate Output, and Scores

| | | $\Delta K_c/K_c$ (%) | | $\Delta Y/Y$ (%) | | Static-Score Offset (%) | |
|---|---|---|---|---|---|---|---|
| Closure | Model | 10Y | Ratio | 10Y | Ratio | 10Y | Ratio |
| PE | NGM | 8.8 | - | 3.25 | - | 37.5 | - |
| | NGMM | 6.4 | 0.73 | 2.37 | 0.73 | 26.6 | 0.71 |
| GE | NGM | 5.5 | - | 0.56 | - | 7.0 | - |
| | NGMM | 4.1 | 0.75 | 0.43 | 0.76 | 5.2 | 0.74 |
| GE + crowding out | NGM | 3.6 | - | 0.21 | - | 2.6 | - |
| | NGMM | 2.3 | 0.65 | 0.10 | 0.46 | 1.1 | 0.44 |
| GE + competitive maintenance sector | NGM | 5.3 | - | 0.53 | - | 6.6 | - |
| | NGMM | 3.8 | 0.72 | 0.61 | 1.15 | 8.9 | 1.36 |

**Notes:** NGM excludes maintenance; NGMM includes maintenance. GE denotes general equilibrium and PE partial equilibrium. Ratios divide the NGMM mean by the NGM mean within each closure. Capital and GDP are changes at year ten; the static-score offset is cumulative growth-induced revenue divided by the static cost over the ten-year window.

Means use 5,000 paired draws with $\omega > 1.3$; no draws are dropped. The ten-year corporate-capital gap in the baseline GE closure is approximately $248B using the 2017 BEA corporate-capital base.

The competitive maintenance sector uses labor and materials with labor elasticity $\eta = 0.30$. Its NGM benchmark matches initial capital, private GDP, and wages; GDP is net of intermediate materials. The existing first-order adjustment-cost calibration rule is retained, with parameters recomputed under this closure's initial steady-state targets.

**Baseline Results (GE).** In the baseline general equilibrium model, the NGM predicts corporate capital rises by 5.5% and aggregate output by 0.56% over ten years, offsetting 7.0% of the static cost. The NGMM predicts capital rises by 4.1% and output by 0.43%, offsetting only 5.2%. The bill does not pay for itself under either model. The ratios are stable across outcomes: NGMM responses are approximately 74–76% as large as NGM responses. Against an $18.3 trillion corporate capital base, the capital gap implies roughly $248 billion less corporate capital under the maintenance model.

**Crowding Out.** This closure incorporates fiscal crowding out. I compute the 2027 debt-GDP ratio implied by TCJA's fiscal cost and endogenous revenue feedback, then calculate the resulting increase in real interest rates. Higher rates reduce private investment. Crowding out affects both

models but exacerbates the proportional difference: the NGMM offset falls to 1.1% while the NGM offset falls to 2.6%. The ratio of NGMM to NGM responses falls from 0.74–0.76 to 0.44–0.65 with crowding out. Both models continue to predict positive capital accumulation, but the absolute capital gap narrows even as the proportional reduction deepens.

**Partial Equilibrium (PE).** This closure assumes perfectly elastic labor supply. Aggregate outcomes are substantially larger. The NGM offset rises to 37.5%, compared with 26.6% in the NGMM. Nevertheless, the ratio of NGMM to NGM responses remains similar to the baseline GE model, ranging from 0.71 to 0.73 across capital, output, and revenue feedback.

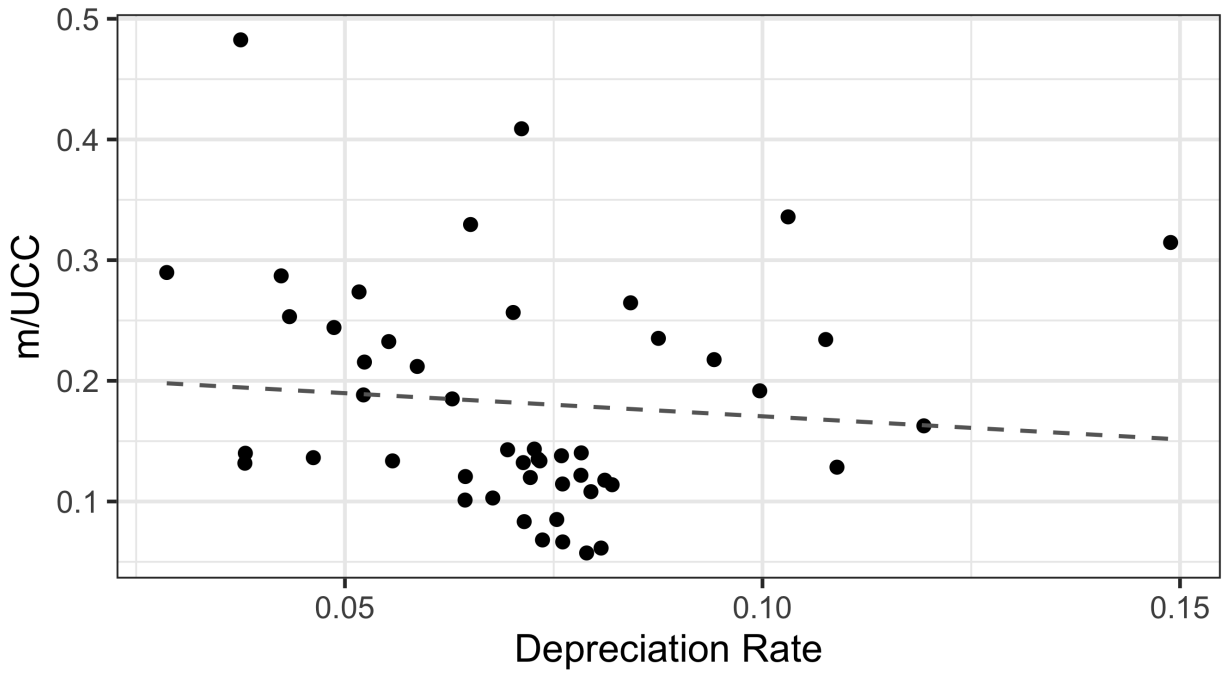
**Competitive Maintenance Sector.** The final closure allows maintenance to be supplied by a competitive sector using labor and materials as defined in Appendix **??**. Accounting for maintenance continues to reduce capital accumulation: corporate capital rises by 5.3% in the NGM and 3.8% in the NGMM, a reduction of about 28%. However, the comparison for output and revenue feedback reverses. Aggregate output rises by 0.53% in the NGM and 0.61% in the NGMM, while the static-score offset rises from 6.6% to 8.9%.

The intuition is straightforward. Reducing maintenance releases labor for corporate and noncorporate production and reduces the materials used to maintain existing capital. These resource effects offset the smaller capital stock. Moreover, maintenance adjusts immediately, so the resources become available before capital has finished adjusting. The capital preservation effect therefore continues to attenuate capital accumulation, but it need not attenuate aggregate output or revenue feedback when maintenance itself uses scarce productive inputs.

**Interpretation.** Given how little of the static cost is offset in the general equilibrium closures, investment-driven growth provides limited self-financing for capital tax cuts. In the baseline model, maintenance reduces this already-small revenue feedback by roughly a quarter, and by more than half once crowding out is included. The competitive maintenance-sector closure shows that this revenue comparison depends on how maintenance is supplied. Across the closures, the smaller capital response remains, while the implications for output and dynamic scoring also reflect the resources released by reduced maintenance.

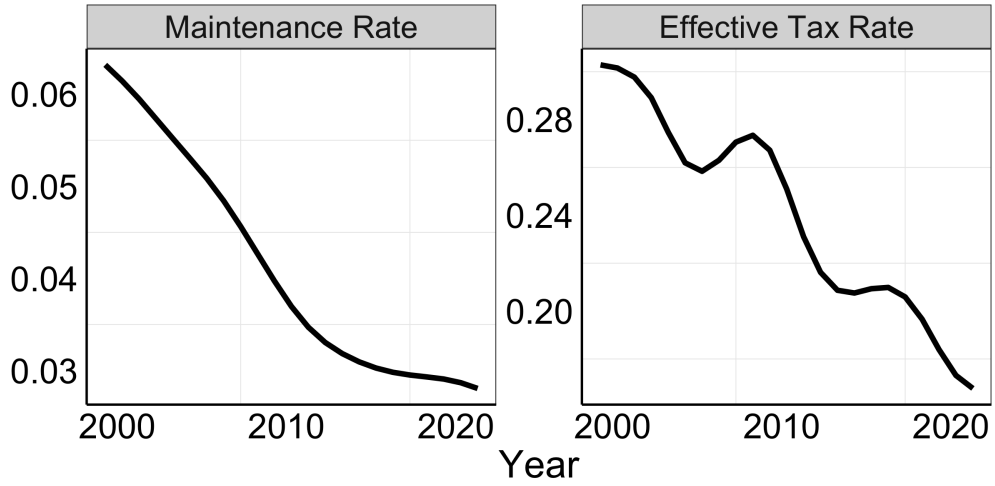
# H Additional Figures and Tables

Figure H.1: Industry maintenance share of user cost against the depreciation rate




**Notes:** Each point is the time series average of maintenance shares and depreciation rates within an industry.

Figure H.2: Trends in Maintenance and Corporate Tax Rates




**Notes:** The maintenance rate is constructed as gross output in NAICS 811 excluding home repair as a share of current cost private equipment capital. The effective tax rate is the ratio of domestic corporate taxes paid to pre-tax profits from BEA Tables 6.16D and 6.17D. The cyclical component of each series has been removed with a Hodrick-Prescott filter.